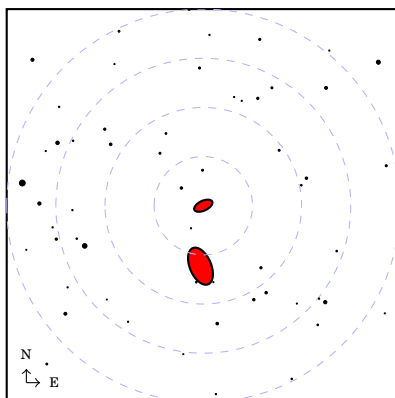
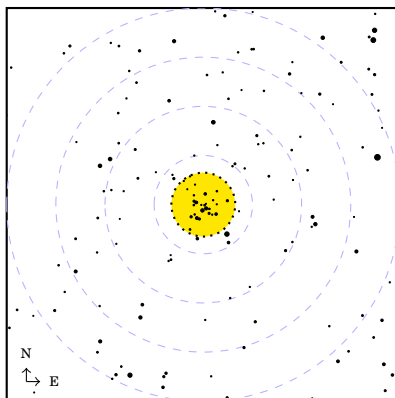


Pocket Finder-Chart Atlas



Inverted-Image Version

Alan Watson Forster

Pocket Finder-Chart Atlas

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Inverted-image version of 2022-03-26

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Preface

This is an atlas of finder charts for the Messier, Caldwell, and Astronomical League Urban Observing Program deep-sky objects.

The charts have a field of 4 degrees at a scale of 13 mm per degree. Circles show fields with diameters of 1, 2, 3, and 4 degrees. The charts show stars to magnitude 9.5, and represent deep-sky objects following the current convention. The correspondence between the magnitude of a star and its size on the charts is designed to better match the view in a telescope eyepiece.

There are versions of the atlas with inverted-image and correct-image charts. This version has inverted-image charts, with north up and east to the right, corresponding to the view through a Newtonian telescope or a refractor or Cassegrain telescope with a conventional diagonal.

The size and binding of this atlas are designed for convenience at the telescope.

Chart Legend

Stars			
• 1 mag		Open Cluster	0° 1° 2° 3° 4°
• 2 mag		Globular Cluster	
• 3 mag		Star Cloud or Asterism	
• 4 mag		Bright Nebula	
• 5 mag		Planetary Nebula	
• 6 mag		Dark Nebula	
• 7 mag		Galaxy	
• 8 mag			
• 9 mag			

I created this atlas to assist me as I observed under the light-polluted skies of Mexico City with a 70 mm f/6 refractor. I don't use a conventional finder with this telescope, but instead star-hop using a 32 mm eyepiece with a true field of about 4 degrees. Under my usual observing conditions, very few objects are immediately obvious, and the charts tell me when I've successfully reached the desired field and where to focus my efforts.

The obvious question is, but why not just use an all-sky atlas? I do indeed use the *Pocket Sky Atlas* for star-hopping, but I find that for confirming a field none of my all-sky atlases combine adequate depth with convenience at the telescope, and none really give me the sense of what I see through the eyepiece.

To some degree, the relation between these charts and the *Pocket Sky Atlas* is similar to the relation between the larger-scale and smaller-scale charts in *The Observer's Sky Atlas*. The smaller-scale charts are for star-hopping to the field and the larger-scale charts are to confirm the field and locate the object. I don't use *The Observer's Sky Atlas* at the telescope, because the smaller-scale charts are too shallow and too narrow for my needs, the large-scale charts aren't inverted, and the binding is inconvenient.

The positions and magnitudes of stars are taken from the Tycho-2 catalog, supplemented by the Hipparcos catalog for the very brightest stars. The positions, types, and sizes of deep sky objects are taken from the CDS SIMBAD database, supplemented by information from the NASA/IPAC Extragalactic Database (NED), O'Meara's *Deep-Sky Companions* volumes, and the Astronomical League's Urban Observing Program list.

One flaw of these charts is that all deep-sky objects are drawn as either ellipses (galaxies and star clouds) or circles (everything else). This means, for example, that they do not correctly show the shapes of irregular nebulae such as M42 and M43.

The Messier Objects

The Messier objects are probably the most famous deep-sky objects. I find O'Meara's *Deep-Sky Companion: The Messier Objects* to be excellent on the origin of the catalog, the appearance of the objects, and their nature.

The following table lists the objects with their J2000 positions (decimal hours of right ascension and degrees of declination), the charts on which they appear in the *Pocket Sky Atlas*, their types, and other names. I follow O'Meara in identifying M102 as NGC 5866.

For completeness, I include finder charts for all of the Messier objects, even bright ones like M45.

Name	Position	PSA	Type	Other Names
M1	05.6 +22	14	BN	NGC 1952 = Crab Nebula
M2	21.6 -01	75/77	GC	U81 = NGC 7089
M3	13.7 +28	43/44	GC	U51 = NGC 5272
M4	16.4 -27	56/58	GC	U53 = NGC 6121
M5	15.3 +02	55/57	GC	U52 = NGC 5904
M6	17.7 -32	58/69	OC	U60 = NGC 6405 = Butterfly Nebula
M7	17.9 -35	58/69	OC	U62 = NGC 6475
M8	18.1 -24	67	BN	U64 = NGC 6523/6530 = Lagoon Nebula
M9	17.3 -19	56	GC	NGC 6333
M10	17.0 -04	54/56	GC	U57 = NGC 6254
M11	18.9 -06	65/67	OC	U69 = NGC 6705 = Wild Duck Cluster
M12	16.8 -02	54/56	GC	U56 = NGC 6218
M13	16.7 +36	52	GC	U54 = NGC 6205 = Hercules Cluster
M14	17.6 -03	54	GC	NGC 6402
M15	21.5 +12	75	GC	U80 = NGC 7078
M16	18.3 -14	67	BN	NGC 6611 = Eagle Nebula
M17	18.3 -16	67	BN	U65 = NGC 6618 = Omega Nebula
M18	18.3 -17	67	OC	NGC 6613
M19	17.0 -26	56	GC	NGC 6273
M20	18.0 -23	67	BN	NGC 6514 = Trifid Nebula
M21	18.1 -23	67	OC	NGC 6531
M22	18.6 -24	67	GC	U67 = NGC 6656
M23	17.9 -19	67	OC	NGC 6494
M24	18.3 -19	67	SC	IC 4715
M25	18.5 -19	67	OC	IC 4725
M26	18.8 -09	67	OC	NGC 6694
M27	20.0 +23	64	PN	U75 = NGC 6853 = Dumbell Nebula
M28	18.4 -25	67/I	GC	NGC 6626
M29	20.4 +38	62	OC	NGC 6913
M30	21.7 -23	77	GC	NGC 7099
M31	00.7 +41	3	Gal	U3 = NGC 224 = Andromeda Galaxy
M32	00.7 +41	3	Gal	U2 = NGC 221
M33	01.6 +31	3	Gal	NGC 598
M34	02.7 +43	2	OC	NGC 1039
M35	06.2 +24	14	OC	U27 = NGC 2168
M36	05.6 +34	12	OC	U23 = NGC 1960
M37	05.9 +33	12	OC	U26 = NGC 2099
M38	05.5 +36	12	OC	U22 = NGC 1912
M39	21.5 +48	73	OC	U82 = NGC 7092
M40	12.4 +58	32	DS	

Name	Position	PSA	Type	Other Names
M41	06.8 −21	27	OC	U33 = NGC 2287
M42	05.6 −05	16/B	BN	U24 = NGC 1976 = Orion Nebula
M43	05.6 −05	16/B	BN	NGC 1982
M44	08.7 +20	24	OC	U39 = NGC 2632 = Beehive Cluster
M45	03.8 +24	15/A	OC	U17 = Mel 22 = Pleiades
M46	07.7 −15	26	OC	NGC 2437
M47	07.6 −14	27	OC	NGC 2422
M48	08.2 −06	26	OC	U38 = NGC 2548
M49	12.5 +08	45/C	Gal	NGC 4472
M50	07.0 −08	27	OC	U35 = NGC 2323
M51	13.5 +47	43	Gal	NGC 5194 = Whirlpool Galaxy
M52	23.4 +62	72	OC	NGC 7654
M53	13.2 +18	45	GC	NGC 5024
M54	18.9 −30	69	GC	NGC 6715
M55	19.7 −31	68	GC	NGC 6809
M56	19.3 +30	63	GC	NGC 6779
M57	18.9 +33	63	PN	U71 = NGC 6720 = Ring Nebula
M58	12.6 +12	45/C	Gal	NGC 4579
M59	12.7 +12	45/C	Gal	NGC 4621
M60	12.7 +12	45/C	Gal	NGC 4649
M61	12.4 +04	45	Gal	NGC 4303
M62	17.0 −30	58	GC	U58 = NGC 6266
M63	13.3 +42	43	Gal	NGC 5055 = Sunflower Galaxy
M64	12.9 +22	45	Gal	U50 = NGC 4826 = Black-Eye Galaxy
M65	11.3 +13	34	Gal	NGC 3623
M66	11.3 +13	34	Gal	NGC 3627
M67	08.9 +12	24	OC	U40 = NGC 2682
M68	12.7 −27	47	GC	NGC 4590
M69	18.5 −32	67	GC	NGC 6637
M70	18.7 −32	67	GC	NGC 6681
M71	19.9 +19	64	GC	NGC 6838
M72	20.9 −13	66/77	GC	NGC 6981
M73	21.0 −13	66/77	AST	NGC 6994
M74	01.6 +16	4/5	Gal	NGC 628
M75	20.1 −22	66	GC	NGC 6864
M76	01.7 +52	2	PN	NGC 651/650 = Little Dumbell
M77	02.7 −00	4	Gal	U12 = NGC 1068
M78	05.8 +00	16	BN	NGC 2068
M79	05.4 −25	16	GC	NGC 1904
M80	16.3 −23	56	GC	NGC 6093

Name	Position	PSA	Type	Other Names
M81	09.9 +69	31	Gal	U41 = NGC 3031
M82	09.9 +70	31	Gal	U42 = NGC 3034
M83	13.6 −30	47/48	Gal	NGC 5236
M84	12.4 +13	45/C	Gal	U45 = NGC 4374
M85	12.4 +18	45/C	Gal	NGC 4382
M86	12.4 +13	45/C	Gal	U46 = NGC 4406
M87	12.5 +12	45/C	Gal	U47 = NGC 4486
M88	12.5 +14	45/C	Gal	NGC 4501
M89	12.6 +13	45/C	Gal	NGC 4552
M90	12.6 +13	45/C	Gal	NGC 4569
M91	12.6 +14	45/C	Gal	NGC 4548
M92	17.3 +43	52	GC	U59 = NGC 6341
M93	07.7 −24	26	OC	NGC 2447
M94	12.8 +41	43	Gal	U49 = NGC 4736
M95	10.7 +12	34	Gal	NGC 3351
M96	10.8 +12	34	Gal	NGC 3368
M97	11.2 +55	32	PN	NGC 3587
M98	12.2 +15	45/C	Gal	NGC 4192
M99	12.3 +14	45/C	Gal	NGC 4254
M100	12.4 +16	45/C	Gal	NGC 4321
M101	14.1 +54	42	Gal	NGC 5457
M102	15.1 +56	42	Gal	NGC 5866
M103	01.6 +61	3	OC	NGC 581
M104	12.7 −12	47	Gal	U48 = NGC 4594 = Sombrero Galaxy
M105	10.8 +13	34	Gal	NGC 3379
M106	12.3 +47	34	Gal	NGC 4258
M107	16.5 −13	56	GC	NGC 6171
M108	11.2 +56	32/33	Gal	NGC 3556
M109	12.0 +53	32	Gal	NGC 3992
M110	00.7 +42	3	Gal	NGC 205

M1

NGC 1952 = Crab Nebula

RA : 05.6^h PSA: 14

Dec: +22° OSA: E3

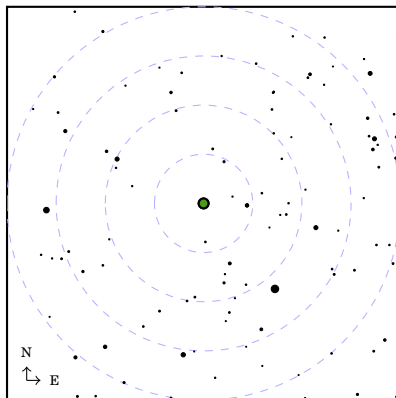
Con: Tau CSA: 3/9

See: DSC:TMO OSA

Type: BN

Mag : 8.0

Size : 6' × 4'



M2

U81 = NGC 7089

RA : 21.6^h PSA: 75/77

Dec: -01° OSA: E24

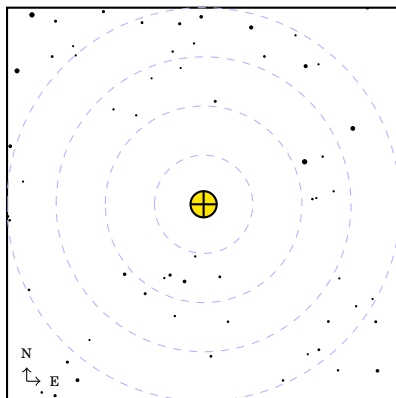
Con: Aqr CSA: 13

See: DSC:TMO OSA

Type: GC

Mag : 6.3

Size : 16'



M3

U51 = NGC 5272

RA : 13.7^h PSA: 43/44

Dec: +28° OSA: E15

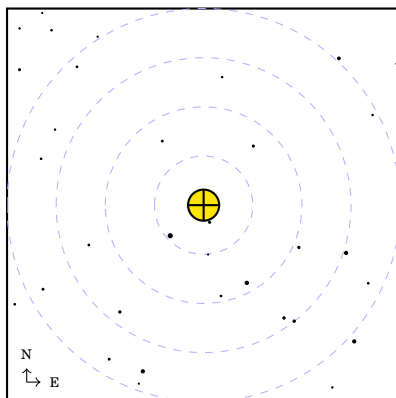
Con: CVn CSA: 5

See: DSC:TMO OSA

Type: GC

Mag : 5.9

Size : 19'



M4

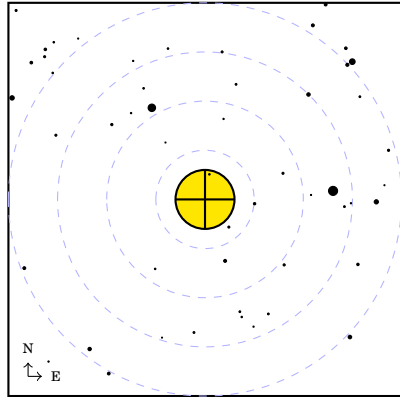
U53 = NGC 6121

RA : 16.4 ^h	PSA: 56/58	Type: GC
Dec: -27°	OSA: E18	Mag : 5.4
Con: Sco	CSA: 11/12/17/18	Size : 36'

See: DSC:TMO OSA

α Sco Antares (mag 0.9–1.1)

→ 1.2° W to M4

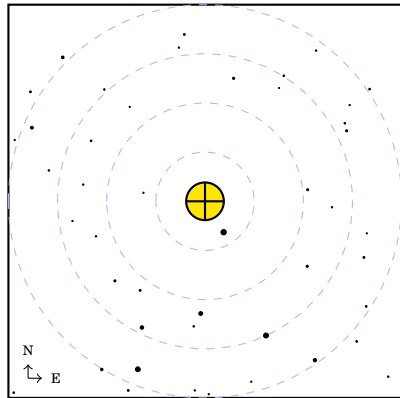


M5

U52 = NGC 5904

RA : 15.3 ^h	PSA: 55/57	Type: GC
Dec: +02°	OSA: E15	Mag : 5.7
Con: Ser	CSA: 11/12	Size : 23'

See: DSC:TMO OSA



M6

U60 = NGC 6405 = Butterfly Nebula

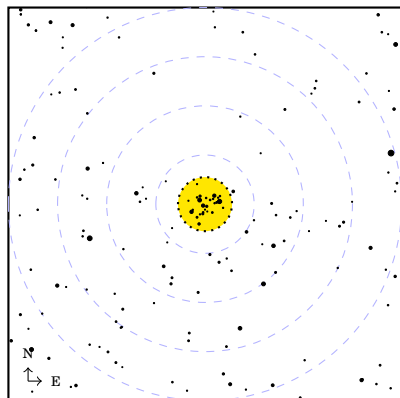
RA : 17.7 ^h	PSA: 58/69	Type: OC
Dec: -32°	OSA: E18	Mag : 4.2
Con: Sco	CSA: 18	Size : 33'

See: DSC:TMO OSA

λ Sco Shaula (mag 1.6)

→ 5° N

→ 1° E to M6/U60



M7

U62 = NGC 6475

RA : 17.9^h PSA: 58/69 Type: OC

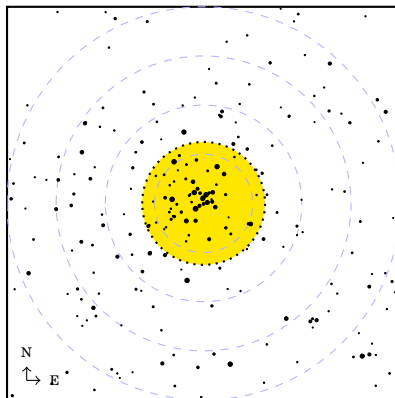
Dec: -35° OSA: E18 Mag : 2.8

Con: Sco CSA: 18 Size : 75'

See: DSC:TMO OSA

ε Sgr Kaus Australis (mag 1.8)

→ 6° W to M7/U62



M8

U64 = NGC 6523/6530 = Lagoon Nebula

RA : 18.1^h PSA: 67 Type: BN

Dec: -24° OSA: E20 Mag : 3.0

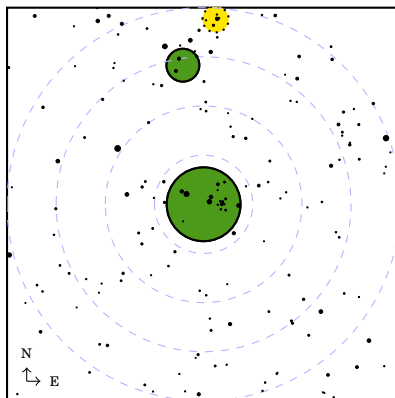
Con: Sgr CSA: 12/18 Size : 45' × 30'

See: DSC:TMO OSA

Notes: M20 is 1.5° N. M21 is 2.0° N.

γ Sgr Alnasl (mag 3.0)

→ 6° N to M8/U64



M9

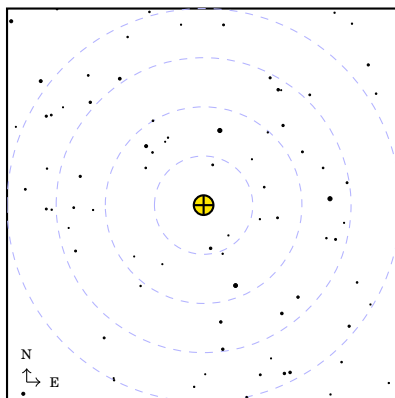
NGC 6333

RA : 17.3^h PSA: 56 Type: GC

Dec: -19° OSA: E17 Mag : 7.8

Con: Oph CSA: 12/18 Size : 12'

See: DSC:TMO OSA



M10

U57 = NGC 6254

RA : 17.0^h PSA: 54/56

Dec: -04° OSA: E17

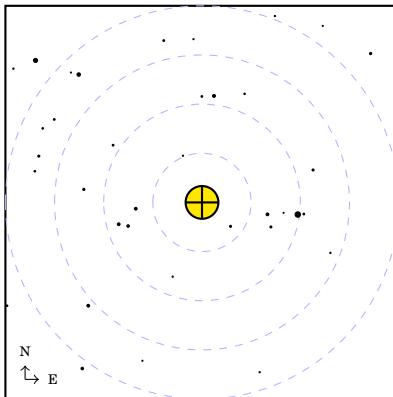
Con: Oph CSA: 12

Type: GC

Mag: 6.6

Size : 20'

See: DSC:TMO OSA



M11

U69 = NGC 6705 = Wild Duck Cluster

RA : 18.9^h PSA: 65/67

Dec: -06° OSA: E19

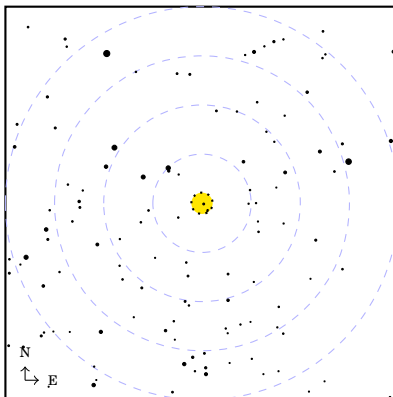
Con: Sct CSA: 12

Type: OC

Mag: 5.3

Size : 13'

See: DSC:TMO OSA



M12

U56 = NGC 6218

RA : 16.8^h PSA: 54/56

Dec: -02° OSA: E17

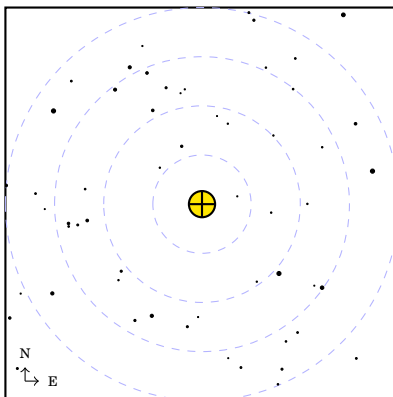
Con: Oph CSA: 12

Type: GC

Mag: 6.1

Size : 16'

See: DSC:TMO OSA



M13

U54 = NGC 6205 = Hercules Cluster

RA : 16.7^h PSA: 52 Type: GC

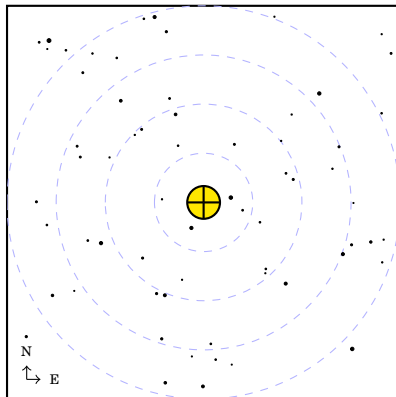
Dec: +36° OSA: N14 Mag : 5.3

Con: Her CSA: 5/6 Size : 20'

See: DSC:TMO OSA

ζ Her (mag 2.8)

→ 5° N (2/3 distance to η Her) to M13/U54



M14

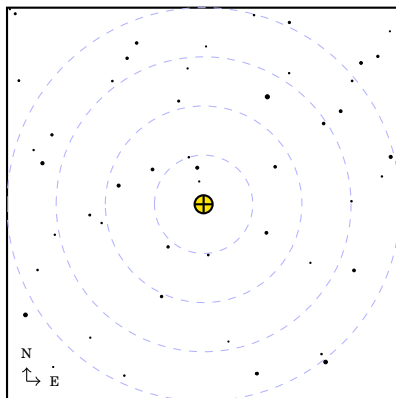
NGC 6402

RA : 17.6^h PSA: 54 Type: GC

Dec: -03° OSA: E17 Mag : 7.6

Con: Oph CSA: 12 Size : 11'

See: DSC:TMO OSA



M15

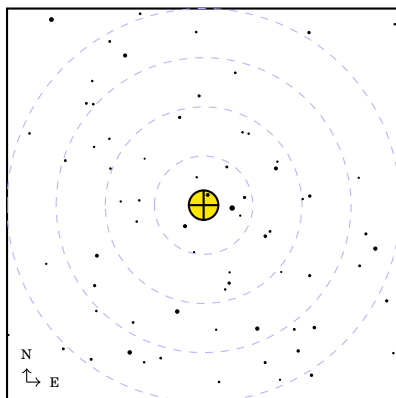
U80 = NGC 7078

RA : 21.5^h PSA: 75 Type: GC

Dec: +12° OSA: E23 Mag : 6.0

Con: Peg CSA: 13 Size : 18'

See: DSC:TMO OSA



M16

NGC 6611 = Eagle Nebula

RA : 18.3 ^h	PSA: 67	Type: BN
Dec: -14°	OSA: E20	Mag : 6.0
Con: Ser	CSA: 12	Size : 35' × 28'

See: DSC:TMO OSA

v Oph (mag 3.3)

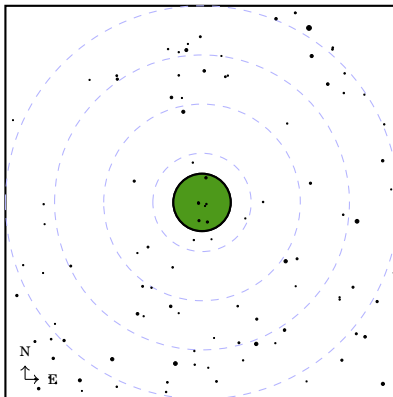
→ 4° E

→ 4° S to M16

μ Sgr (mag 3.8)

→ 7° N

→ 1° E to M16



M17

U65 = NGC 6618 = Omega Nebula

RA : 18.3 ^h	PSA: 67	Type: BN
Dec: -16°	OSA: E20	Mag : 6.0
Con: Sgr	CSA: 12	Size : 20' × 15'

See: DSC:TMO OSA

Notes: M18 is 1° S. M24 is 3° SW.

v Oph (mag 3.3)

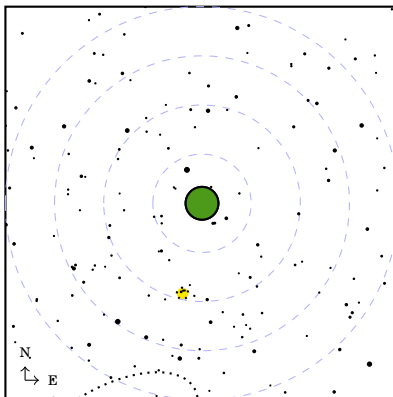
→ 5° E

→ 6° S to M17/U65

μ Sgr (mag 3.8)

→ 5° N passing M24 and M18

→ 2° E to M17



M18

NGC 6613

RA : 18.3 ^h	PSA: 67	Type: OC
Dec: -17°	OSA: E20	Mag : 6.9
Con: Sgr	CSA: 12/18	Size : 7'

See: DSC:TMO OSA

Notes: M17/U65 is 1° N. M24 is 2° SW.

v Oph (mag 3.3)

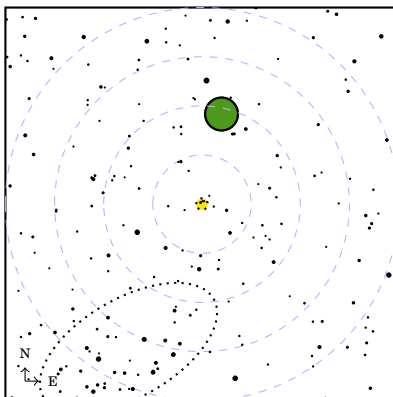
→ 5° E

→ 7° S passing M17/U65 to M18

μ Sgr (mag 3.8)

→ 4° N passing M24

→ 2° E to M18



M19

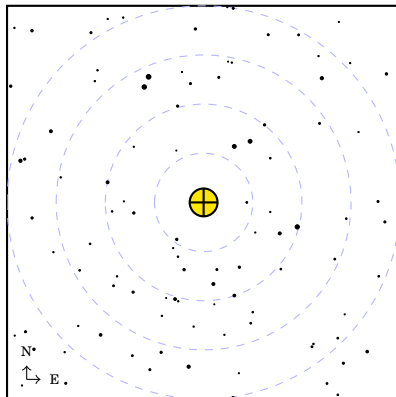
NGC 6273

RA : 17.0^h PSA: 56 Type: GC

Dec: -26° OSA: E18 Mag : 6.8

Con: Oph CSA: 12/18 Size : 17'

See: DSC:TMO OSA



M20

NGC 6514 = Trifid Nebula

RA : 18.0^h PSA: 67 Type: BN

Dec: -23° OSA: E20 Mag : 6.3

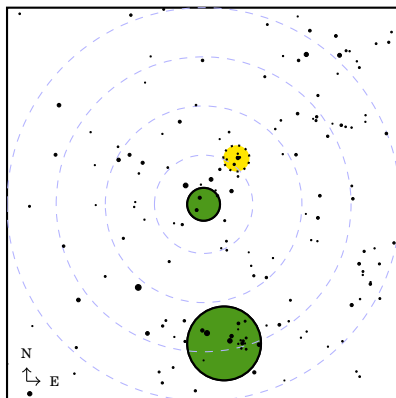
Con: Sgr CSA: 12/18 Size : 20'

See: DSC:TMO OSA

Notes: M8/U64 is 1.5° S. M21 is 0.5° NE.

γ Sgr Alnasl (mag 3.0)

→ 7.5° N passing M8/U64 to M20



M21

NGC 6531

RA : 18.1^h PSA: 67 Type: OC

Dec: -23° OSA: E20 Mag : 5.9

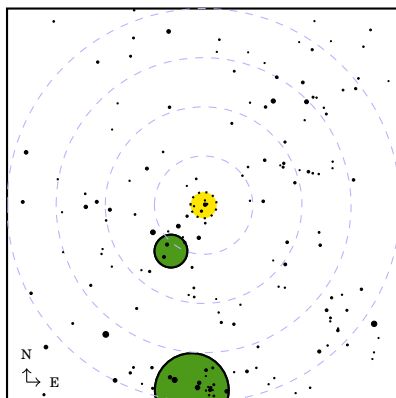
Con: Sgr CSA: 12/18 Size : 16'

See: DSC:TMO OSA

Notes: M8/U64 is 2.0° S. M20 is 0.5° SW.

γ Sgr Alnasl (mag 3.0)

→ 8.0° N passing M8/U64 and M20 to M21



M22

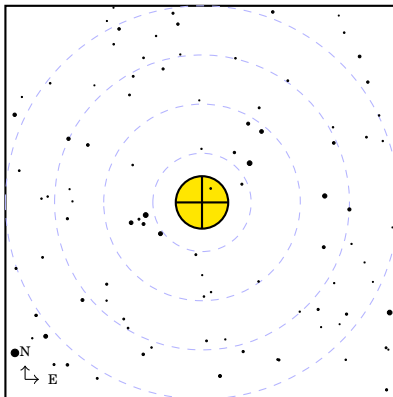
U67 = NGC 6656

RA : 18.6^h PSA: 67 Type: GC

Dec: -24° OSA: E20 Mag: 5.2

Con: Sgr CSA: 12/18 Size : 32'

See: DSC:TMO OSA



M23

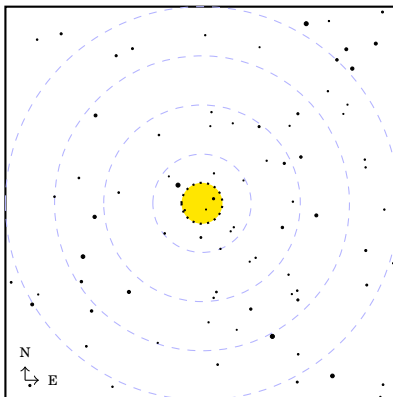
NGC 6494

RA : 17.9^h PSA: 67 Type: OC

Dec: -19° OSA: E20 Mag: 5.5

Con: Sgr CSA: 12/18 Size : 25'

See: DSC:TMO OSA



M24

IC 4715

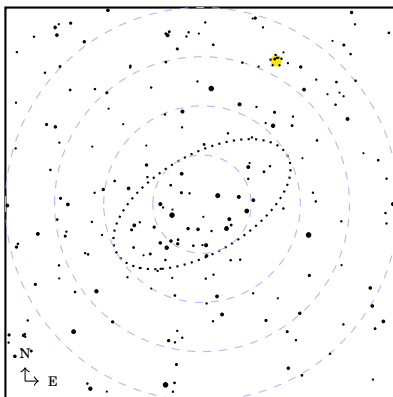
RA : 18.3^h PSA: 67 Type: SC

Dec: -19° OSA: E20 Mag: 2.5

Con: Sgr CSA: 12/18 Size : 120' × 60'

See: DSC:TMO OSA

Notes: M18 is 1.5° NE. M17/U65 is 3° NE.



M25

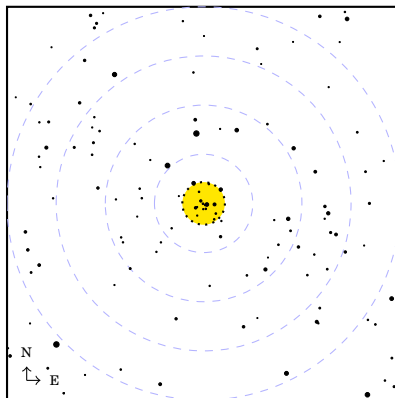
IC 4725

RA : 18.5^h PSA: 67 Type: OC

Dec: -19° OSA: E20 Mag : 4.6

Con: Sgr CSA: 12/18 Size : 26'

See: DSC:TMO OSA



M26

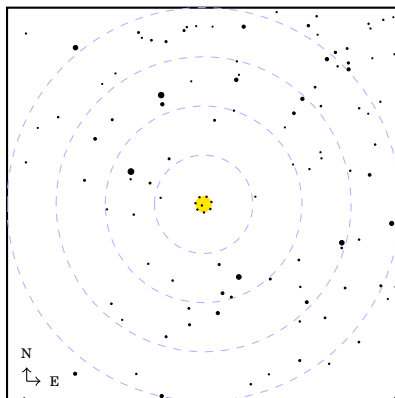
NGC 6694

RA : 18.8^h PSA: 67 Type: OC

Dec: -09° OSA: E19 Mag : 8.0

Con: Sct CSA: 12 Size : 10'

See: DSC:TMO OSA



M27

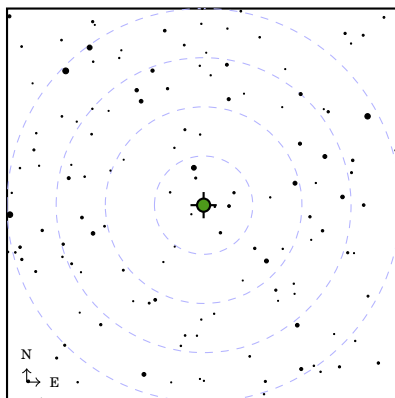
U75 = NGC 6853 = Dumbell Nebula

RA : 20.0^h PSA: 64 Type: PN

Dec: +23° OSA: E21 Mag : 7.3

Con: Vul CSA: 6/7/12/13 Size : 8' × 6'

See: DSC:TMO OSA



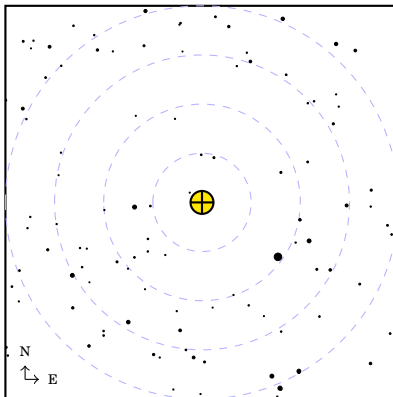
M28

NGC 6626

RA : 18.4^h PSA: 67/1
Dec: -25° OSA: E20
Con: Sgr CSA: 12/18

Type: GC
Mag: 6.9
Size : 14'

See: DSC:TMO OSA



M29

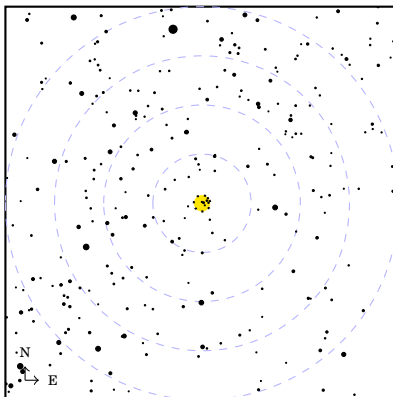
NGC 6913

RA : 20.4^h PSA: 62
Dec: +38° OSA: N20
Con: Cyg CSA: 6/7

Type: OC
Mag: 6.6
Size : 10'

See: DSC:TMO OSA

γ Cyg Sadr (mag 2.2)
→ 1.8° S to M29



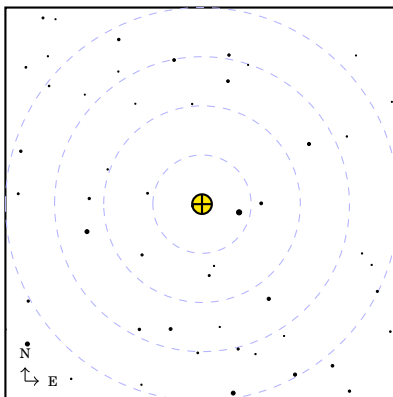
M30

NGC 7099

RA : 21.7^h PSA: 77
Dec: -23° OSA: E22
Con: Cap CSA: 13/19

Type: GC
Mag: 6.9
Size : 12'

See: DSC:TMO OSA



M31

U3 = NGC 224 = Andromeda Galaxy

RA : 00.7 ^h	PSA: 3	Type: Gal
Dec: +41°	OSA: N0	Mag : 3.4
Con: And	CSA: 2/7	Size : 3.0° × 1.0°

See: DSC:TMO OSA

Notes: M32/U2 is 0.5° S. M100 is 0.5° NW.

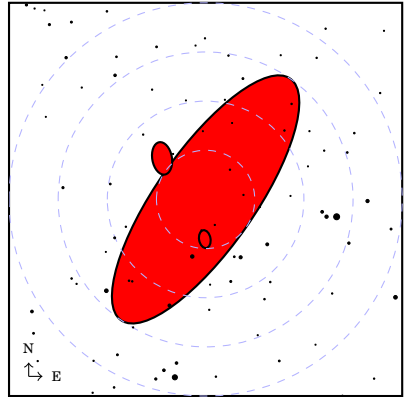
β And Mirach (mag 2.1)

→ 2.5° W

→ 3.0° N to μ And (mag 3.9)

→ 2.5° W

→ 3.0° N to M31/U3



M32

U2 = NGC 221

RA : 00.7 ^h	PSA: 3	Type: Gal
Dec: +41°	OSA: N0	Mag : 8.2
Con: And	CSA: 2/7	Size : 11' × 7'

See: DSC:TMO OSA

Notes: M31/U3 is 0.5° N. M110 is 1° NW.

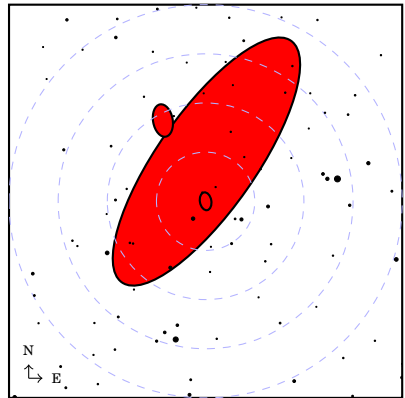
β And Mirach (mag 2.1)

→ 2.5° W

→ 3.0° N to μ And (mag 3.9)

→ 2.5° W

→ 2.5° N to M32/U2



M33

NGC 598

RA : 01.6 ^h	PSA: 3	Type: Gal
Dec: +31°	OSA: N0	Mag : 5.7
Con: Tri	CSA: 2	Size : 67' × 42'

See: DSC:TMO OSA

β And Mirach (mag 2.1)

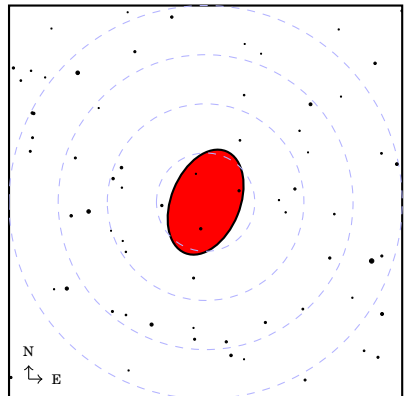
→ 5.5° E

→ 5.0° S to M33

α Tri (mag 3.4)

→ 1.0° N

→ 4.0° W to M33



M34

NGC 1039

RA : 02.7^h PSA: 2 Type: OC

Dec: +43° OSA: N4 Mag: 5.2

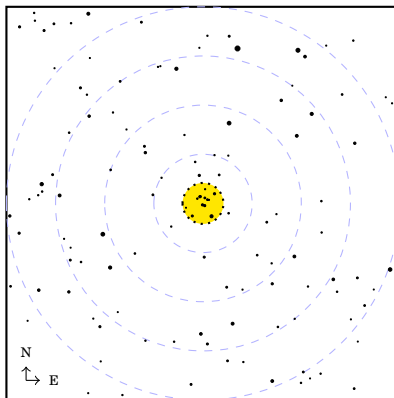
Con: Per CSA: 2/3 Size : 25'

See: DSC:TMO OSA

β Per Algol (mag 2.1–3.4)

→ 2.0° N (1/2 distance to κ Per)

→ 4.5° W (2/5 distance to γ And) to M34



M35

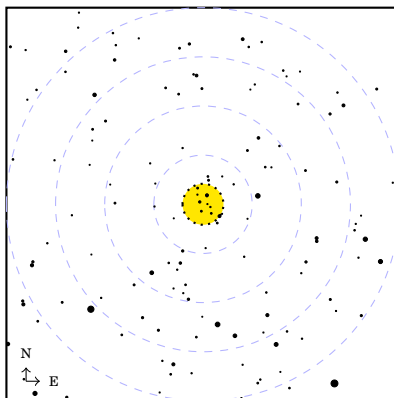
U27 = NGC 2168

RA : 06.2^h PSA: 14 Type: OC

Dec: +24° OSA: E7 Mag: 5.1

Con: Gem CSA: 3/9 Size : 25'

See: DSC:TMO OSA



M36

U23 = NGC 1960

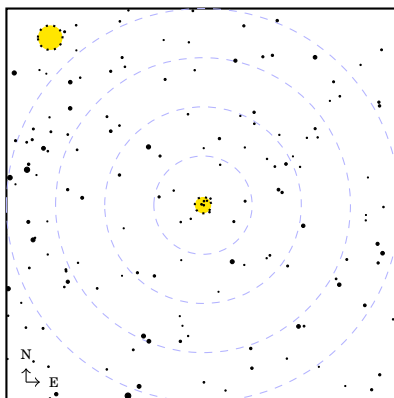
RA : 05.6^h PSA: 12 Type: OC

Dec: +34° OSA: N6 Mag: 6.0

Con: Aur CSA: 3 Size : 10'

See: DSC:TMO OSA

Notes: M38/U22 is 2.5° NW.



M37

U26 = NGC 2099

RA : 05.9^h PSA: 12

Dec: +33° OSA: N6

Con: Aur CSA: 3

Type: OC

Mag : 5.6

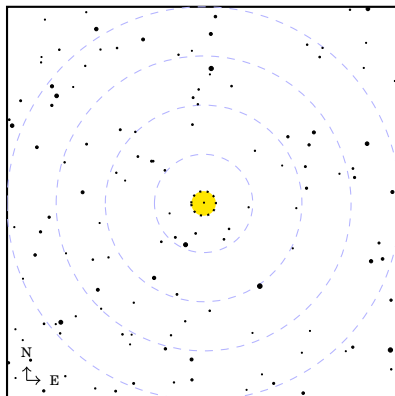
Size : 15'

See: DSC:TMO OSA

β Tau Elnath (mag 1.7)

→ 6° W to point 1° N of 136 Tau

→ 4° N to M37/U26



M38

U22 = NGC 1912

RA : 05.5^h PSA: 12

Dec: +36° OSA: N6

Con: Aur CSA: 3

Type: OC

Mag : 6.4

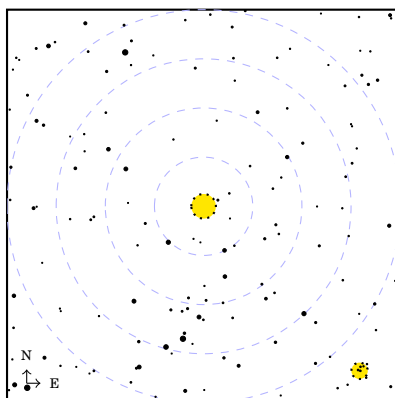
Size : 15'

See: DSC:TMO OSA

Notes: M36/U23 is 2.5° SE.

β Tau Elnath (mag 1.7)

→ 7° N to M38/U23



M39

U82 = NGC 7092

RA : 21.5^h PSA: 73

Dec: +48° OSA: N24

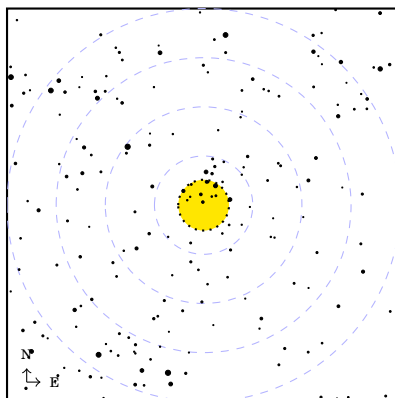
Con: Cyg CSA: 1/6/7

Type: OC

Mag : 4.6

Size : 31'

See: DSC:TMO OSA

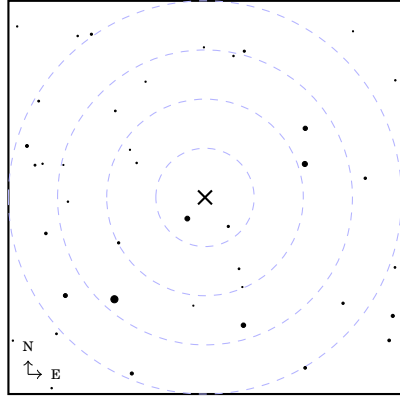


M40

RA : 12.4^h PSA: 32
Dec: +58° OSA: N10
Con: UMa CSA: 1/4/5

Type: DS
Mag : 9.0/9.6
Size : 0.9'

See: DSC:TMO OSA



M41

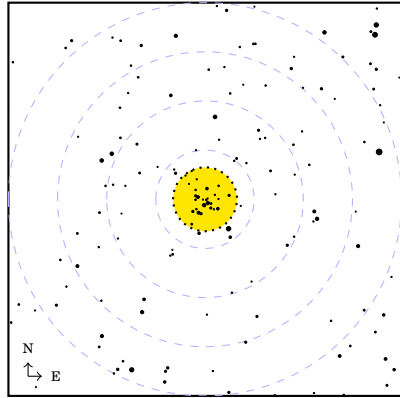
U33 = NGC 2287

RA : 06.8^h PSA: 27
Dec: -21° OSA: E6
Con: CMa CSA: 9/15

Type: OC
Mag : 4.5
Size : 39'

See: DSC:TMO OSA

α CMa Sirius (mag -1.5)
→ 4° S to M41/U33



M42

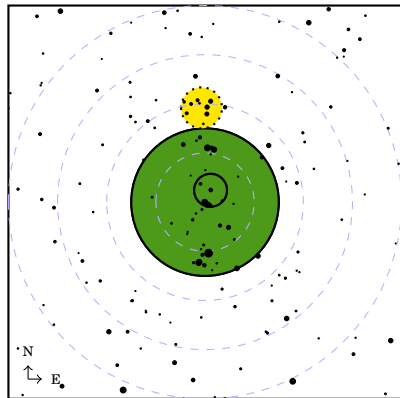
U24 = NGC 1976 = Orion Nebula

RA : 05.6^h PSA: 16/B
Dec: -05° OSA: E4
Con: Ori CSA: 9

Type: BN
Mag : 3.7
Size : 90' × 60'

See: DSC:TMO OSA

Notes: M42/U24 surrounds M43. U25 is 1° N.



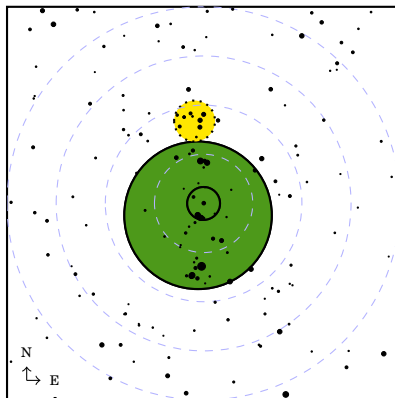
M43

NGC 1982

RA : 05.6 ^h	PSA: 16/B	Type: BN
Dec: -05°	OSA: E4	Mag : 6.8
Con: Ori	CSA: 9	Size : 20' × 15'

See: DSC:TMO OSA

Notes: M42/U24 surrounds M43. U25 is 1° N.

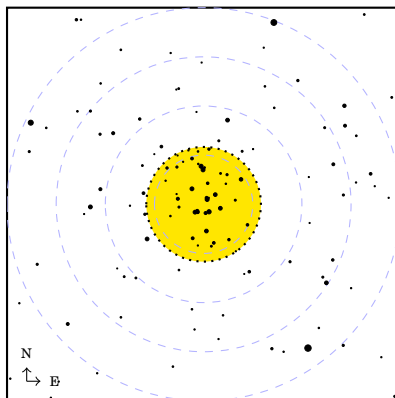


M44

U39 = NGC 2632 = Beehive Cluster

RA : 08.7 ^h	PSA: 24	Type: OC
Dec: +20°	OSA: E9	Mag : 3.1
Con: Cnc	CSA: 3/4/9/10	Size : 70'

See: DSC:TMO OSA

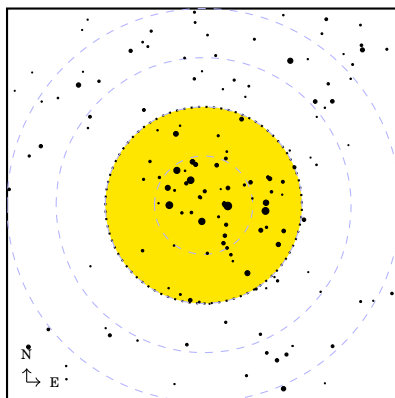


M45

U17 = Mel 22 = Pleiades

RA : 03.8 ^h	PSA: 15/A	Type: OC
Dec: +24°	OSA: E3	Mag : 1.5
Con: Tau	CSA: 2/3/8/9	Size : 120'

See: DSC:TMO OSA



M46

NGC 2437

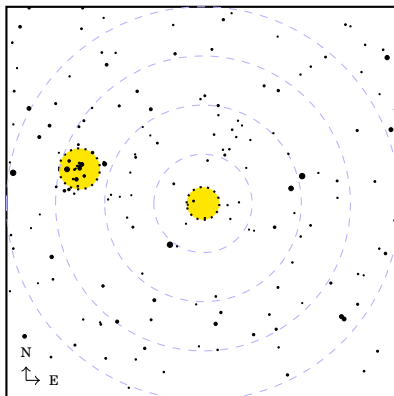
RA : 07.7^h PSA: 26 Type: OC

Dec: -15° OSA: E8 Mag: 6.1

Con: Pup CSA: 9/10/15/16 Size: 20'

See: DSC:TMO OSA

Notes: M47 is 1.3° W.



M47

NGC 2422

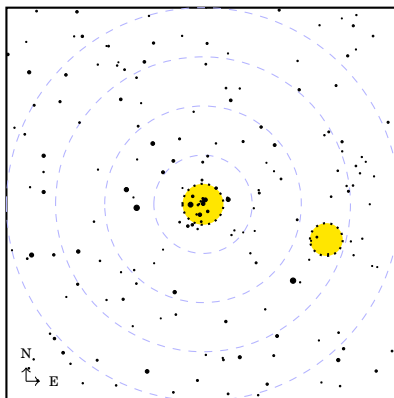
RA : 07.6^h PSA: 27 Type: OC

Dec: -14° OSA: E8 Mag: 4.4

Con: Pup CSA: 9/10/15/16 Size: 25'

See: DSC:TMO OSA

Notes: M46 is 1.3° E.



M48

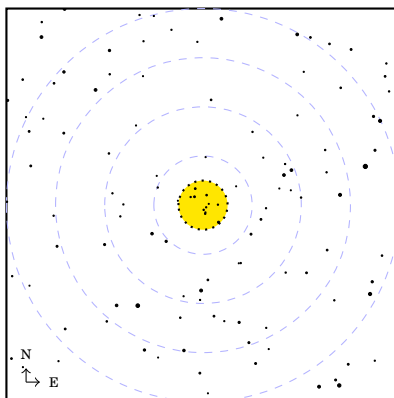
U38 = NGC 2548

RA : 08.2^h PSA: 26 Type: OC

Dec: -06° OSA: E10 Mag: 5.8

Con: Hya CSA: 9/10 Size: 30'

See: DSC:TMO OSA



M49

NGC 4472

RA : 12.5^h PSA: 45/C

Dec: +08° OSA: E14

Con: Vir CSA: 10/11

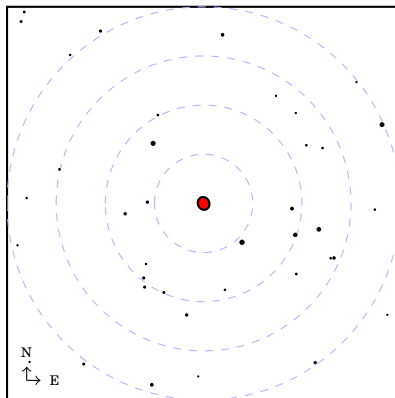
Type: Gal

Mag : 8.4

Size : 8' × 7'

See: DSC:TMO OSA

Notes: In the Virgo Cluster.



M50

U35 = NGC 2323

RA : 07.0^h PSA: 27

Dec: -08° OSA: E8

Con: Mon CSA: 9

Type: OC

Mag : 5.9

Size : 15'

See: DSC:TMO OSA

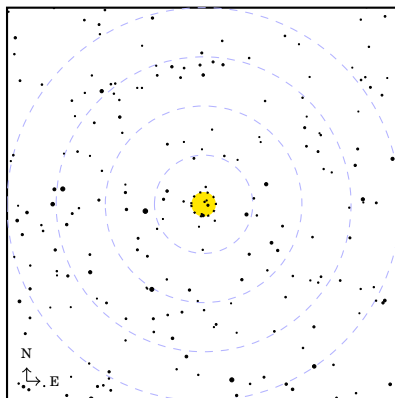
α CMa Sirius (mag -1.5)

→ 3° E to ι CMa

→ 5° N passing μ CMa to θ CMa

→ 4° N to asterism of three stars

→ 2° E to M50/U35



M51

NGC 5194 = Whirlpool Galaxy

RA : 13.5^h PSA: 43

Dec: +47° OSA: N12

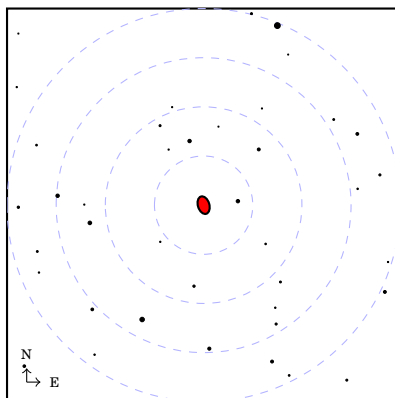
Con: CVn CSA: 4/5

Type: Gal

Mag : 8.4

Size : 11' × 7'

See: DSC:TMO OSA



M52

NGC 7654

RA : 23.4^h

PSA: 72

Type: OC

Dec: +62°

OSA: N22

Mag: 6.9

Con: Cas

CSA: 2/7

Size : 16'

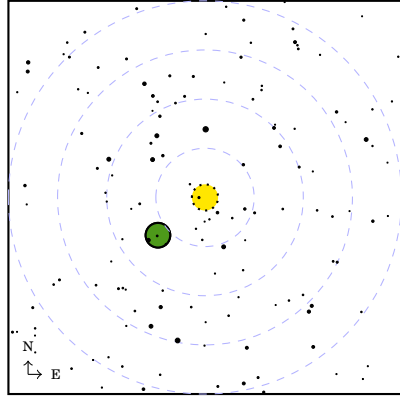
See: DSC:TMO OSA

Notes: C11 is 0.6° to SW.

β Cas Chaph (mag 2.3)

→ 2° N

→ 6° W to M52



M53

NGC 5024

RA : 13.2^h

PSA: 45

Type: GC

Dec: +18°

OSA: E13

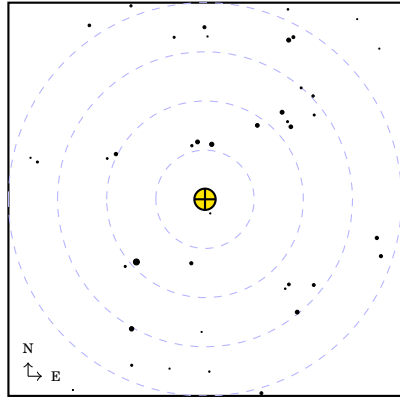
Mag: 7.7

Con: Com

CSA: 5/11

Size : 13'

See: DSC:TMO OSA



M54

NGC 6715

RA : 18.9^h

PSA: 69

Type: GC

Dec: -30°

OSA: E20

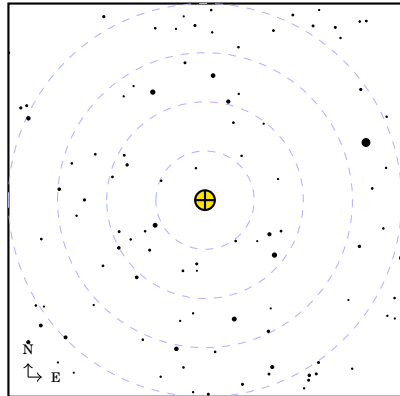
Mag: 7.2

Con: Sgr

CSA: 18/19

Size : 12'

See: DSC:TMO OSA



M55

NGC 6809

RA : 19.7^h

PSA: 68

Type: GC

Dec: -31°

OSA: E22

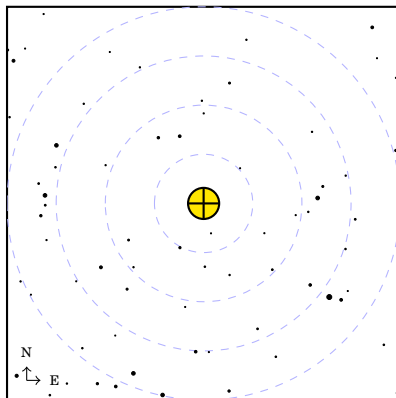
Mag : 6.3

Con: Sgr

CSA: 18/19

Size : 19'

See: DSC:TMO OSA



M56

NGC 6779

RA : 19.3^h

PSA: 63

Type: GC

Dec: +30°

OSA: N18

Mag : 8.3

Con: Lyr

CSA: 6/7

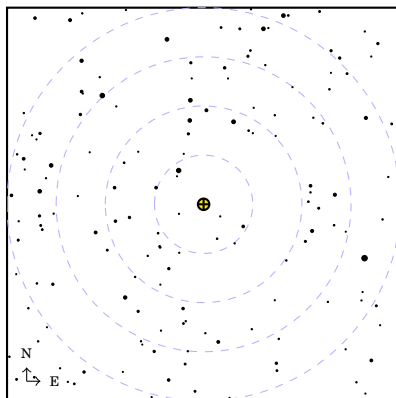
Size : 7'

See: DSC:TMO OSA

γ Lyr Sulaphat (mag 3.2)

→ 3.5° E

→ 2.5° S to M56



M57

U71 = NGC 6720 = Ring Nebula

RA : 18.9^h

PSA: 63

Type: PN

Dec: +33°

OSA: N18

Mag : 8.8

Con: Lyr

CSA: 6/7

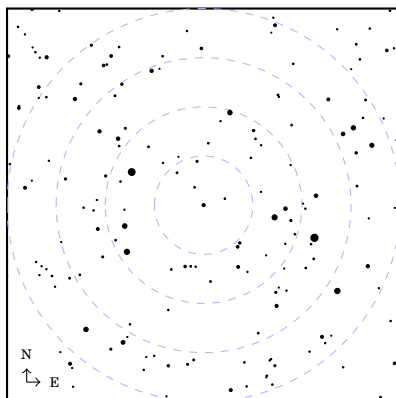
Size : 1.1'

See: DSC:TMO OSA

γ Lyr Sulaphat (mag 3.2)

→ 1.2° W

→ 0.4° N to M57/U71



M58

NGC 4579

RA : 12.6^h PSA: 45/C

Dec: +12° OSA: E14

Con: Vir CSA: 10/11

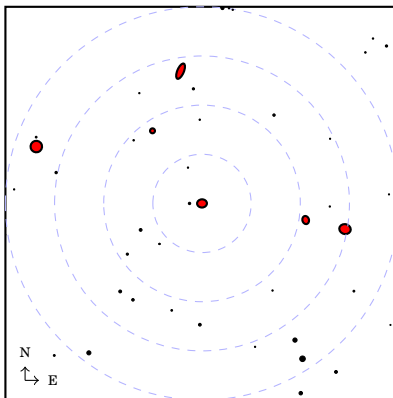
Type: Gal

Mag : 9.6

Size : 6' × 5'

See: DSC:TMO OSA

Notes: In the Virgo Cluster.



M59

NGC 4621

RA : 12.7^h PSA: 45/C

Dec: +12° OSA: E14

Con: Vir CSA: 11

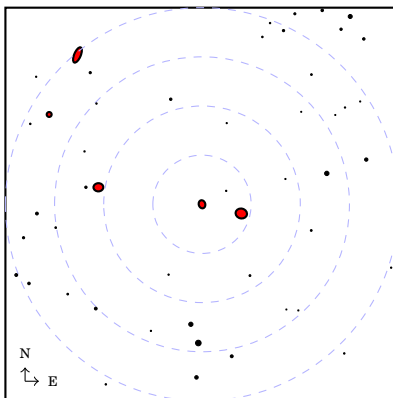
Type: Gal

Mag : 9.6

Size : 5' × 4'

See: DSC:TMO OSA

Notes: In the Virgo Cluster.



M60

NGC 4649

RA : 12.7^h PSA: 45/C

Dec: +12° OSA: E14

Con: Vir CSA: 11

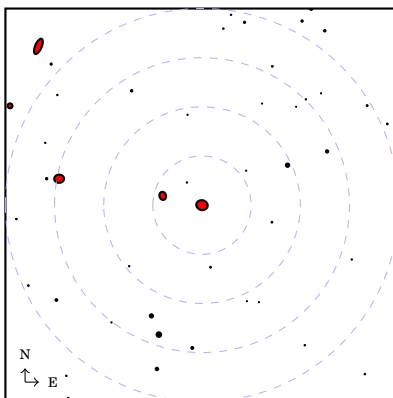
Type: Gal

Mag : 8.8

Size : 7' × 6'

See: DSC:TMO OSA

Notes: In the Virgo Cluster.



M61

NGC 4303

RA : 12.4^h

PSA: 45

Type: Gal

Dec: +04°

OSA: E14

Mag : 9.7

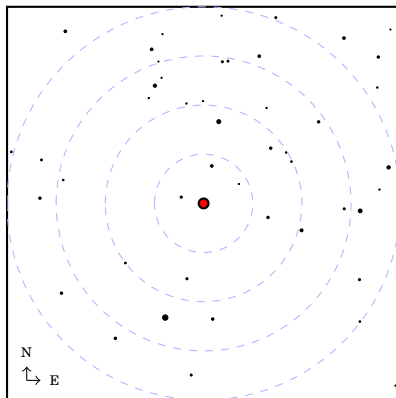
Con: Vir

CSA: 10/11

Size : 6'

See: DSC:TMO OSA

Notes: In the Virgo Cluster.



M62

U58 = NGC 6266

RA : 17.0^h

PSA: 58

Type: GC

Dec: -30°

OSA: E18

Mag : 6.7

Con: Oph

CSA: 17/18

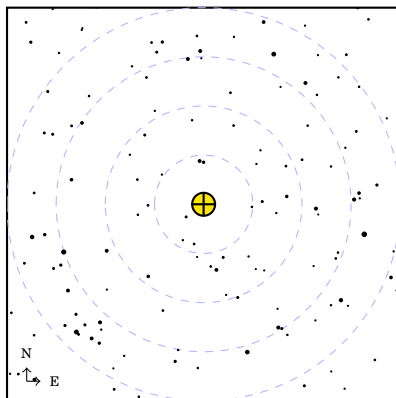
Size : 14'

See: DSC:TMO OSA

ε Sco (mag 2.3)

→ 2° E

→ 4° N to M63



M63

NGC 5055 = Sunflower Galaxy

RA : 13.3^h

PSA: 43

Type: Gal

Dec: +42°

OSA: N12

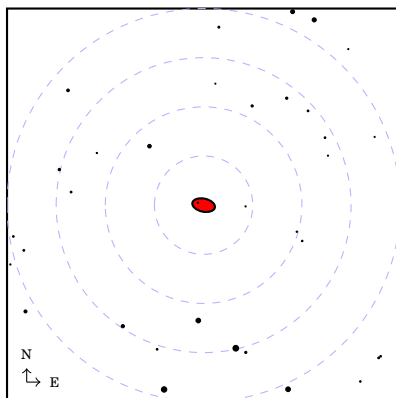
Mag : 8.6

Con: CVn

CSA: 4/5

Size : 14' × 8'

See: DSC:TMO OSA

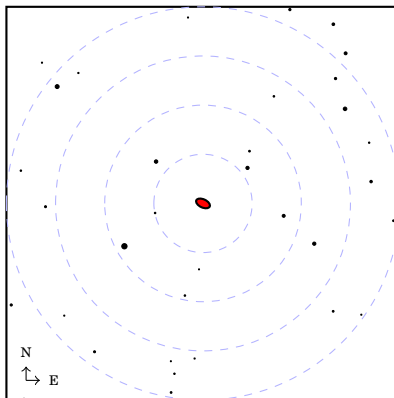


M64

U50 = NGC 4826 = Black-Eye Galaxy

RA : 12.9 ^h	PSA: 45	Type: Gal
Dec: +22°	OSA: E13	Mag : 8.5
Con: Com	CSA: 5/11	Size : 9' × 5'

See: DSC:TMO OSA



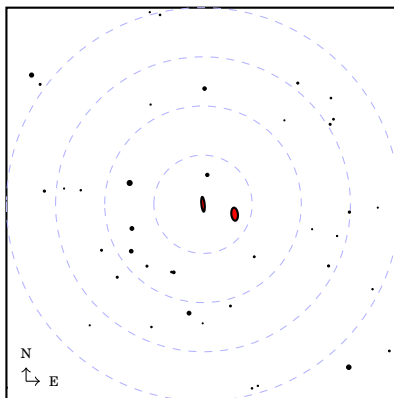
M65

NGC 3623

RA : 11.3 ^h	PSA: 34	Type: Gal
Dec: +13°	OSA: E11	Mag : 8.8
Con: Leo	CSA: 10/11	Size : 9' × 2'

See: DSC:TMO OSA

Notes: M66 is 20' E.



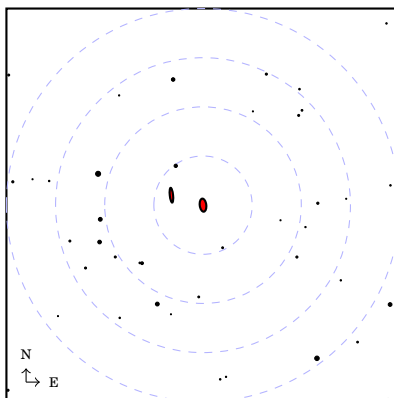
M66

NGC 3627

RA : 11.3 ^h	PSA: 34	Type: Gal
Dec: +13°	OSA: E11	Mag : 8.9
Con: Leo	CSA: 10/11	Size : 8' × 4'

See: DSC:TMO OSA

Notes: M65 is 20' W.



M67

U40 = NGC 2682

RA : 08.9^h PSA: 24

Dec: +12° OSA: E9

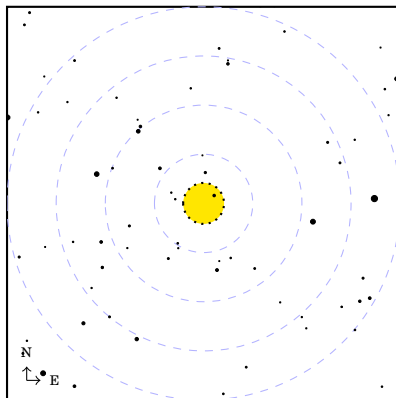
Con: Cnc CSA: 10

Type: OC

Mag : 6.0

Size : 25'

See: DSC:TMO OSA



M68

NGC 4590

RA : 12.7^h PSA: 47

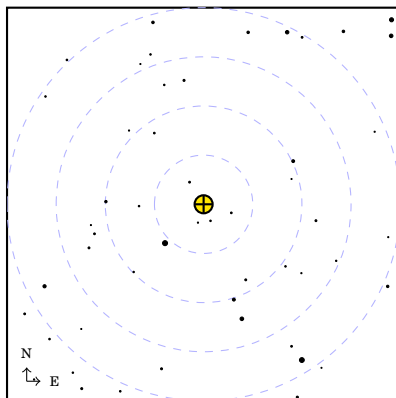
Dec: -27° OSA: E12

Con: Hya CSA: 10/11/16/17 Size : 11'

Type: GC

Mag : 7.6

See: DSC:TMO OSA



M69

NGC 6637

RA : 18.5^h PSA: 67

Dec: -32° OSA: E20

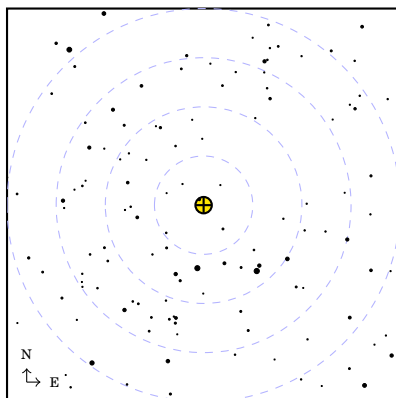
Con: Sgr CSA: 18

Type: GC

Mag : 7.4

Size : 10'

See: DSC:TMO OSA



M70

NGC 6681

RA : 18.7^h

PSA: 67

Type: GC

Dec: -32°

OSA: E20

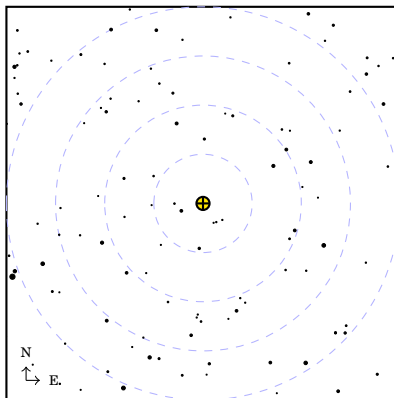
Mag: 7.8

Con: Sgr

CSA: 18

Size : 8'

See: DSC:TMO OSA



M71

NGC 6838

RA : 19.9^h

PSA: 64

Type: GC

Dec: +19°

OSA: E21

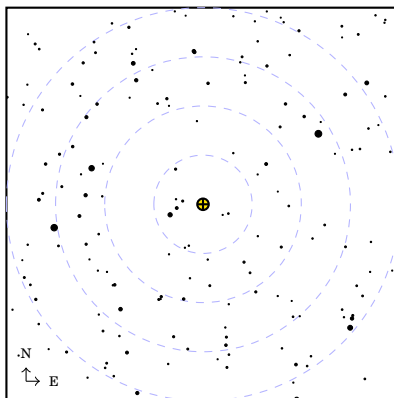
Mag: 8.0

Con: Sge

CSA: 6/7/12/13

Size : 7'

See: DSC:TMO OSA



M72

NGC 6981

RA : 20.9^h

PSA: 66/77

Type: GC

Dec: -13°

OSA: E24

Mag: 9.2

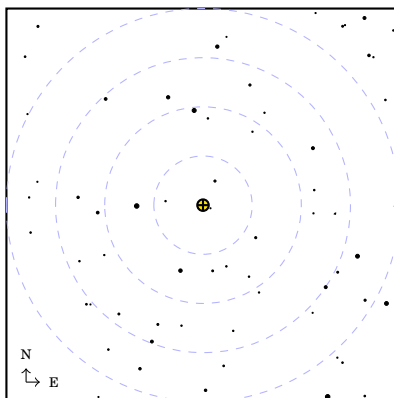
Con: Aqr

CSA: 13

Size : 7'

See: DSC:TMO OSA

Notes: M73 is 1.4° W.



M73

NGC 6994

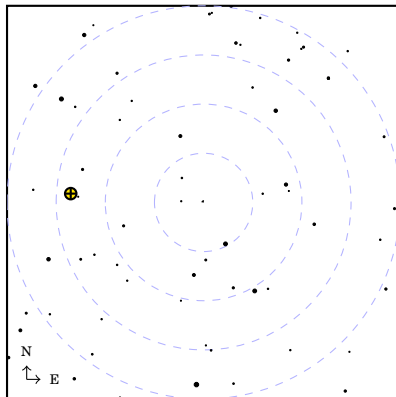
RA : 21.0^h PSA: 66/77 Type: AST

Dec: -13° OSA: E24 Mag : 9.7

Con: Aqr CSA: 13 Size : 1.4'

See: DSC:TMO OSA

Notes: Y-shaped asterism of four stars. M72 is 1.4° E.



M74

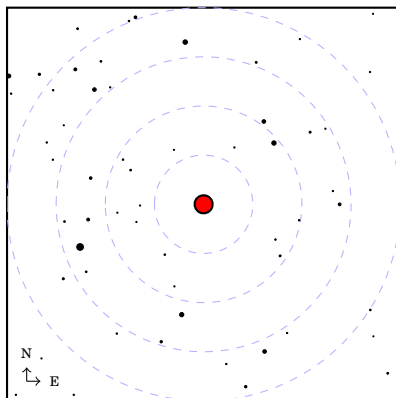
NGC 628

RA : 01.6^h PSA: 4/5 Type: Gal

Dec: +16° OSA: E1 Mag : 8.5

Con: Psc CSA: 8 Size : 11'

See: DSC:TMO OSA



M75

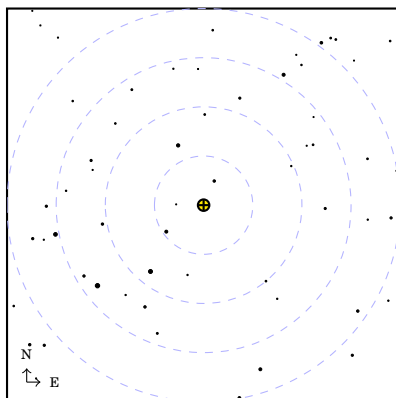
NGC 6864

RA : 20.1^h PSA: 66 Type: GC

Dec: -22° OSA: E22 Mag : 8.6

Con: Sgr CSA: 12/13/18/19 Size : 7'

See: DSC:TMO OSA

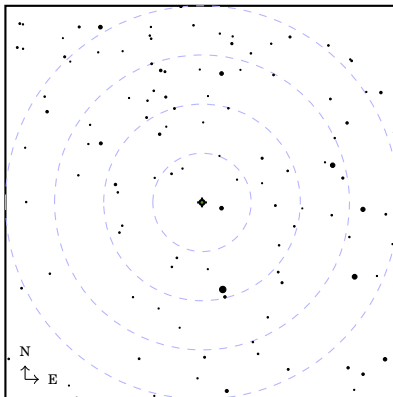


M76

NGC 651/650 = Little Dumbell

RA : 01.7 ^h	PSA: 2	Type: PN
Dec: +52°	OSA: N0	Mag: 10.4
Con: Per	CSA: 1/2/3/7	Size: 3' × 2'

See: DSC:TMO OSA

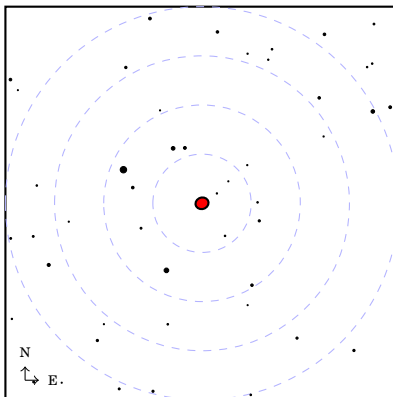


M77

U12 = NGC 1068

RA : 02.7 ^h	PSA: 4	Type: Gal
Dec: -00°	OSA: E0	Mag: 8.9
Con: Cet	CSA: 8	Size: 8' × 7'

See: DSC:TMO OSA

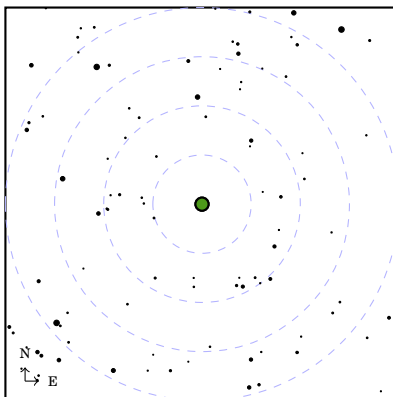


M78

NGC 2068

RA : 05.8 ^h	PSA: 16	Type: BN
Dec: +00°	OSA: E5	Mag: 8.0
Con: Ori	CSA: 9	Size: 8' × 6'

See: DSC:TMO OSA



M79

NGC 1904

RA : 05.4^h

PSA: 16

Type: GC

Dec: -25°

OSA: E4

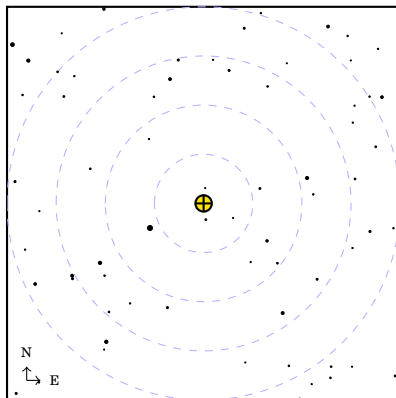
Mag : 7.7

Con: Lep

CSA: 9/15

Size : 10'

See: DSC:TMO OSA



M80

NGC 6093

RA : 16.3^h

PSA: 56

Type: GC

Dec: -23°

OSA: E18

Mag : 7.3

Con: Sco

CSA: 11/12/17/18

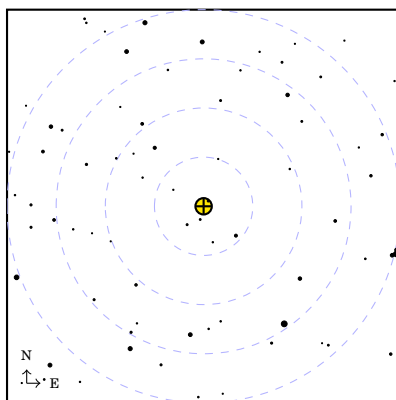
Size : 10'

See: DSC:TMO OSA

α Sco Antares (mag 0.9–1.1)

→ 3.0° W

→ 3.5° N to M80



M81

U41 = NGC 3031

RA : 09.9^h

PSA: 31

Type: Gal

Dec: +69°

OSA: N8

Mag : 6.9

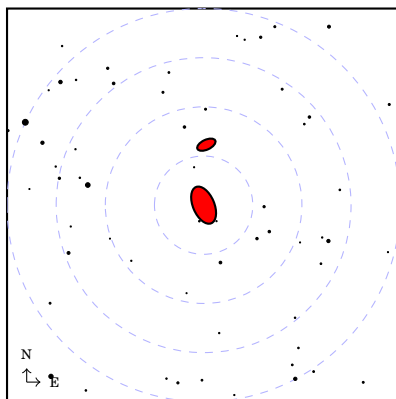
Con: UMa

CSA: 1/4

Size : 24' × 13'

See: DSC:TMO OSA

Notes: M82/U42 is 0.6° N.



M82

U42 = NGC 3034

RA : 09.9^h PSA: 31

Dec: +70° OSA: N8

Con: UMa CSA: 1/4

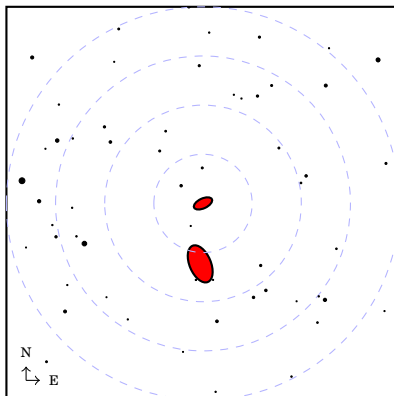
Type: Gal

Mag : 8.4

Size : 12' × 6'

See: DSC:TMO OSA

Notes: M81/U41 is 0.6° S.



M83

NGC 5236

RA : 13.6^h PSA: 47/48

Dec: -30° OSA: E16

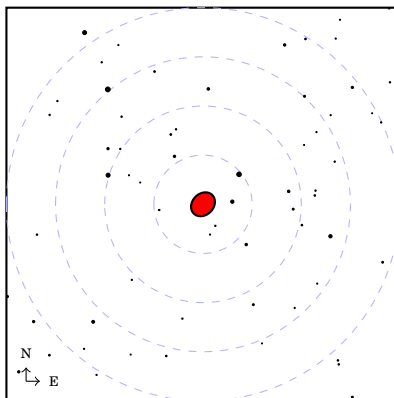
Con: Hya CSA: 17

Type: Gal

Mag : 7.6

Size : 16' × 13'

See: DSC:TMO OSA



M84

U45 = NGC 4374

RA : 12.4^h PSA: 45/C

Dec: +13° OSA: E14

Con: Vir CSA: 4/10/11

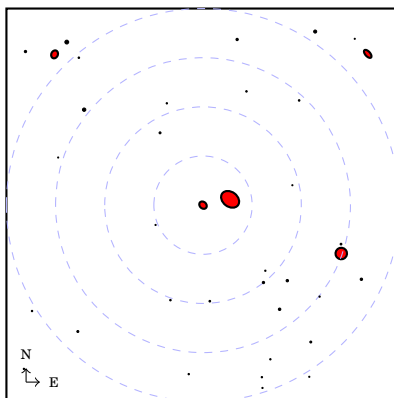
Type: Gal

Mag : 9.1

Size : 5' × 4'

See: DSC:TMO OSA

Notes: In the Virgo Cluster.



M85

NGC 4382

RA : 12.4^h PSA: 45/C

Dec: +18° OSA: E14

Con: Com CSA: 4/5/10/11

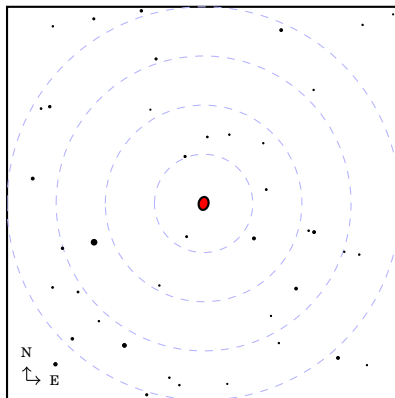
Type: Gal

Mag : 9.1

Size : 8' × 6'

See: DSC:TMO OSA

Notes: In the Virgo Cluster.



M86

U46 = NGC 4406

RA : 12.4^h PSA: 45/C

Dec: +13° OSA: E14

Con: Vir CSA: 4/10/11

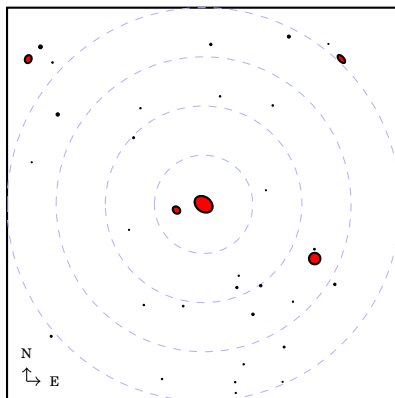
Type: Gal

Mag : 8.9

Size : 12' × 9'

See: DSC:TMO OSA

Notes: In the Virgo Cluster.



M87

U47 = NGC 4486

RA : 12.5^h PSA: 45/C

Dec: +12° OSA: E14

Con: Vir CSA: 10/11

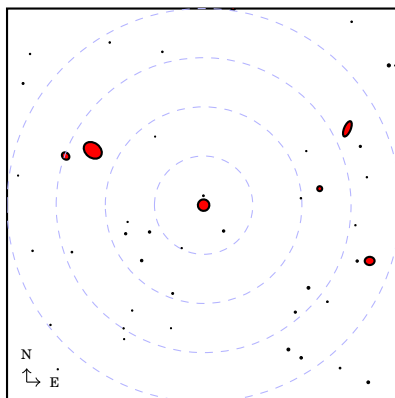
Type: Gal

Mag : 8.6

Size : 7'

See: DSC:TMO OSA

Notes: In the Virgo Cluster.



M88

NGC 4501

RA : 12.5^h PSA: 45/C

Dec: +14° OSA: E14

Con: Com CSA: 5/10/11

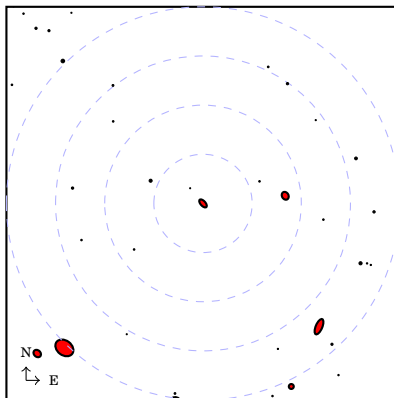
Type: Gal

Mag: 9.6

Size : 6' × 3'

See: DSC:TMO OSA

Notes: In the Virgo Cluster.



M89

NGC 4552

RA : 12.6^h PSA: 45/C

Dec: +13° OSA: E14

Con: Vir CSA: 10/11

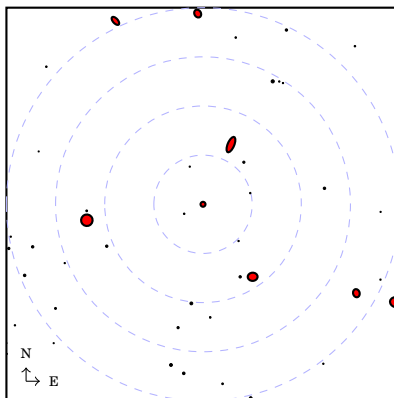
Type: Gal

Mag: 9.8

Size : 3'

See: DSC:TMO OSA

Notes: In the Virgo Cluster.



M90

NGC 4569

RA : 12.6^h PSA: 45/C

Dec: +13° OSA: E14

Con: Vir CSA: 10/11

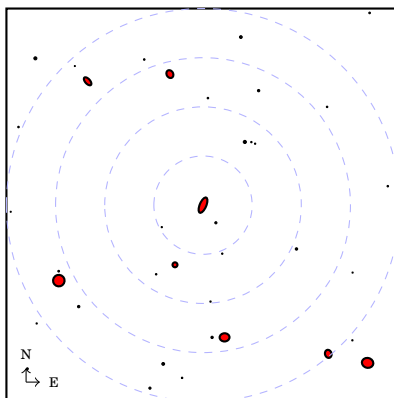
Type: Gal

Mag: 9.5

Size : 10' × 4'

See: DSC:TMO OSA

Notes: In the Virgo Cluster.



M91

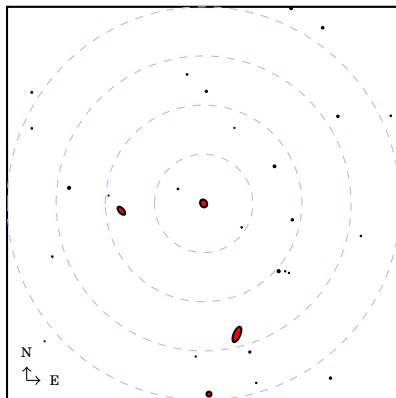
NGC 4548

RA : 12.6^h PSA: 45/C
Dec: +14° OSA: E14
Con: Com CSA: 5/10/11

Type: Gal
Mag : 10.2
Size : 5' × 4'

See: DSC:TMO OSA

Notes: In the Virgo Cluster.



M92

U59 = NGC 6341

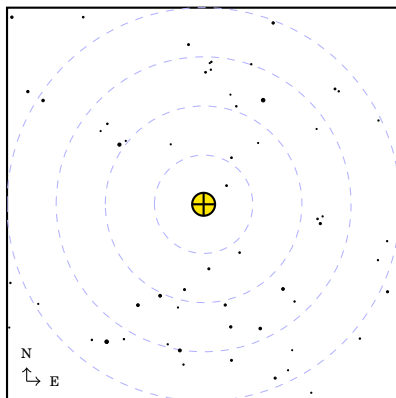
RA : 17.3^h PSA: 52
Dec: +43° OSA: N14
Con: Her CSA: 5/6

Type: GC
Mag : 6.5
Size : 14'

See: DSC:TMO OSA

π Her (mag 3.2)

→ 6.5° N to M92/U59



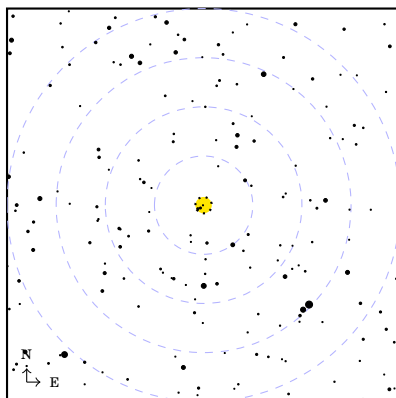
M93

NGC 2447

RA : 07.7^h PSA: 26
Dec: -24° OSA: E6
Con: Pup CSA: 9/10/15/16

Type: OC
Mag : 6.2
Size : 10'

See: DSC:TMO OSA



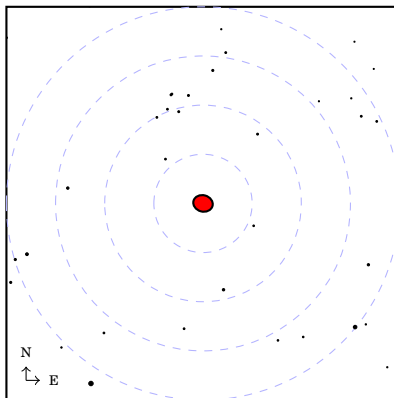
M94

U49 = NGC 4736

RA : 12.8^h PSA: 43
Dec: +41° OSA: N12
Con: CVn CSA: 4/5

Type: Gal
Mag: 8.2
Size : 12' × 10'

See: DSC:TMO OSA



M95

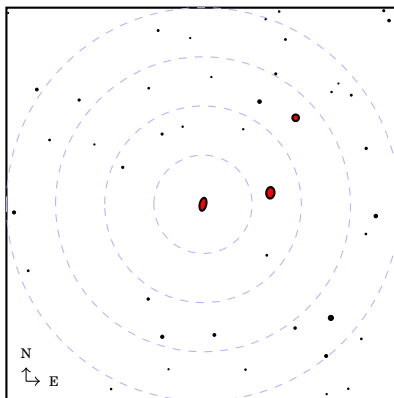
NGC 3351

RA : 10.7^h PSA: 34
Dec: +12° OSA: E11
Con: Leo CSA: 10

Type: Gal
Mag: 9.7
Size : 8' × 4'

See: DSC:TMO OSA

Notes: M96 is 0.7° E. M105 is 1.3° NE.



M96

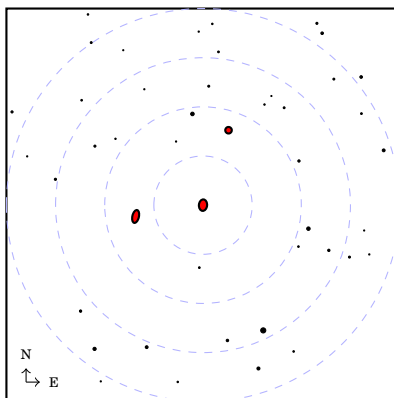
NGC 3368

RA : 10.8^h PSA: 34
Dec: +12° OSA: E11
Con: Leo CSA: 10

Type: Gal
Mag: 9.2
Size : 7' × 5'

See: DSC:TMO OSA

Notes: M96 is 0.7° W. M105 is 0.8° NNE.



M97

NGC 3587

RA : 11.2^h

PSA: 32

Type: PN

Dec: +55°

OSA: N10

Mag : 9.8

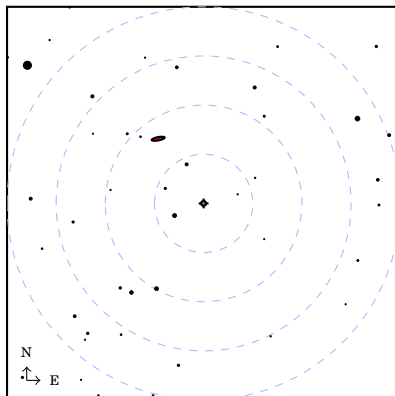
Con: UMa

CSA: 1/4/5

Size : 3'

See: DSC:TMO OSA

Notes: M108 is 0.8° to NW.



M98

NGC 4192

RA : 12.2^h

PSA: 45/C

Type: Gal

Dec: +15°

OSA: E14

Mag : 10.1

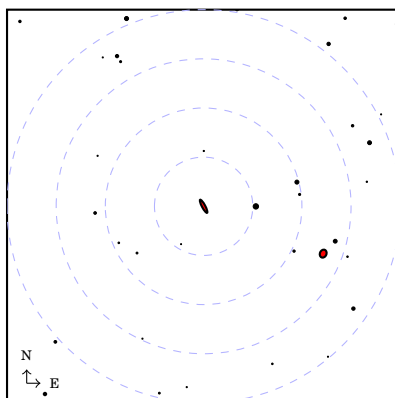
Con: Com

CSA: 4/5/10/11

Size : 9' × 2'

See: DSC:TMO OSA

Notes: In the Virgo Cluster.



M99

NGC 4254

RA : 12.3^h

PSA: 45/C

Type: Gal

Dec: +14°

OSA: E14

Mag : 9.9

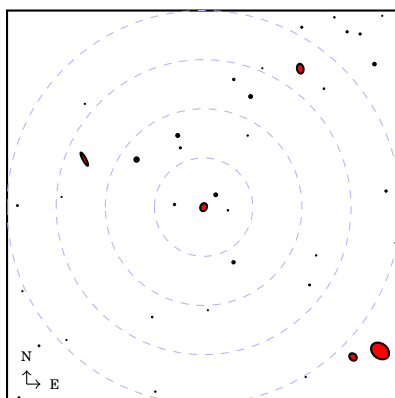
Con: Com

CSA: 4/5/10/11

Size : 5' × 4'

See: DSC:TMO OSA

Notes: In the Virgo Cluster.



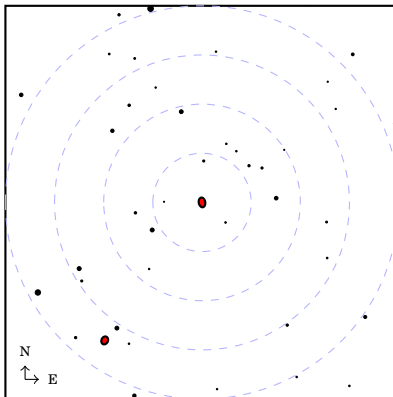
M100

NGC 4321

RA : 12.4 ^h	PSA: 45/C	Type: Gal
Dec: +16°	OSA: E14	Mag : 9.3
Con: Com	CSA: 4/5/10/11	Size : 6' × 4'

See: DSC:TMO OSA

Notes: In the Virgo Cluster.

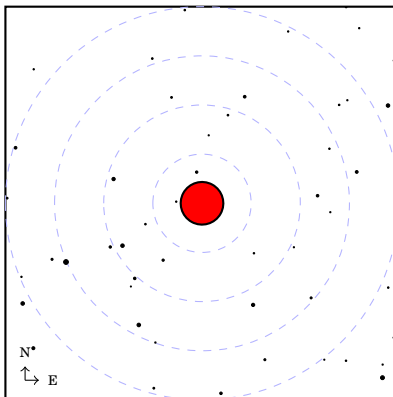


M101

NGC 5457

RA : 14.1 ^h	PSA: 42	Type: Gal
Dec: +54°	OSA: N10	Mag : 7.9
Con: UMa	CSA: 1/4/5/6	Size : 26'

See: DSC:TMO OSA

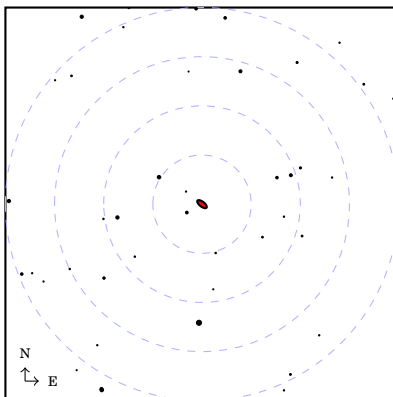


M102

NGC 5866

RA : 15.1 ^h	PSA: 42	Type: Gal
Dec: +56°	OSA: N16	Mag : 9.9
Con: Dra	CSA: 1/5/6	Size : 7' × 3'

See: DSC:TMO OSA



M103

NGC 581

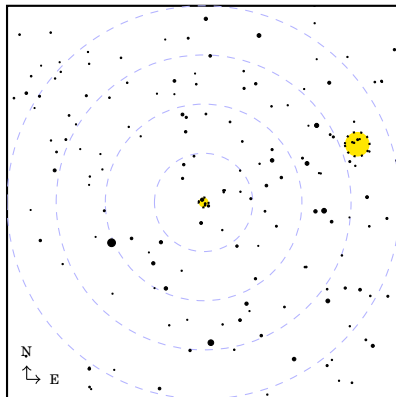
RA : 01.6^h PSA: 3 Type: OC

Dec: +61° OSA: N2 Mag : 7.4

Con: Cas CSA: 1/2/3/7 Size : 6'

See: DSC:TMO OSA

Notes: C10 is 1.7° ENE.



M104

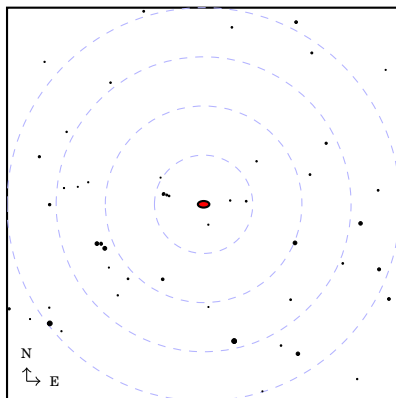
U48 = NGC 4594 = Sombrero Galaxy

RA : 12.7^h PSA: 47 Type: Gal

Dec: -12° OSA: E12 Mag : 8.0

Con: Vir CSA: 10/11 Size : 7' × 4'

See: DSC:TMO OSA



M105

NGC 3379

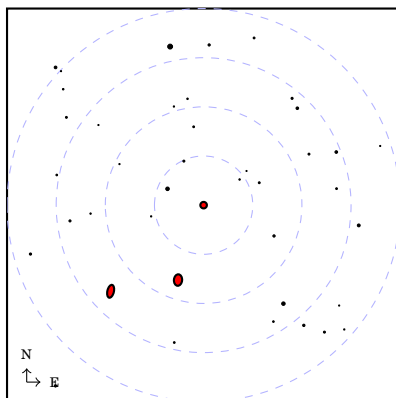
RA : 10.8^h PSA: 34 Type: Gal

Dec: +13° OSA: E11 Mag : 9.3

Con: Leo CSA: 10 Size : 4'

See: DSC:TMO OSA

Notes: M95 is 1.3° SW. M96 is 0.8° SSW.



M106

NGC 4258

RA : 12.3^h

PSA: 34

Dec: +47°

OSA: N12

Con: CVn

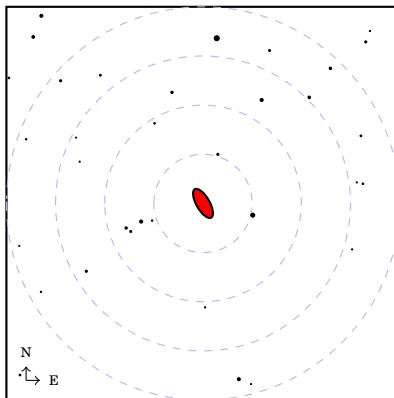
CSA: 4/5

Type: Gal

Mag: 8.4

Size : 20' × 8'

See: DSC:TMO OSA



M107

NGC 6171

RA : 16.5^h

PSA: 56

Dec: -13°

OSA: E17

Con: Oph

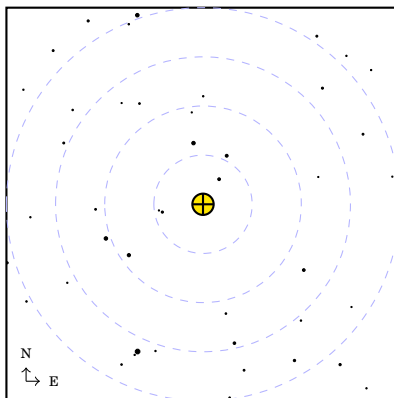
CSA: 11/12

Type: GC

Mag: 7.8

Size : 13'

See: DSC:TMO OSA



M108

NGC 3556

RA : 11.2^h

PSA: 32/33

Dec: +56°

OSA: N10

Con: UMa

CSA: 1/4/5

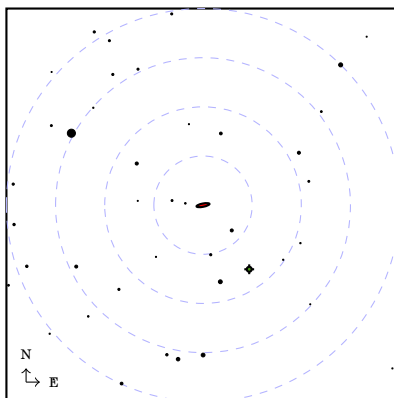
Type: Gal

Mag: 10.0

Size : 8' × 2'

See: DSC:TMO OSA

Notes: M97 is 0.8° to SE.



M109

NGC 3992

RA : 12.0^h

PSA: 32

Type: Gal

Dec: +53°

OSA: N10

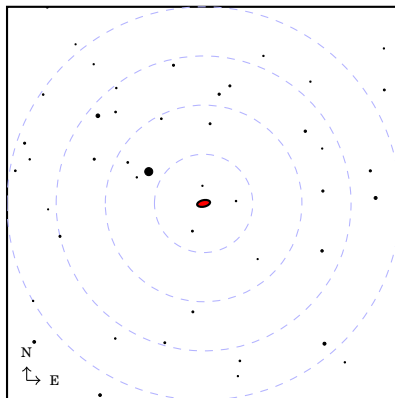
Mag : 9.8

Con: UMa

CSA: 1/4/5

Size : 8' × 4'

See: DSC:TMO OSA



M110

NGC 205

RA : 00.7^h

PSA: 3

Type: Gal

Dec: +42°

OSA: N0

Mag : 8.1

Con: And

CSA: 2/7

Size : 20' × 12'

See: DSC:TMO OSA

Notes: M31/U3 is 0.5° SE. M32/U2 is 1° SE.

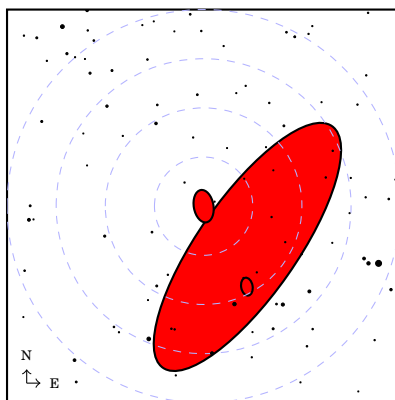
β And Mirach (mag 2.1)

→ 2.5° W

→ 3.0° N to μ And (mag 3.9)

→ 3.0° W

→ 3.5° N to M31/U3



The Caldwell Objects

The Caldwell objects were selected by Patrick Moore as a complement to the Messier objects. Again, I find O'Meara's *Deep-Sky Companion: The Caldwell Objects* to be excellent on the origin of the catalog, the appearance of the objects, and their nature.

I use C1 to C109 to label the Caldwell objects. This is common practice, but does not confirm to the IAU's recommendation on nomenclature, as O'Meara rightly notes (pp. 14–15).

The following table lists the objects with their J2000 positions (decimal hours of right ascension and degrees of declination), the charts on which they appear in the *Pocket Sky Atlas*, their types, and other names. I follow the corrections given by O'Meara (p. 15) with regards to C37, C49, C89, and C100.

For completeness, I include finder charts for all of the Caldwell objects, even bright ones like C41 (the Hyades).

I note that C41 (the Hyades) and C99 (the Coalsack Nebula) are much bigger than the finder charts. They are really binocular objects and are better located using a small-scale all-sky atlas.

Name	Position	PSA	Type	Other Names
C1	00.8 +85	1	OC	NGC 188
C2	00.2 +73	1/71	PN	NGC 40 = Bow-Tie Nebula
C3	12.3 +69	31/41	Gal	NGC 4236
C4	21.0 +68	61/71	BN	NGC 7023 = Iris Nebula
C5	03.8 +68	1/11	Gal	IC 342 = Hidden Galaxy
C6	18.0 +67	51/61	PN	NGC 6543 = Cat's Eye Nebula
C7	07.6 +66	21	Gal	NGC 2403
C8	01.5 +63	1/3	OC	NGC 559
C9	23.0 +63	71/72	BN	Cave Nebula
C10	01.8 +61	1/2	OC	NGC 663
C11	23.3 +61	71/72	BN	NGC 7635 = Bubble Nebula
C12	20.6 +60	61/62	Gal	NGC 6946 = Firecracker Galaxy
C13	01.3 +58	1/3	OC	NGC 457 = Owl Cluster
C14	02.3 +57	1/2	FC	NGC 869/884 = Double Cluster
C15	19.7 +51	62	PN	NGC 6826 = Blinking Planetary
C16	22.3 +50	73	OC	NGC 7243
C17	00.6 +49	3	Gal	NGC 147
C18	00.6 +48	3	Gal	NGC 185
C19	21.9 +47	73	BN	IC 5146 = Cocoon Nebula
C20	21.0 +44	62	BN	NGC 7000 = North America Nebula
C21	12.5 +44	32/43	Gal	NGC 4449
C22	23.4 +43	72	PN	NGC 7662 = Light Blue Snowball
C23	02.4 +42	2	Gal	NGC 891
C24	03.3 +42	13	Gal	NGC 1275 = Per A
C25	07.6 +39	23	GC	NGC 2419
C26	12.3 +38	32/43	Gal	NGC 4244
C27	20.2 +38	62	BN	NGC 6888 = Crescent Nebula
C28	01.9 +38	2	OC	NGC 752
C29	13.2 +37	43	Gal	NGC 5005
C30	22.6 +34	72/74	Gal	NGC 7331
C31	05.3 +34	12	BN	IC 405 = Flaming Star Nebula
C32	12.7 +33	43/45	Gal	NGC 4631 = Whale Galaxy
C33	20.9 +32	62	BN	NGC 6992 = Eastern Veil Nebula
C34	20.8 +31	62	BN	NGC 6960 = Western Veil Nebula
C35	13.0 +28	43/45	Gal	NGC 4889
C36	12.6 +28	43/45	Gal	NGC 4559
C37	20.2 +26	62/64	OC	NGC 6885
C38	12.6 +26	43/45	Gal	NGC 4565 = Needle Galaxy
C39	07.5 +21	25	PN	NGC 2392 = Eskimo Nebula
C40	11.3 +18	34	Gal	NGC 3626

Name	Position	PSA	Type	Other Names
C41	04.4 +16	15	OC	Mel 25 = Hyades
C42	21.0 +16	64/75	GC	NGC 7006
C43	00.1 +16	5/74	Gal	NGC 7814
C44	23.1 +12	74	Gal	NGC 7479
C45	13.6 +09	44	Gal	NGC 5248
C46	06.7 +09	25/E	BN	NGC 2261 = Hubble's Variable Nebula
C47	20.6 +07	64	GC	NGC 6934
C48	09.2 +07	24/35	Gal	NGC 2775
C49	06.5 +05	25/E	BN	NGC 2237/38/46 = Rosette Nebula
C50	06.5 +05	25/E	OC	NGC 2244
C51	01.1 +02	5/7	Gal	IC 1613
C52	12.8 -06	47	Gal	NGC 4697
C53	10.1 -08	37	Gal	NGC 3115 = The Spindle
C54	08.0 -11	26	OC	NGC 2506
C55	21.1 -11	77	PN	NGC 7009 = Saturn Nebula
C56	00.8 -12	7	PN	NGC 246
C57	19.7 -15	66	Gal	NGC 6822 = Barnard's Galaxy
C58	07.3 -16	27	OC	NGC 2360 = Caroline's Cluster
C59	10.4 -19	36/37	PN	NGC 3242 = Ghost of Jupiter
C60	12.0 -19	36/47	Gal	NGC 4038 = NW Antennae Galaxy
C61	12.0 -19	36/47	Gal	NGC 4039 = SE Antennae Galaxy
C62	00.8 -21	7	Gal	NGC 247
C63	22.5 -21	76/77	PN	NGC 7293 = Helix Nebula
C64	07.3 -25	27	OC	NGC 2362 = τ CMa Cluster
C65	00.8 -25	7/9	Gal	NGC 253 = Sculptor Galaxy
C66	14.7 -27	46	GC	NGC 5694
C67	02.8 -30	6/8	Gal	NGC 1097
C68	19.0 -37	69	BN	NGC 6729 = R CrA Nebula
C69	17.2 -37	58	PN	NGC 6302 = Bug Nebula
C70	00.9 -38	9	Gal	NGC 300
C71	07.9 -39	28	OC	NGC 2477
C72	00.2 -39	9	Gal	NGC 55
C73	05.2 -40	18	GC	NGC 1851
C74	10.1 -40	39	PN	NGC 3132
C75	16.4 -41	58	OC	NGC 6124
C76	16.9 -42	58	OC	NGC 6231
C77	13.4 -43	48/49	Gal	NGC 5128 = Cen A
C78	18.1 -44	69	GC	NGC 6541
C79	10.3 -46	39	GC	NGC 3201
C80	13.4 -47	48/49	GC	NGC 5139 = ω Cen

Name	Position	PSA	Type	Other Names
C81	17.4 −48	58	GC	NGC 6352
C82	16.7 −49	58	OC	NGC 6193
C83	13.1 −49	49	Gal	NGC 4945
C84	13.8 −51	48	GC	NGC 5286
C85	08.7 −53	28	OC	IC 2391 = σ Vel Cluster
C86	17.7 −54	58	GC	NGC 6397
C87	03.2 −55	19	GC	NGC 1261
C88	15.1 −56	59	OC	NGC 5823
C89	16.3 −58	58/60	OC	NGC 6087 = S Nor Cluster
C90	09.4 −58	39	PN	NGC 2867
C91	11.1 −59	38/40	OC	NGC 3532
C92	10.8 −60	38/40	BN	NGC 3372 = η Car Nebula
C93	19.2 −60	69/70	GC	NGC 6752
C94	12.9 −60	49/50	OC	NGC 4755 = Jewel Box
C95	16.1 −60	59/60	OC	NGC 6025
C96	08.0 −61	28/30	OC	NGC 2516 = Southern Beehive Cluster
C97	11.6 −62	38/40	OC	NGC 3766
C98	12.7 −63	49/50	OC	NGC 4609
C99	12.5 −64	49/50	DN	Coalsack
C100	11.6 −63	38/40	OC	Cr 249 = λ Cen Cluster
C101	19.2 −64	70	Gal	NGC 6744
C102	10.7 −64	38/40	OC	IC 2602 = θ Car Cluster
C103	05.6 −69	20/30/D	BN	NGC 2070 = Tarantula Nebula
C104	01.1 −71	10/80	GC	NGC 362
C105	13.0 −71	50	GC	NGC 4833
C106	00.4 −72	10/80	GC	NGC 104 = 47 Tuc
C107	16.4 −72	60	GC	NGC 6101
C108	12.4 −73	40/50	GC	NGC 4372
C109	10.2 −81	40	PN	NGC 3195

C1

NGC 188

RA : 00.8^h PSA: 1

Dec: +85° OSA:

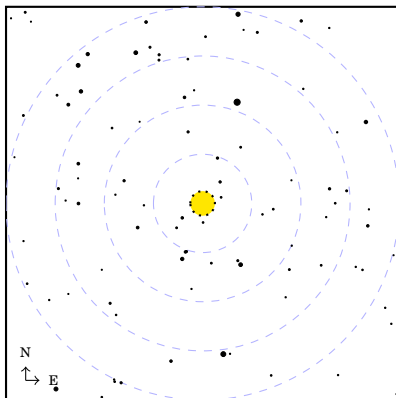
Con: Cep CSA: 1

Type: OC

Mag : 7.1

Size : 15'

See: DSC:TCO



C2

NGC 40 = Bow-Tie Nebula

RA : 00.2^h PSA: 1/71

Dec: +73° OSA:

Con: Cep CSA: 1

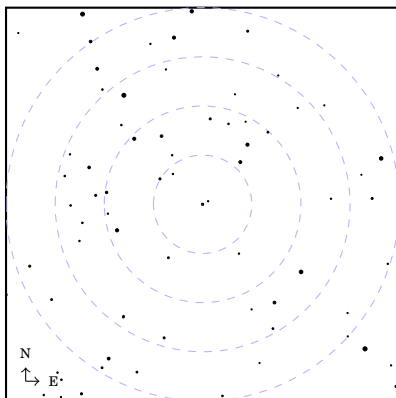
Type: PN

Mag :

Size : 0.6'

See: DSC:TCO 12.3

Notes: Central star is magnitude 11.6 (DSC:TCO).



C3

NGC 4236

RA : 12.3^h PSA: 31/41

Dec: +69° OSA:

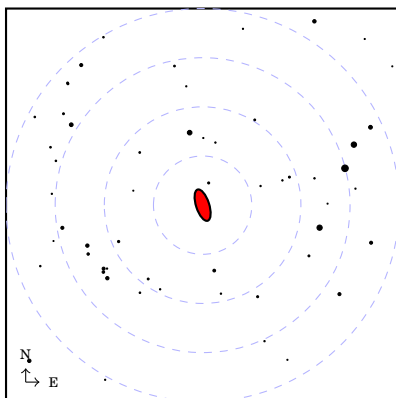
Con: Dra CSA: 1/5

Type: Gal

Mag : 9.6

Size : 20' × 8'

See: DSC:TCO



C4

NGC 7023 = Iris Nebula

RA : 21.0^h PSA: 61/71

Dec: +68° OSA:

Con: Cep CSA: 1/6/7

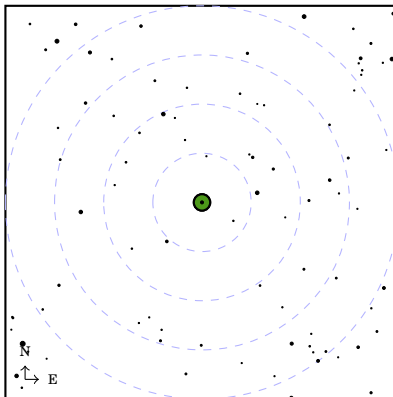
Type: BN

Mag : 7.7

Size : 10' × 8'

See: DSC:TCO

Notes: Central star is magnitude 6.8 (DSC:TCO).



C5

IC 342 = Hidden Galaxy

RA : 03.8^h PSA: 1/11

Dec: +68° OSA:

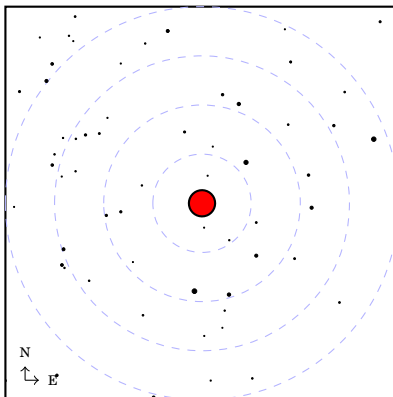
Con: Cam CSA: 1/2/3

Type: Gal

Mag : 7.9

Size : 16'

See: DSC:TCO



C6

NGC 6543 = Cat's Eye Nebula

RA : 18.0^h PSA: 51/61

Dec: +67° OSA: N16

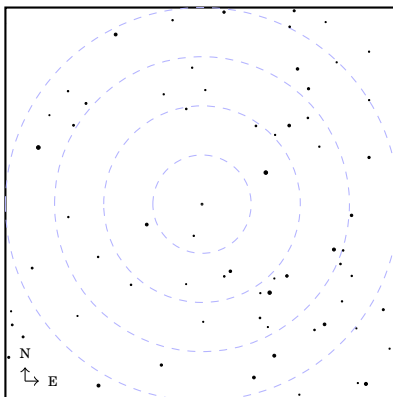
Con: Dra CSA: 1/6

Type: PN

Mag : 8.1

Size : 0.4' × 0.3'

See: DSC:TCO OSA



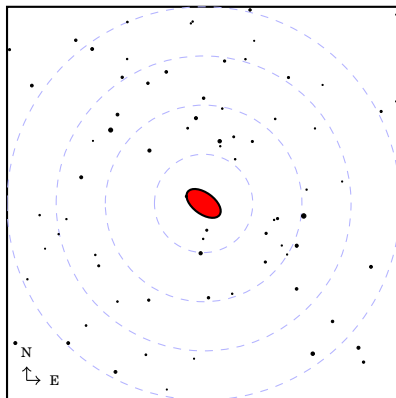
C7

NGC 2403 = Fireworks Galaxy

RA : 07.6^h PSA: 21
Dec: +66° OSA:
Con: Cam CSA: 1/3/4

Type: Gal
Mag : 7.3
Size : 24' × 13'

See: DSC:TCO OSA



C8

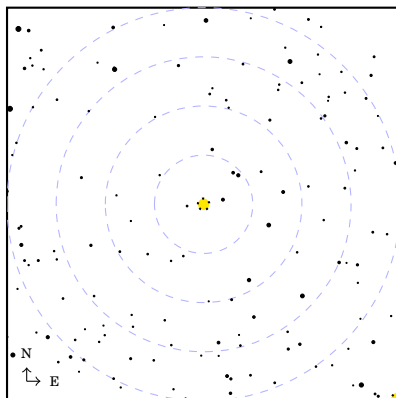
NGC 559 = Ghost's Goblet

RA : 01.5^h PSA: 1/3
Dec: +63° OSA: N2
Con: Cas CSA: 1/2

Type: OC
Mag : 9.5
Size : 7'

See: DSC:TCO OSA

Notes: C10 is 3° to SE.



C9

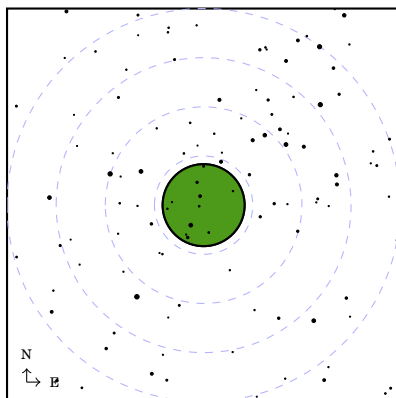
Sh2-155 = Cave Nebula

RA : 23.0^h PSA: 71/72
Dec: +63° OSA:
Con: Cep CSA: 1/7

Type: BN
Mag : 10.0
Size : 50' × 30'

See: DSC:TCO

Notes: Not shown in the CSA.



C10

U5 = NGC 663 = Horseshoe Cluster

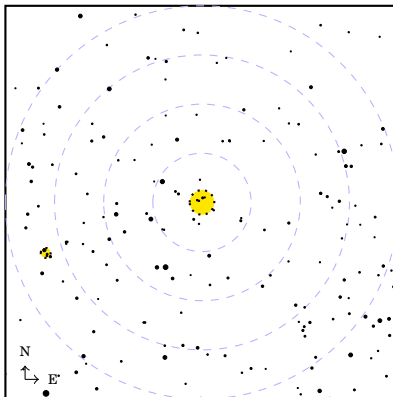
RA : 01.8^h PSA: 1/2 Type: OC

Dec: +61° OSA: N2 Mag : 6.7

Con: Cas CSA: 1/2 Size : 15'

See: DSC:TCO OSA

Notes: M103 is 1.7° WSW. C8 is 3° NW.



C11

NGC 7635 = Bubble Nebula

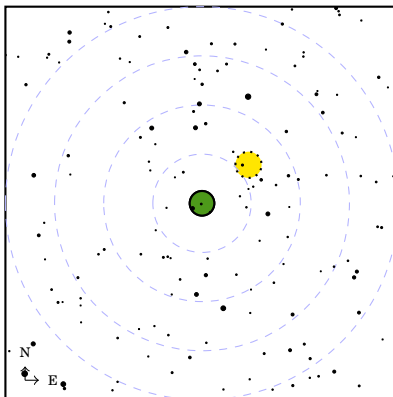
RA : 23.3^h PSA: 71/72 Type: BN

Dec: +61° OSA: Mag : 10

Con: Cas CSA: 1/2/7 Size : 15' × 8'

See: DSC:TCO

Notes: M52 is 0.6° to NE.



C12

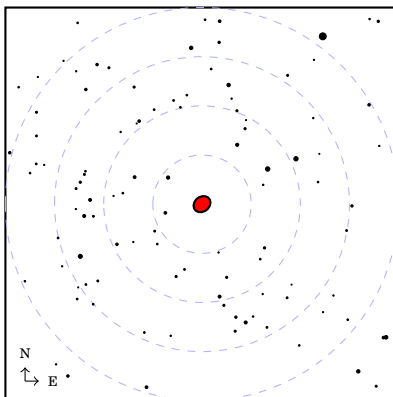
NGC 6946 = Firecracker Galaxy

RA : 20.6^h PSA: 61/62 Type: Gal

Dec: +60° OSA: N22 Mag : 8.8

Con: Cep/Cyg CSA: 1/6/7 Size : 11' × 9'

See: DSC:TCO OSA

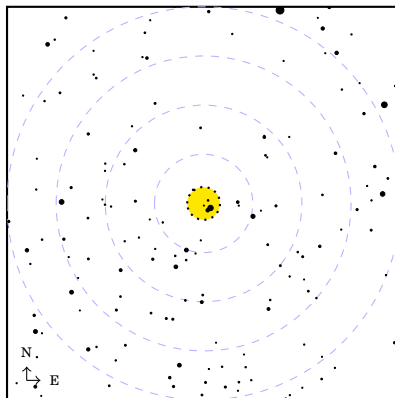


C13

U4 = NGC 457 = E.T. Cluster = Owl Cluster

RA : 01.3 ^h	PSA: 1/3	Type: OC
Dec: +58°	OSA: N2	Mag : 6.4
Con:	CSA:	Size : 20'

See: DSC:TCO OSA



C14

U9/10 = NGC 869/884 = Double Cluster = h/χ Per Clusters

RA : 02.3 ^h	PSA: 1/2	Type: OC
Dec: +57°	OSA: N2	Mag : 4.5/5.7
Con:	CSA:	Size : 18'

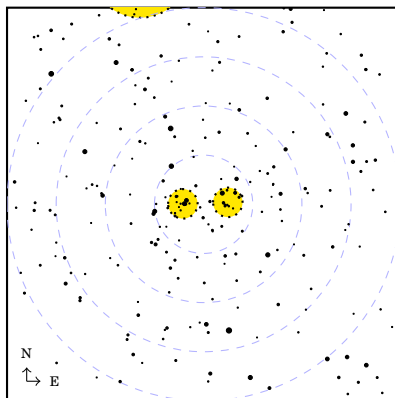
See: DSC:TCO OSA

Notes: NGC 869 (h Per) is to W and brighter;
NGC 884 (χ Per) is to E and fainter. Both are
about 18' in diameter (DSC:TCO). U8 is 2° N.
U11 is 2.5° SE.

α Per Mirfak (mag 1.8)

→ 9° E passing ι Per and θ Per to 64/65 And

→ 7° N to C14/U9/U10

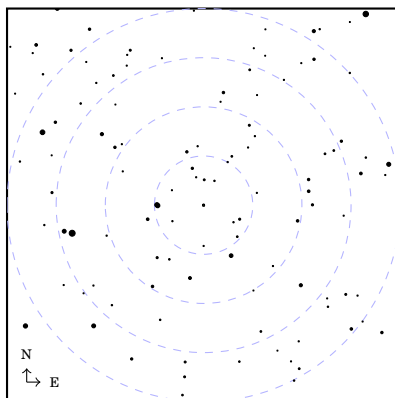


C15

NGC 6826 = Blinking Planetary

RA : 19.7 ^h	PSA: 62	Type: PN
Dec: +51°	OSA: N18	Mag : 8.5
Con:	CSA:	Size : 0.5' × 0.4'

See: DSC:TCO OSA



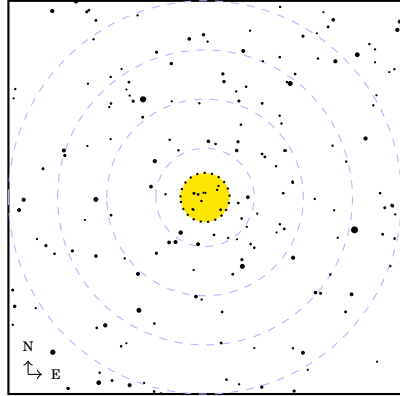
C16

U85 = NGC 7243

RA : 22.3^h PSA: 73
Dec: +50° OSA: N24
Con: CSA:

Type: OC
Mag : 6.4
Size : 30'

See: DSC:TCO OSA



C17

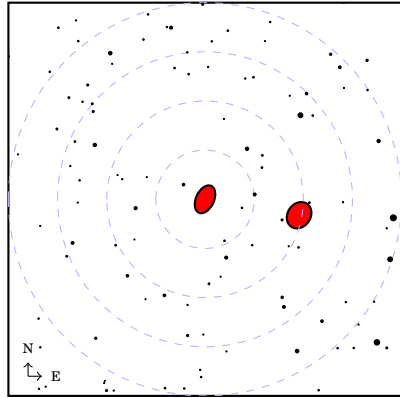
NGC 147

RA : 00.6^h PSA: 3
Dec: +49° OSA:
Con: CSA:

Type: Gal
Mag : 9.5
Size : 18' × 11'

See: DSC:TCO

Notes: C18 is 1° E.



C18

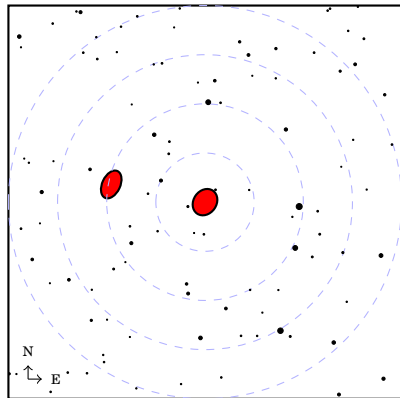
NGC 185

RA : 00.6^h PSA: 3
Dec: +48° OSA:
Con: CSA:

Type: Gal
Mag : 9.2
Size : 17' × 14'

See: DSC:TCO

Notes: C17 is 1° W.



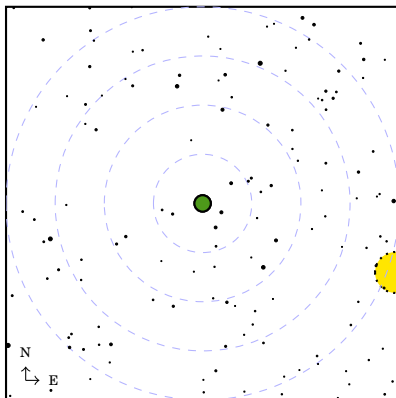
C19

IC 5146 = Cocoon Nebula

RA : 21.9 ^h	PSA: 73	Type: BN
Dec: +47°	OSA:	Mag : 9.3
Con:	CSA:	Size : 10'

See: DSC:TCO

Notes: Cluster is magnitude 7.2 (DSC:TCO). U84 is 2° ESE.



C20

NGC 7000 = North America Nebula

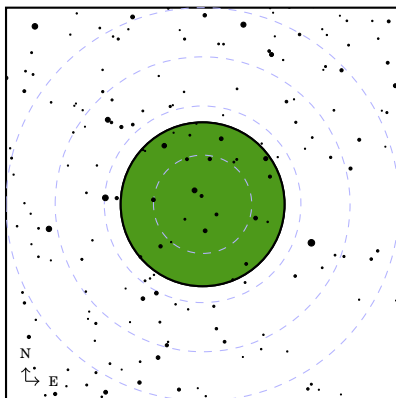
RA : 21.0 ^h	PSA: 62	Type: BN
Dec: +44°	OSA: N20	Mag : 3.8
Con:	CSA:	Size : 100' × 60'

See: DSC:TCO OSA

α Cyg Deneb (mag 1.3)

→ 1° S

→ 3° E to C20

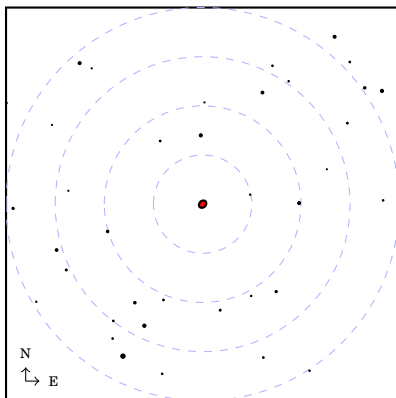


C21

NGC 4449 = Box Galaxy

RA : 12.5 ^h	PSA: 32/43	Type: Gal
Dec: +44°	OSA: N12	Mag : 9.6
Con:	CSA:	Size : 5' × 4'

See: DSC:TCO OSA

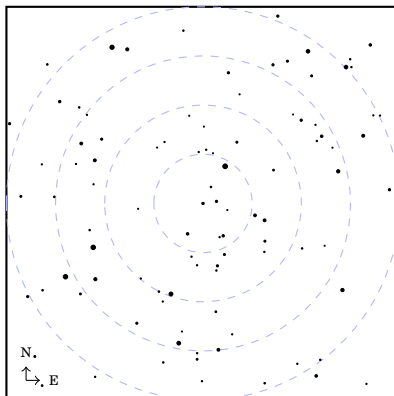


C22

NGC 7662 = Light Blue Snowball

RA : 23.4 ^h	PSA: 72	Type: PN
Dec: +43°	OSA: N24	Mag: 8.3
Con:	CSA:	Size : 0.5'

See: DSC:TCO OSA

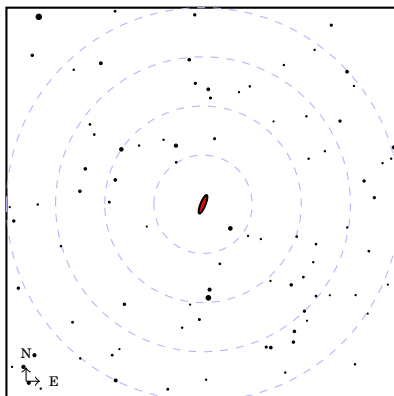


C23

NGC 891

RA : 02.4 ^h	PSA: 2	Type: Gal
Dec: +42°	OSA: N0	Mag: 9.9
Con:	CSA:	Size : 12' × 3'

See: DSC:TCO OSA

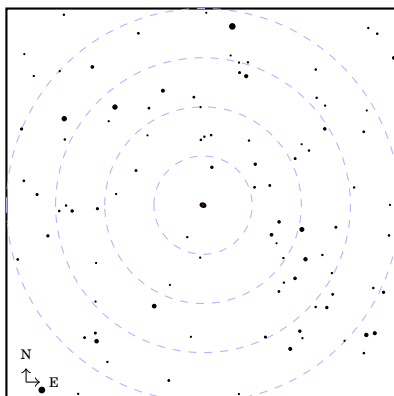


C24

NGC 1275 = Perseus A

RA : 03.3 ^h	PSA: 13	Type: Gal
Dec: +42°	OSA:	Mag: 11.9
Con:	CSA:	Size : 3' × 2'

See: DSC:TCO



C25

NGC 2419

RA : 07.6^h PSA: 23

Dec: +39° OSA:

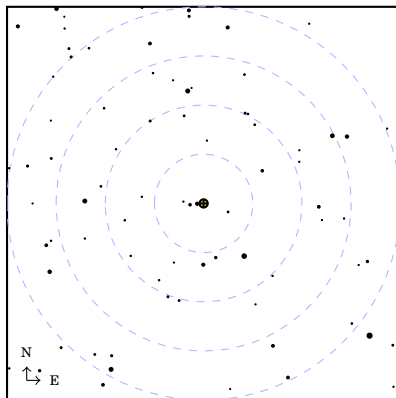
Con: CSA:

Type: GC

Mag : 10.4

Size : 5'

See: DSC:TCO



C26

NGC 4244

RA : 12.3^h PSA: 32/43

Dec: +38° OSA: N12

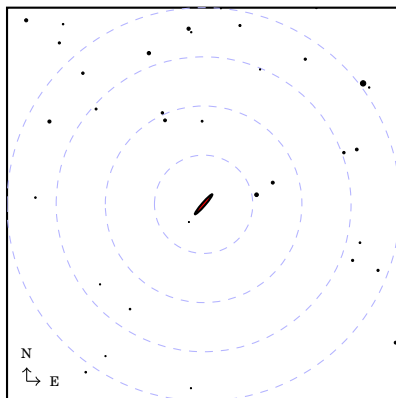
Con: CSA:

Type: Gal

Mag : 10.4

Size : 16' × 2'

See: DSC:TCO OSA



C27

NGC 6888 = Crescent Nebula

RA : 20.2^h PSA: 62

Dec: +38° OSA:

Con: CSA:

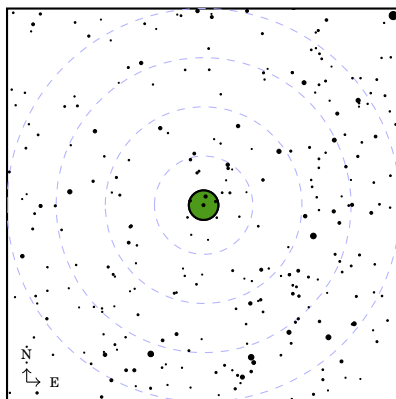
Type: BN

Mag : 8.8

Size : 18' × 13'

See: DSC:TCO

Notes: Central star is magnitude 7.7 (DSC:TCO).



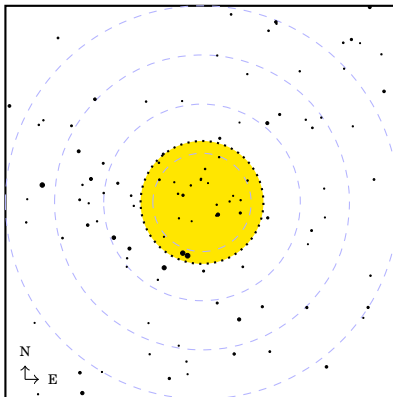
C28

U7 = NGC 752

RA : 01.9^h PSA: 2
Dec: +38° OSA: N0
Con: CSA:

Type: OC
Mag : 5.7
Size : 75'

See: DSC:TCO OSA



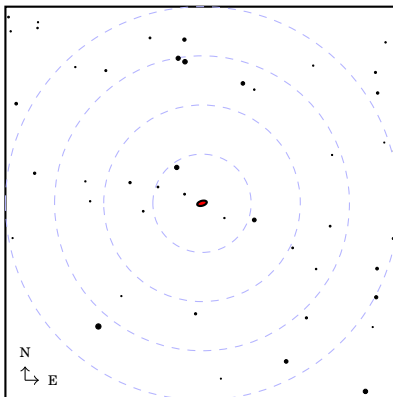
C29

NGC 5005

RA : 13.2^h PSA: 43
Dec: +37° OSA:
Con: CSA:

Type: Gal
Mag : 9.8
Size : 6' × 3'

See: DSC:TCO



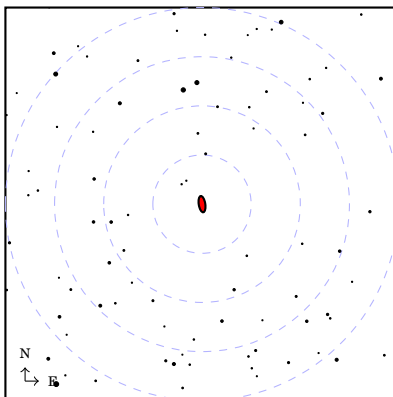
C30

NGC 7331

RA : 22.6^h PSA: 72/74
Dec: +34° OSA: E23
Con: CSA:

Type: Gal
Mag : 9.5
Size : 10' × 4'

See: DSC:TCO OSA



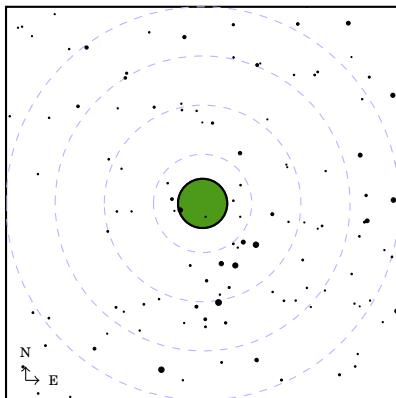
C31

IC 405 = Flaming Star Nebula

RA : 05.3 ^h	PSA: 12	Type: BN
Dec: +34°	OSA:	Mag : 9.2
Con:	CSA:	Size : 30' × 20'

See: DSC:TCO

Notes: AE Aur (towards the SW edge) is magnitude 6 and variable (DSC:TCO).

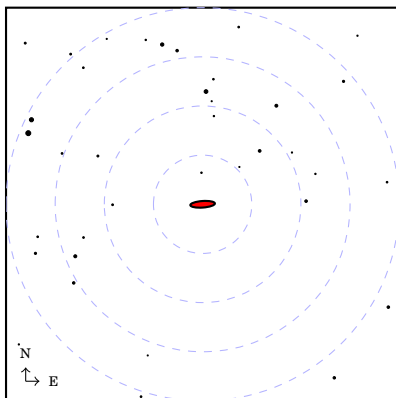


C32

NGC 4631 = Whale Galaxy

RA : 12.7 ^h	PSA: 43/45	Type: Gal
Dec: +33°	OSA: E13	Mag : 9.2
Con:	CSA:	Size : 15' × 4'

See: DSC:TCO OSA



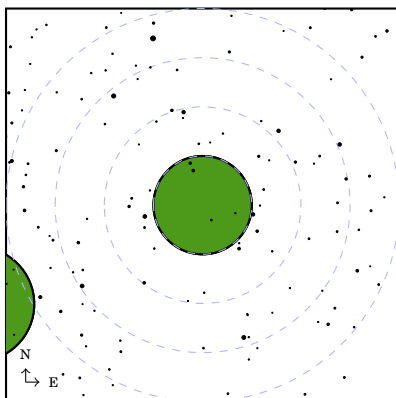
C33

NGC 6992 = Eastern Veil Nebula

RA : 20.9 ^h	PSA: 62	Type: BN
Dec: +32°	OSA: N20	Mag : 7.5
Con:	CSA:	Size : 60' × 8'

See: DSC:TCO OSA

Notes: C34 is 2.5° SW.



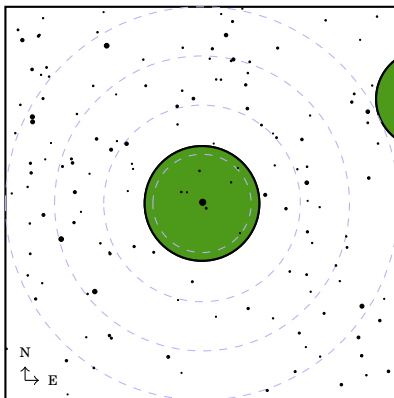
C34

NGC 6960 = Western Veil Nebula

RA : 20.8 ^h	PSA: 62	Type: BN
Dec: +31°	OSA: N20	Mag : 7.9
Con:	CSA:	Size : 70' × 6'

See: DSC:TCO OSA

Notes: C33 is 2.5° NE.

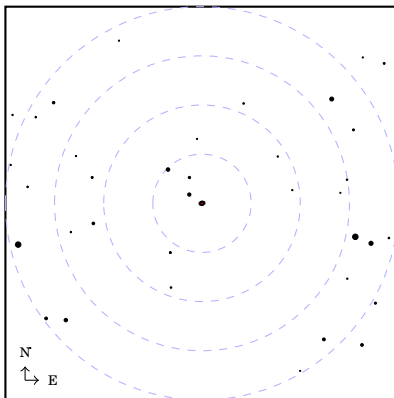


C35

NGC 4889

RA : 13.0 ^h	PSA: 43/45	Type: Gal
Dec: +28°	OSA:	Mag : 11.5
Con:	CSA:	Size : 3' × 2'

See: DSC:TCO



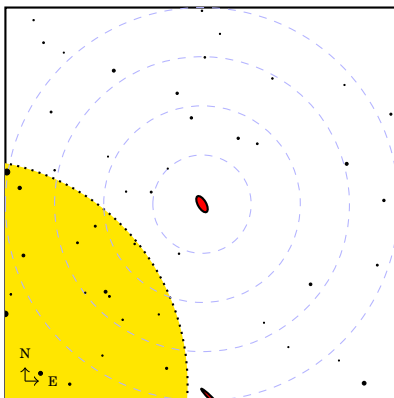
C36

NGC 4559

RA : 12.6 ^h	PSA: 43/45	Type: Gal
Dec: +28°	OSA: E13	Mag : 10.0
Con:	CSA:	Size : 11' × 5'

See: DSC:TCO OSA

Notes: C38 is 2° S. U44 is 4° to SW.



C37

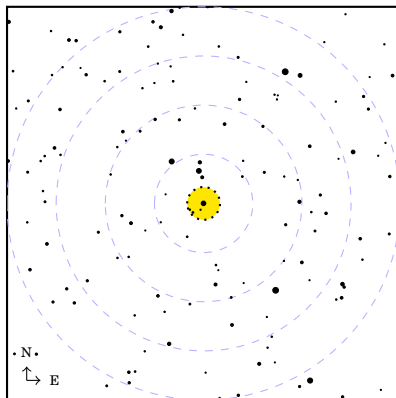
NGC 6885

RA : 20.2^h PSA: 62/64 Type: OC

Dec: +26° OSA: Mag : 8.1

Con: CSA: Size : 20'

See: DSC:TCO



C38

NGC 4565 = Needle Galaxy

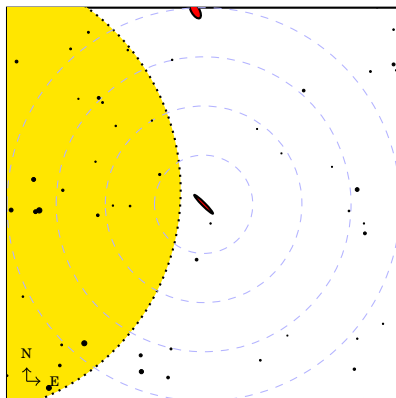
RA : 12.6^h PSA: 43/45 Type: Gal

Dec: +26° OSA: E13 Mag : 9.6

Con: CSA: Size : 16' × 2'

See: DSC:TCO OSA

Notes: C36 is 2° N. U44 is 3° to W.



C39

U36 = NGC 2392 = Eskimo Nebula

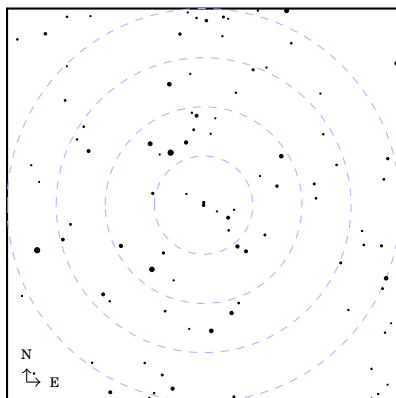
RA : 07.5^h PSA: 25 Type: PN

Dec: +21° OSA: E7 Mag : 9.2

Con: CSA: Size : 0.8' × 0.7'

See: DSC:TCO DSC:TMO OSA

Notes: The open cluster NGC 2420 is 2° N (see DSC:TCO).



C40

NGC 3626

RA : 11.3^h PSA: 34

Dec: +18° OSA:

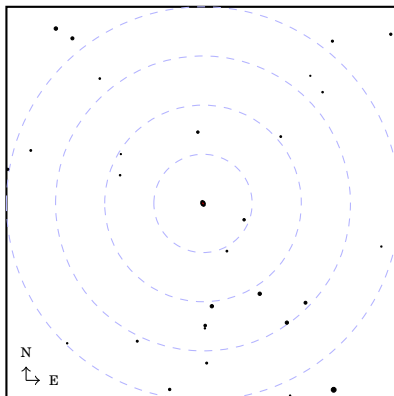
Con: CSA:

Type: Gal

Mag : 11.0

Size : 3' × 2'

See: DSC:TCO



C41

U18 = Mel 25 = Hyades

RA : 04.4^h PSA: 15

Dec: +16° OSA: E3

Con: CSA:

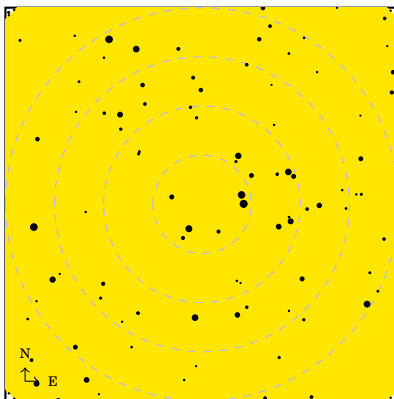
Type: OC

Mag : 0.5

Size : 5.5°

See: DSC:TCO OSA

Notes: Much bigger than the finder chart.



C42

NGC 7006

RA : 21.0^h PSA: 64/75

Dec: +16° OSA:

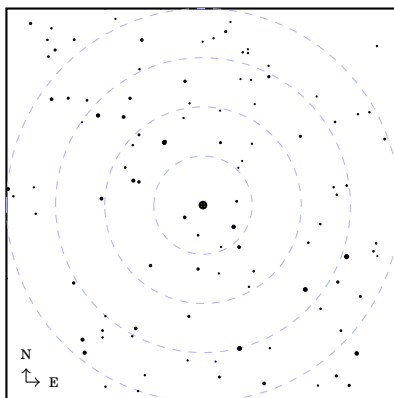
Con: CSA:

Type: GC

Mag : 10.6

Size : 4'

See: DSC:TCO



C43

NGC 7814

RA : 00.1^h PSA: 5/74

Dec: +16° OSA:

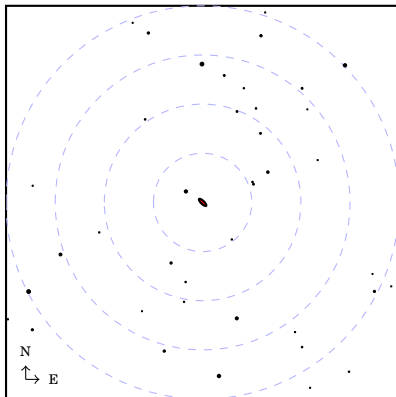
Con: CSA:

Type: Gal

Mag : 10.6

Size : 6' × 2'

See: DSC:TCO



C44

NGC 7479

RA : 23.1^h PSA: 74

Dec: +12° OSA:

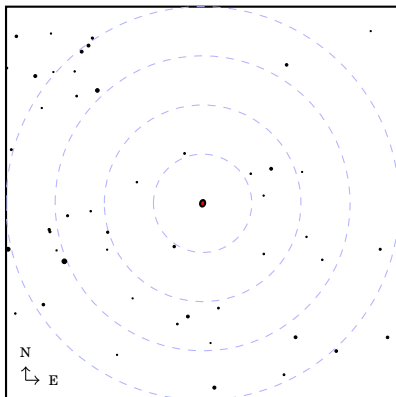
Con: CSA:

Type: Gal

Mag : 10.8

Size : 4' × 3'

See: DSC:TCO



C45

NGC 5248

RA : 13.6^h PSA: 44

Dec: +09° OSA:

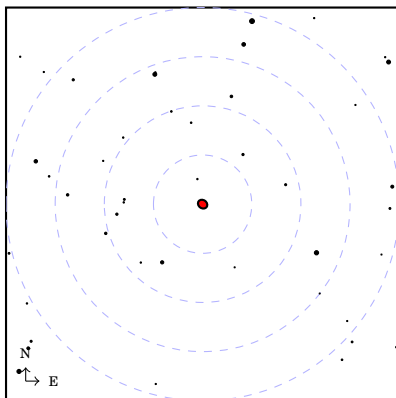
Con: CSA:

Type: Gal

Mag : 10.3

Size : 6' × 5'

See: DSC:TCO



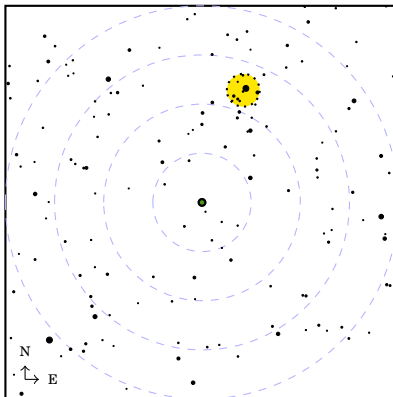
C46

NGC 2261 = Hubble's Variable Nebula

RA : 06.7 ^h	PSA: 25/E	Type: BN
Dec: +09°	OSA: E7	Mag: 10
Con:	CSA:	Size : 4' × 2'

See: DSC:TCO OSA

Notes: The magnitude of the nebula is variable (DSC:TCO). U31 is 1.25° NNE.

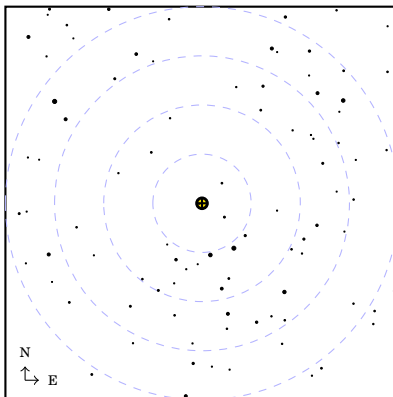


C47

NGC 6934

RA : 20.6 ^h	PSA: 64	Type: GC
Dec: +07°	OSA:	Mag: 8.8
Con:	CSA:	Size : 7'

See: DSC:TCO

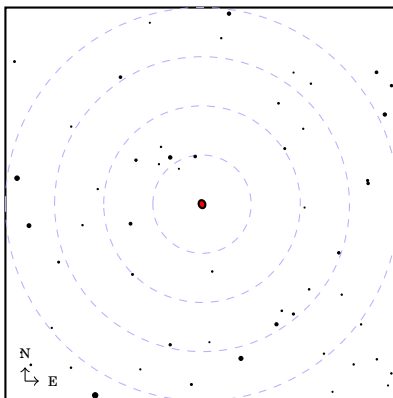


C48

NGC 2775

RA : 09.2 ^h	PSA: 24/35	Type: Gal
Dec: +07°	OSA:	Mag: 10.1
Con:	CSA:	Size : 5' × 4'

See: DSC:TCO



C49

NGC 2237/38/46 = Rosette Nebula

RA : 06.5 ^h	PSA: 25/E	Type: BN
Dec: +05°	OSA: E9	Mag : 5
Con:	CSA:	Size : 80' × 60'

See: DSC:TCO OSA

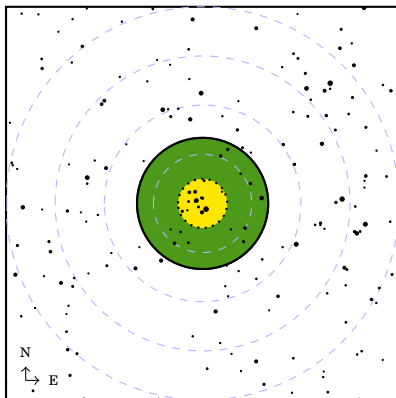
Notes: Surrounding C50.

α Ori Betelgeuse (mag 0.3–0.9)

→ 3° S

→ 7° E to 8 Mon

→ 2° E to C49



C50

U30 = NGC 2244

RA : 06.5 ^h	PSA: 25/E	Type: OC
Dec: +05°	OSA: E9	Mag : 4.4
Con:	CSA:	Size : 30'

See: DSC:TCO OSA

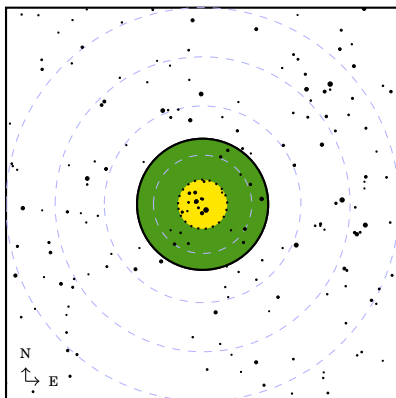
Notes: Surrounded by C49.

α Ori Betelgeuse (mag 0.3–0.9)

→ 3° S

→ 7° E to 8 Mon

→ 2° E to C50/U30

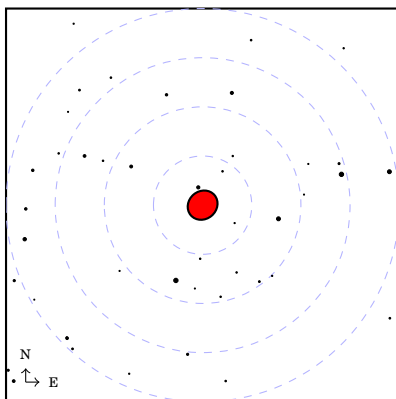


C51

IC 1613

RA : 01.1 ^h	PSA: 5/7	Type: Gal
Dec: +02°	OSA:	Mag : 9.2
Con:	CSA:	Size : 19' × 17'

See: DSC:TCO



C52

NGC 4697

RA : 12.8^h PSA: 47

Dec: -06° OSA: E12

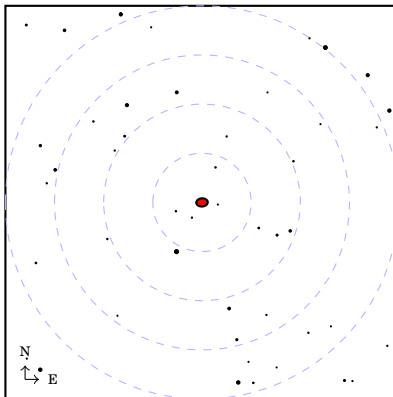
Con: CSA:

Type: Gal

Mag : 9.0

Size : 7' × 5'

See: DSC:TCO OSA



C53

NGC 3115 = The Spindle

RA : 10.1^h PSA: 37

Dec: -08° OSA: E10

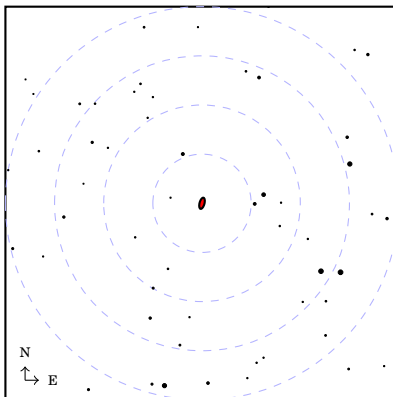
Con: CSA:

Type: Gal

Mag : 8.9

Size : 7' × 3'

See: DSC:TCO OSA



C54

NGC 2506

RA : 08.0^h PSA: 26

Dec: -11° OSA:

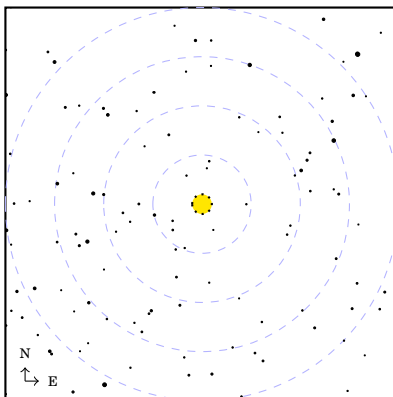
Con: CSA:

Type: OC

Mag : 7.6

Size : 12'

See: DSC:TCO



C55

NGC 7009 = Saturn Nebula

RA : 21.1^h

PSA: 77

Dec: -11°

OSA: E24

Con:

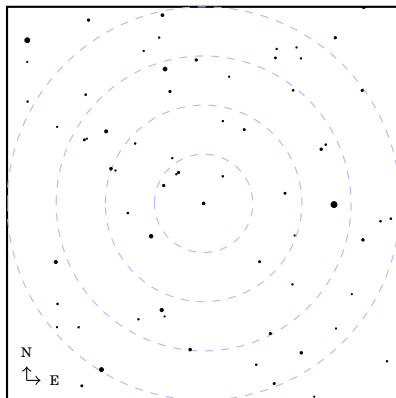
CSA:

Type: PN

Mag : 8.0

Size : 0.7' × 0.4'

See: DSC:TCO OSA



C56

NGC 246

RA : 00.8^h

PSA: 7

Dec: -12°

OSA:

Con:

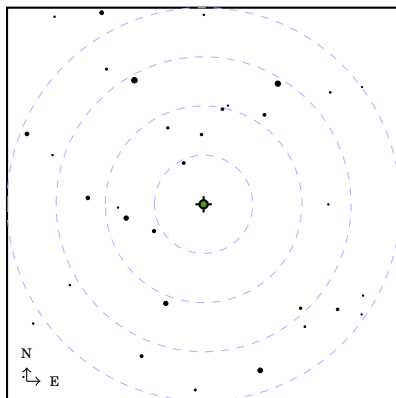
CSA:

Type: PN

Mag : 10.9

Size : 5' × 4'

See: DSC:TCO



C57

NGC 6822 = Barnard's Galaxy

RA : 19.7^h

PSA: 66

Dec: -15°

OSA: E22

Con:

CSA:

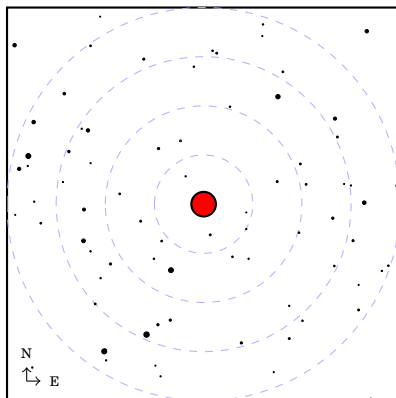
Type: Gal

Mag : 7.9

Size : 15'

See: DSC:TCO OSA

Notes: U73 is 0.7° NNW.

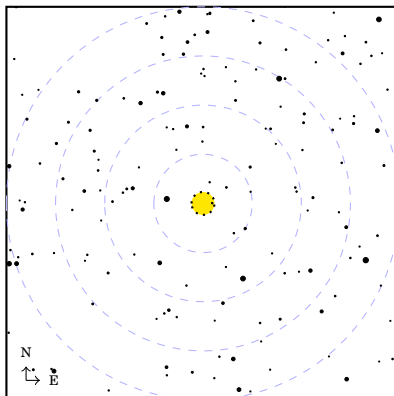


C58

NGC 2360 = Caroline's Cluster

RA : 07.3 ^h	PSA: 27	Type: OC
Dec: -16°	OSA: E8	Mag : 7.0
Con:	CSA:	Size : 14'

See: DSC:TCO OSA

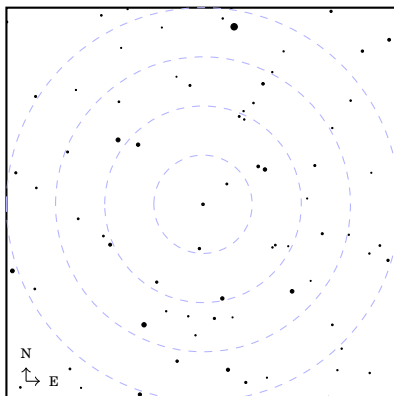


C59

U43 = NGC 3242 = Ghost of Jupiter

RA : 10.4 ^h	PSA: 36/37	Type: PN
Dec: -19°	OSA: E10	Mag : 7.3
Con:	CSA:	Size : 0.8' × 0.6'

See: DSC:TCO DSC:TMO OSA



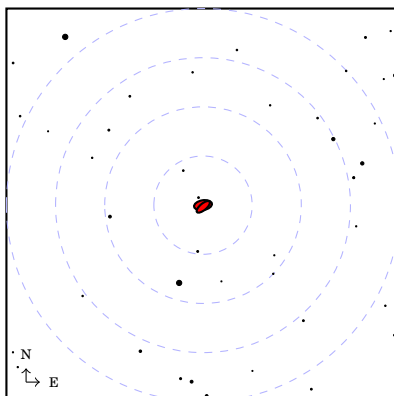
C60

NGC 4038 = NW Antennae Galaxy

RA : 12.0 ^h	PSA: 36/47	Type: Gal
Dec: -19°	OSA:	Mag : 10.5
Con:	CSA:	Size : 11' × 6'

See: DSC:TCO

Notes: C61 is 1.2' SW.



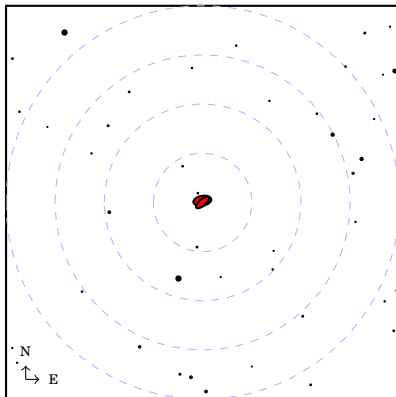
C61

NGC 4039 = SE Antennae Galaxy

RA : 12.0 ^h	PSA: 36/47	Type: Gal
Dec: -19°	OSA:	Mag : 11.2
Con:	CSA:	Size : 10' × 4'

See: DSC:TCO

Notes: C60 is 1.2' NW

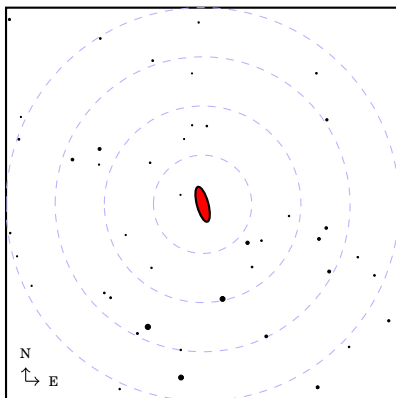


C62

NGC 247

RA : 00.8 ^h	PSA: 7	Type: Gal
Dec: -21°	OSA: E0	Mag : 8.9
Con:	CSA:	Size : 22' × 7'

See: DSC:TCO OSA

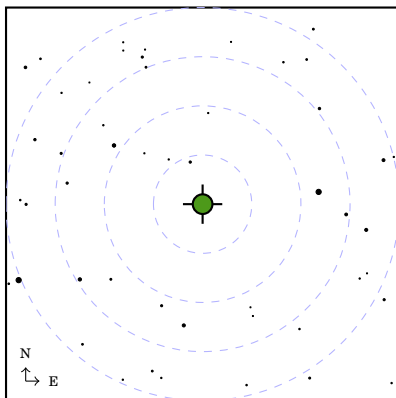


C63

NGC 7293 = Helix Nebula

RA : 22.5 ^h	PSA: 76/77	Type: PN
Dec: -21°	OSA: E24	Mag : 6.0
Con:	CSA:	Size : 12' × 10'

See: DSC:TCO OSA



C64

NGC 2362 = τ CMa Cluster

RA : 07.3^h PSA: 27

Dec: -25° OSA: E6

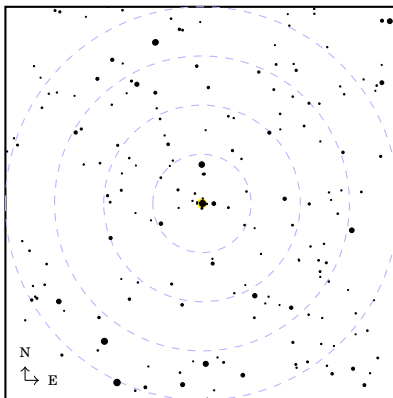
Con: CSA:

Type: OC

Mag : 4.1

Size : 6'

See: DSC:TCO OSA



C65

NGC 253 = Sculptor Galaxy

RA : 00.8^h PSA: 7/9

Dec: -25° OSA: E0

Con: CSA:

Type: Gal

Mag : 7.1

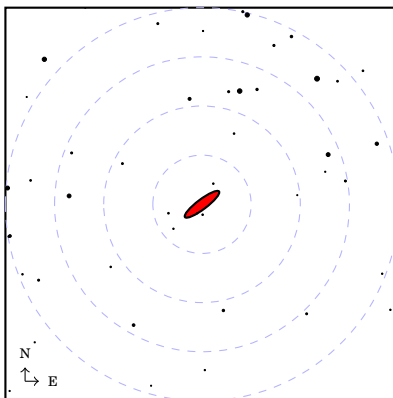
Size : 26' × 6'

See: DSC:TCO OSA

β Cet Deneb Kaitos (mag 2.0)

→ 7° S

→ 1° E to C65



C66

NGC 5694

RA : 14.7^h PSA: 46

Dec: -27° OSA:

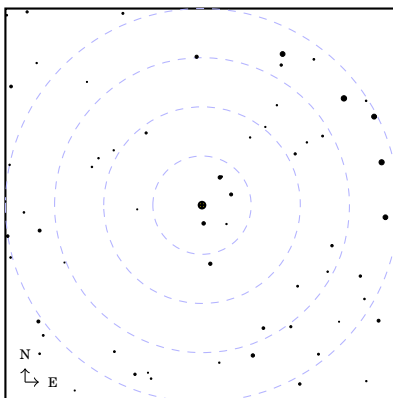
Con: CSA:

Type: GC

Mag : 10.2

Size : 4'

See: DSC:TCO



C67

NGC 1097

RA : 02.8^h PSA: 6/8

Dec: -30° OSA:

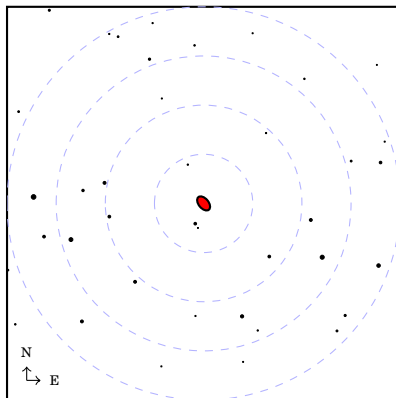
Con: CSA:

Type: Gal

Mag: 9.2

Size : 10' × 6'

See: DSC:TCO



C68

NGC 6729 = R CrA Nebula

RA : 19.0^h PSA: 69

Dec: -37° OSA:

Con: CSA:

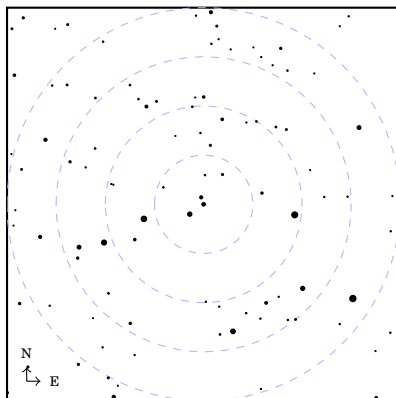
Type: BN

Mag: 9.5

Size : 1.3'

See: DSC:TCO

Notes: The morphology and magnitude of the nebula are variable (DSC:TCO).



C69

NGC 6302 = Bug Nebula

RA : 17.2^h PSA: 58

Dec: -37° OSA:

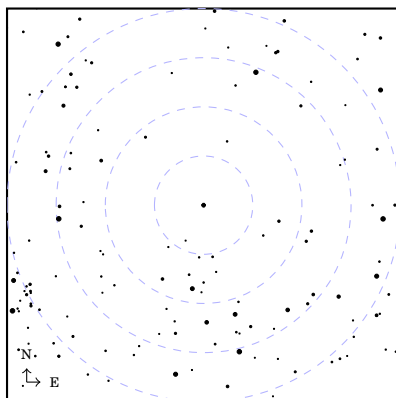
Con: CSA:

Type: PN

Mag: 9.6

Size : 1.4' × 0.4'

See: DSC:TCO



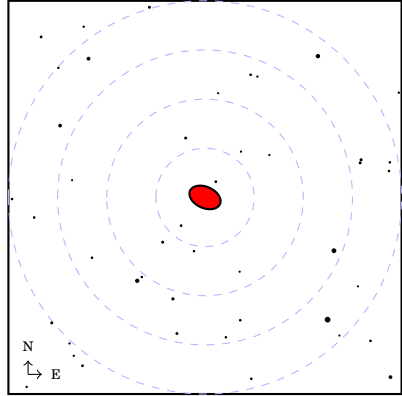
C70

NGC 300

RA : 00.9^h PSA: 9
Dec: -38° OSA:
Con: CSA:

Type: Gal
Mag : 7.2
Size : 20' × 13'

See: DSC:TCO



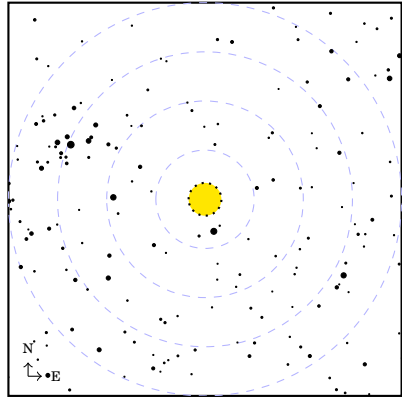
C71

NGC 2477

RA : 07.9^h PSA: 28
Dec: -39° OSA: S6
Con: CSA:

Type: OC
Mag : 5.8
Size : 20'

See: DSC:TCO OSA



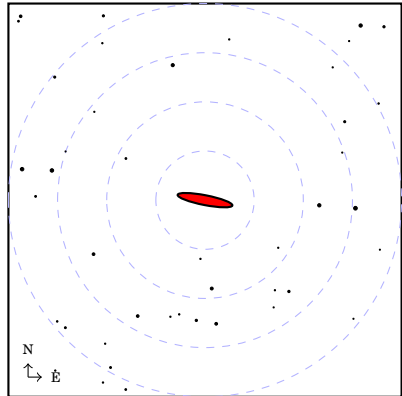
C72

NGC 55

RA : 00.2^h PSA: 9
Dec: -39° OSA: S0
Con: CSA:

Type: Gal
Mag : 7.5
Size : 34' × 6'

See: DSC:TCO OSA



C73

NGC 1851

RA : 05.2^h PSA: 18

Dec: -40° OSA: S0

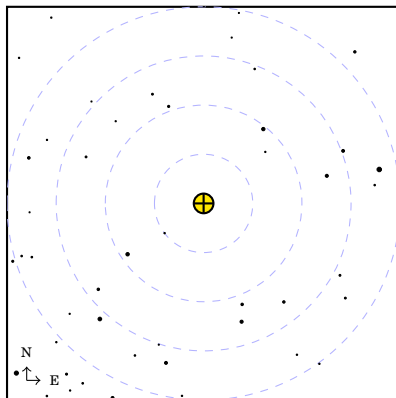
Con: CSA:

See: DSC:TCO OSA

Type: GC

Mag : 7.1

Size : 12'



C74

NGC 3132

RA : 10.1^h PSA: 39

Dec: -40° OSA: S6

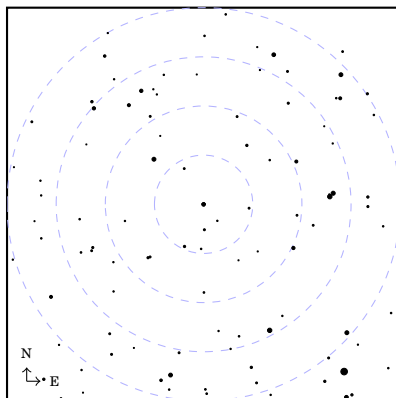
Con: CSA:

See: DSC:TCO OSA

Type: PN

Mag : 9.2

Size : 1.4' × 0.9'



C75

NGC 6124

RA : 16.4^h PSA: 58

Dec: -41° OSA: S21

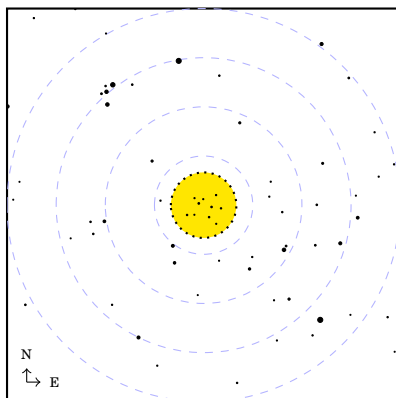
Con: CSA:

See: DSC:TCO OSA

Type: OC

Mag : 5.3

Size : 40'



C76

NGC 6231

RA : 16.9^h PSA: 58

Dec: -42° OSA: S21

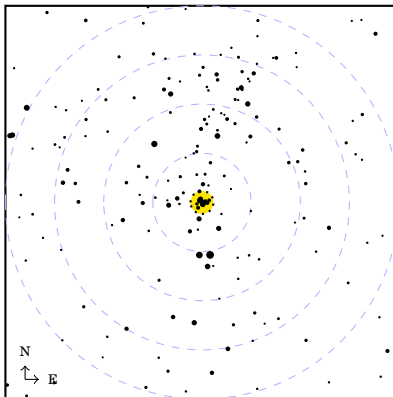
Con: CSA:

Type: OC

Mag: 2.6

Size : 14'

See: DSC:TCO OSA



C77

NGC 5128 = Cen A

RA : 13.4^h PSA: 48/49

Dec: -43° OSA: S15

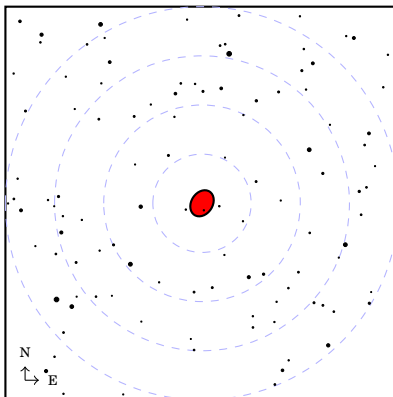
Con: CSA:

Type: Gal

Mag: 6.6

Size : 17' × 13'

See: DSC:TCO OSA



C78

NGC 6541

RA : 18.1^h PSA: 69

Dec: -44° OSA: S21

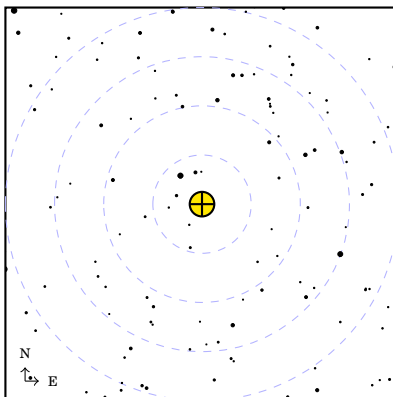
Con: CSA:

Type: GC

Mag: 6.3

Size : 15'

See: DSC:TCO OSA



C79

NGC 3201

RA : 10.3^h PSA: 39

Dec: -46° OSA: S6

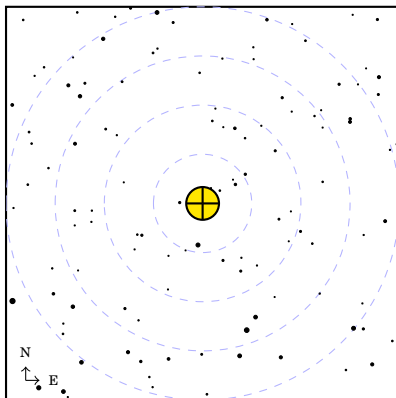
Con: CSA:

See: DSC:TCO OSA

Type: GC

Mag : 6.7

Size : 20'



C80

NGC 5139 = ω Cen

RA : 13.4^h PSA: 48/49

Dec: -47° OSA: S15

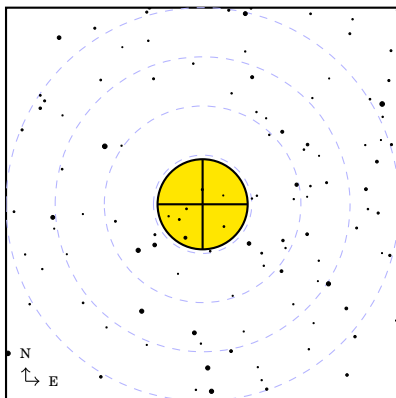
Con: CSA:

See: DSC:TCO OSA

Type: GC

Mag : 3.9

Size : 55'



C81

NGC 6352

RA : 17.4^h PSA: 58

Dec: -48° OSA:

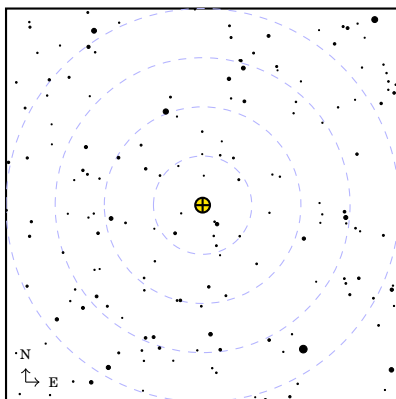
Con: CSA:

See: DSC:TCO

Type: GC

Mag : 7.8

Size : 9'



C82

NGC 6193

RA : 16.7^h PSA: 58

Dec: -49° OSA:

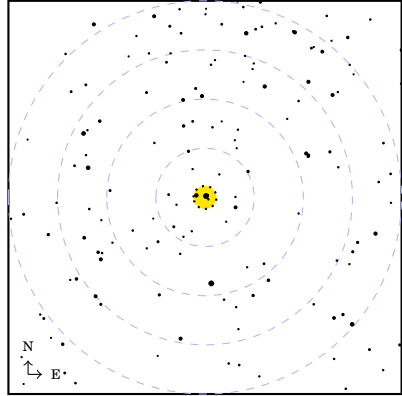
Con: CSA:

Type: OC

Mag : 4.8

Size : 14'

See: DSC:TCO



C83

NGC 4945

RA : 13.1^h PSA: 49

Dec: -49° OSA:

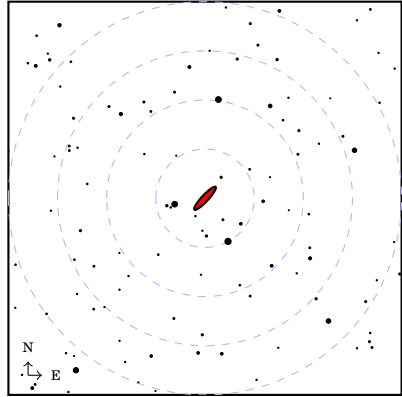
Con: CSA:

Type: Gal

Mag : 8.8

Size : 19' × 4'

See: DSC:TCO



C84

NGC 5286

RA : 13.8^h PSA: 48

Dec: -51° OSA:

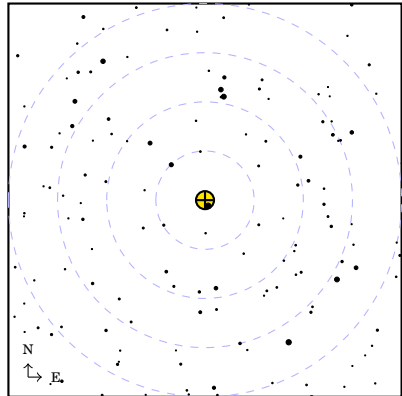
Con: CSA:

Type: GC

Mag : 7.2

Size : 11'

See: DSC:TCO



C85

IC 2391 = *o* Vel Cluster

RA : 08.7^h PSA: 28

Dec: -53° OSA: S6

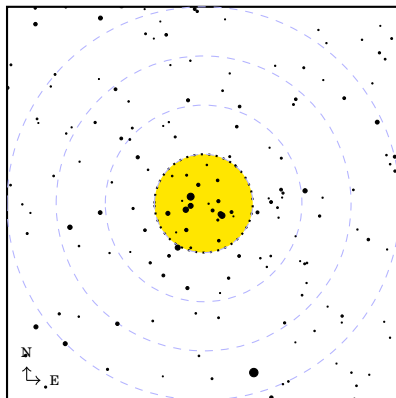
Con: CSA:

See: DSC:TCO OSA

Type: OC

Mag : 2.8

Size : 60'



C86

NGC 6397

RA : 17.7^h PSA: 58

Dec: -54° OSA: S18

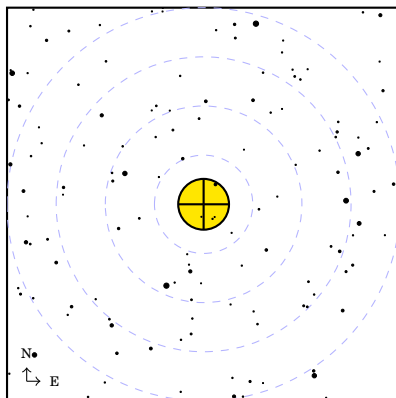
Con: CSA:

See: DSC:TCO OSA

Type: GC

Mag : 5.3

Size : 31'



C87

NGC 1261

RA : 03.2^h PSA: 19

Dec: -55° OSA:

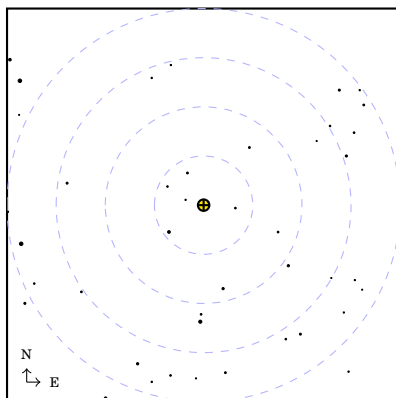
Con: CSA:

See: DSC:TCO

Type: GC

Mag : 8.3

Size : 7'



C88

NGC 5823

RA : 15.1^h PSA: 59

Dec: -56° OSA:

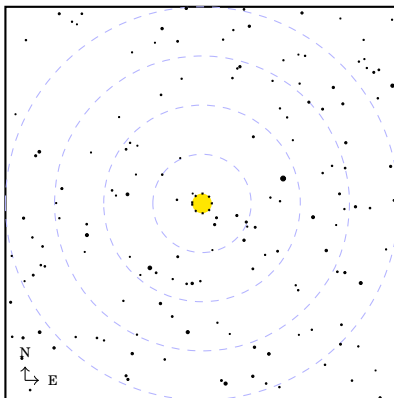
Con: CSA:

Type: OC

Mag : 7.9

Size : 12'

See: DSC:TCO



C89

NGC 6087 = S Nor Cluster

RA : 16.3^h PSA: 58/60

Dec: -58° OSA: S18

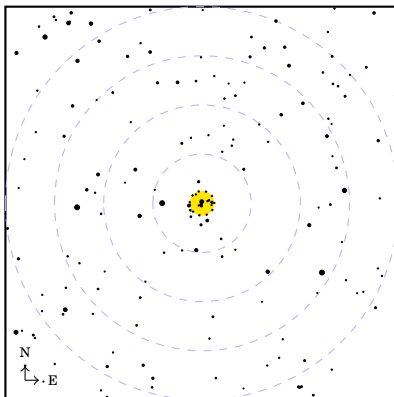
Con: CSA:

Type: OC

Mag : 5.2

Size : 15'

See: DSC:TCO OSA



C90

NGC 2867

RA : 09.4^h PSA: 39

Dec: -58° OSA:

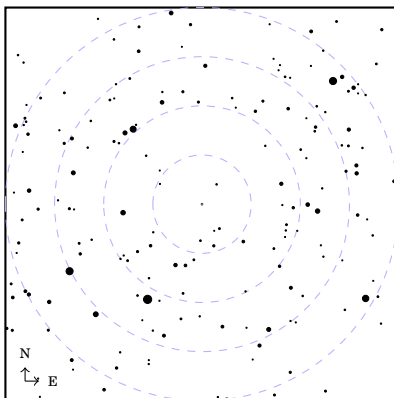
Con: CSA:

Type: PN

Mag : 9.7

Size : 0.2'

See: DSC:TCO



C91

NGC 3532

RA : 11.1^h PSA: 38/40

Dec: -59° OSA: S9

Con: CSA:

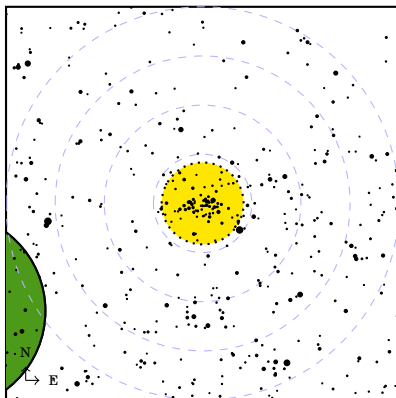
Type: OC

Mag : 3.2

Size : 50'

See: DSC:TCO OSA

Notes: C90 is 2.5° SW.



C92

NGC 3372 = η Car Nebula

RA : 10.8^h PSA: 38/40

Dec: -60° OSA: S9

Con: CSA:

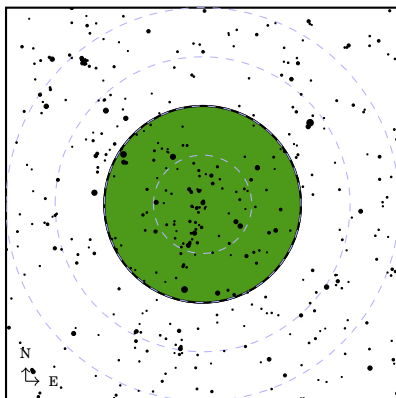
Type: BN

Mag : 4.5

Size : 120'

See: DSC:TCO OSA

Notes: The nebula is variable in magnitude. C91 is 2.5° NE.



C93

NGC 6752

RA : 19.2^h PSA: 69/70

Dec: -60° OSA: S24

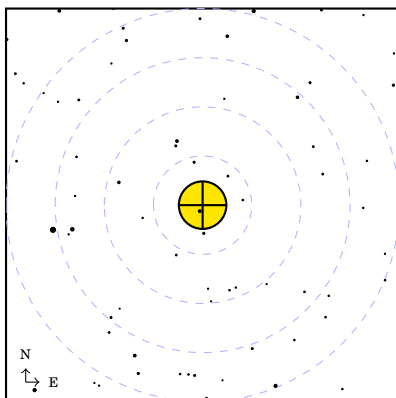
Con: CSA:

Type: GC

Mag : 5.4

Size : 29'

See: DSC:TCO OSA



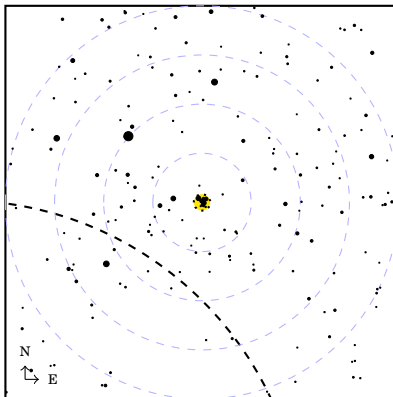
C94

NGC 4755 = Jewel Box

RA : 12.9 ^h	PSA: 49/50	Type: OC
Dec: -60°	OSA: S12	Mag: 4.2
Con:	CSA:	Size : 10'

See: DSC:TCO OSA

Notes: The northern edge of C99 is 1° S.

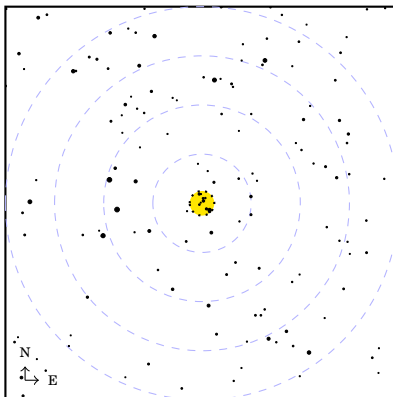


C95

NGC 6025

RA : 16.1 ^h	PSA: 59/60	Type: OC
Dec: -60°	OSA:	Mag: 5.5
Con:	CSA:	Size : 15'

See: DSC:TCO

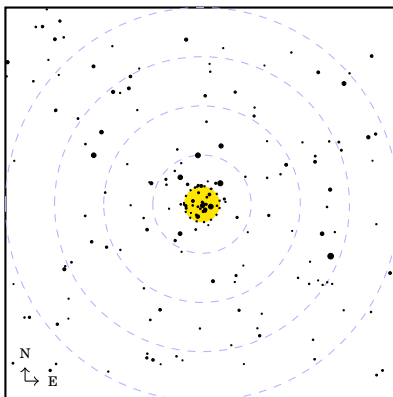


C96

NGC 2516 = Southern Beehive Cluster

RA : 08.0 ^h	PSA: 28/30	Type: OC
Dec: -61°	OSA: S3	Mag: 3.5
Con:	CSA:	Size : 22'

See: DSC:TCO OSA



C97

NGC 3766

RA : 11.6^h PSA: 38/40

Dec: -62° OSA: S12

Con: CSA:

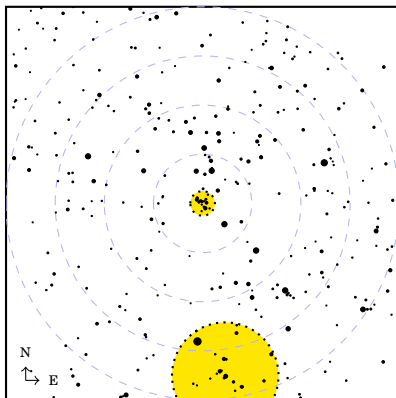
Type: OC

Mag : 5.3

Size : 15'

See: DSC:TCO OSA

Notes: C100 is 1.8° S.



C98

NGC 4609 = Coalsack Cluster

RA : 12.7^h PSA: 49/50

Dec: -63° OSA:

Con: CSA:

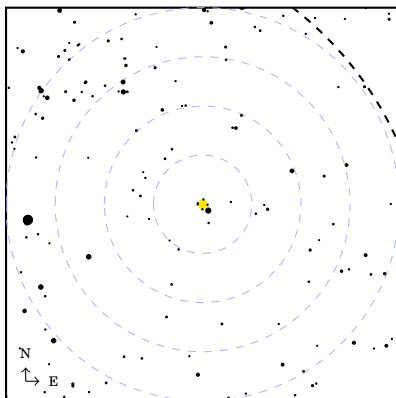
Type: OC

Mag : 6.9

Size : 6'

See: DSC:TCO

Notes: Surrounded by C99.



C99

Coalsack

RA : 12.5^h PSA: 49/50

Dec: -64° OSA: S12

Con: CSA:

Type: DN

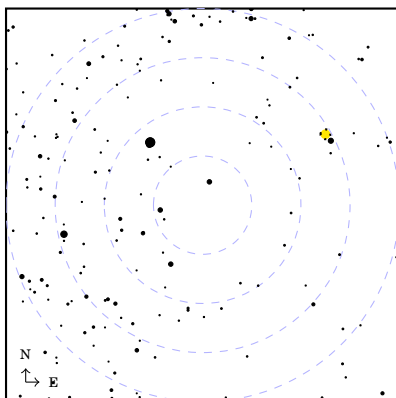
Mag :

Size : 7.0° × 5.0°

See: DSC:TCO OSA

Notes: Much bigger than the finder chart.

Surrounds C98. C94 is 1° N of the northern edge.



C100

Cr 249 = λ Cen Cluster

RA : 11.6^h PSA: 38/40

Dec: -63° OSA:

Con: CSA:

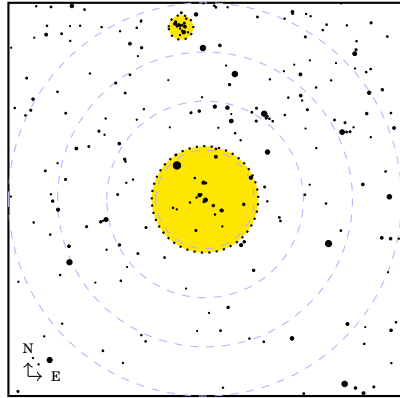
Type: OC

Mag: 2.9

Size : 65' × 40'

See: DSC:TCO

Notes: C97 is 1.8° N.



C101

NGC 6744

RA : 19.2^h PSA: 70

Dec: -64° OSA:

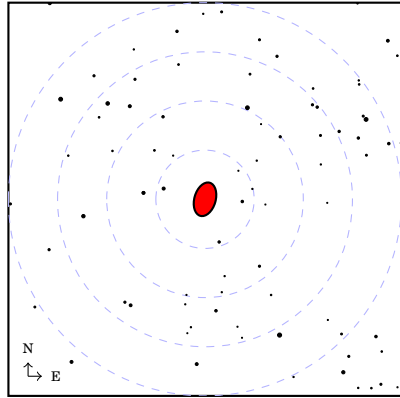
Con: CSA:

Type: Gal

Mag: 8.6

Size : 21' × 13'

See: DSC:TCO



C102

IC 2602 = θ Car Cluster

RA : 10.7^h PSA: 38/40

Dec: -64° OSA: S9

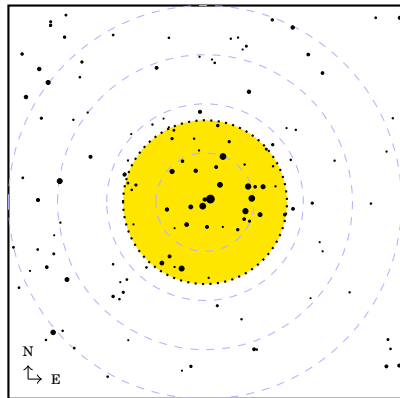
Con: CSA:

Type: OC

Mag: 1.9

Size : 100'

See: DSC:TCO OSA



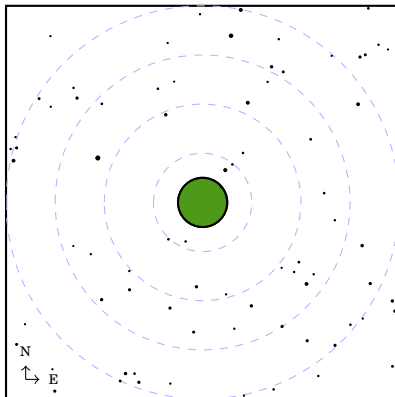
C103

NGC 2070 = Tarantula Nebula

RA : 05.6 ^h	PSA: 20/30/D	Type: BN
Dec: -69°	OSA: S3	Mag : 4.0/8.3
Con:	CSA:	Size : 30' × 20'

See: DSC:TCO OSA

Notes: The nebula is magnitude 4.0 (DSC:TCO).
the cluster is magnitude 8.3 and has a diameter of
5' (DSC:TCO).

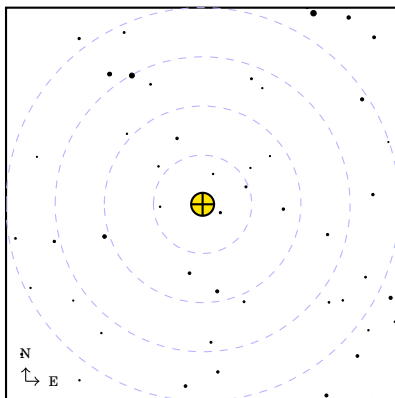


C104

NGC 362

RA : 01.1 ^h	PSA: 10/80	Type: GC
Dec: -71°	OSA: S0	Mag : 6.4
Con:	CSA:	Size : 14'

See: DSC:TCO OSA

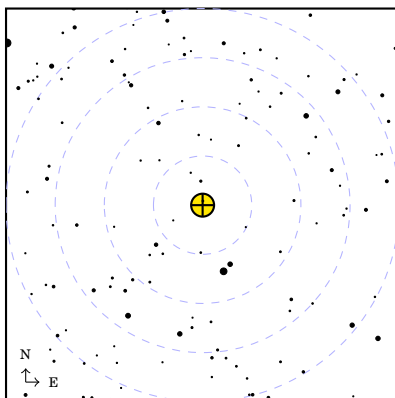


C105

NGC 4833

RA : 13.0 ^h	PSA: 50	Type: GC
Dec: -71°	OSA:	Mag : 8.4
Con:	CSA:	Size : 14'

See: DSC:TCO



C106

NGC 104 = 47 Tuc

RA : 00.4^h PSA: 10/80

Dec: -72° OSA: S0

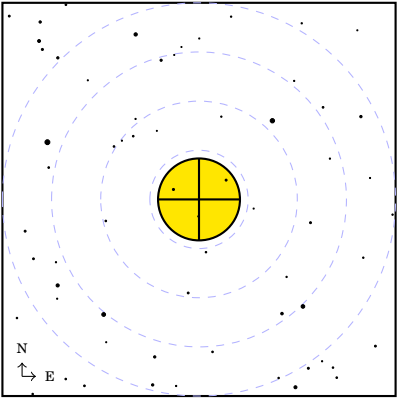
Con: CSA:

Type: GC

Mag : 4.0

Size : 50'

See: DSC:TCO OSA



C107

NGC 6101

RA : 16.4^h PSA: 60

Dec: -72° OSA:

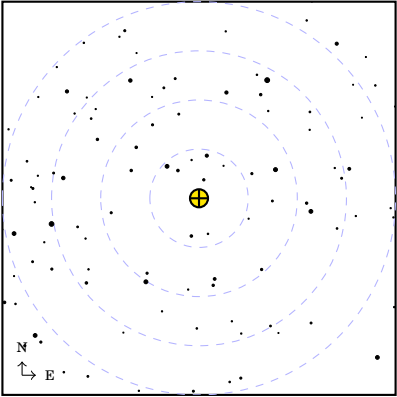
Con: CSA:

Type: GC

Mag : 9.2

Size : 11'

See: DSC:TCO



C108

NGC 4372

RA : 12.4^h PSA: 40/50

Dec: -73° OSA:

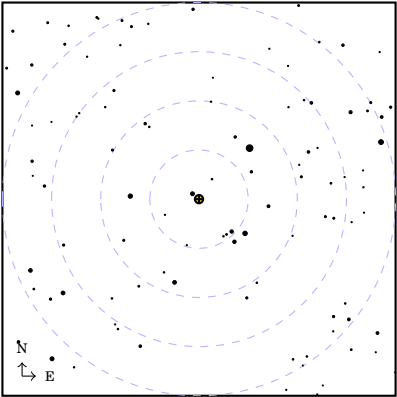
Con: CSA:

Type: GC

Mag : 7.2

Size : 5'

See: DSC:TCO



C109

NGC 3195

RA : 10.2^h PSA: 40

Dec: -81° OSA:

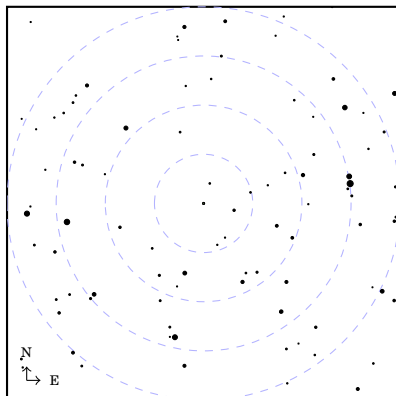
Con: CSA:

Type: PN

Mag : 11.6

Size : 0.6'

See: DSC:TCO



The Urban Observing Program

Objects

The Urban Observing Program of the Astronomical League consists of bright objects selected by Terry Trees for northern observers under light-polluted skies. There are 87 deep-sky objects, 12 double stars, and 1 variable star. The deep-sky objects range in declination from -35 to $+72$ degrees, cover all seasons, and include 41 Messier objects and 14 Caldwell objects.

I give finder charts for the deep-sky objects only; the stars are all no fainter than magnitude 5.2 and have Bayer designations, and so a standard all-sky atlas is adequate. I use U1 to U87 to label the deep-sky objects, following the order in right ascension given by Trees, but this is not a standard designation. For completeness, I include all of the deep-sky objects, even bright ones like the Hyades (U18).

The following table lists the deep-sky objects with their J2000 positions (decimal hours of right ascension and degrees of declination), the charts on which they appear in the *Pocket Sky Atlas*, their types, other names, and references to O'Meara's *Deep-Sky Companions* volumes (*The Messier Objects*, *The Caldwell Objects*, *The Secret Deep*, and *Hidden Treasures*).

I note that the Hyades (U18) and Coma Star Cluster (U44) are much bigger than the finder charts. They are really binocular objects and better located using a small-scale all-sky atlas. That said, be aware that the α Per Cluster (U15) and Coma Star Cluster (U44) are not labelled in the *Pocket Sky Atlas*.

Name	Position	PSA	Type	Other Names	DSC
U1	00.5 +60	3	OC	NGC 129	TSD
U2	00.7 +41	3	Gal	M32 = NGC 221	TMO
U3	00.7 +41	3	Gal	M31 = NGC 224 = Andromeda Galaxy	TMO
U4	01.3 +58	1/3	OC	C13 = NGC 457 = E.T. Cluster = Owl Cluster	TCO
U5	01.8 +61	1/2	OC	C10 = NGC 663 = Horseshoe Cluster	TCO
U6	01.8 +72	1	OC	Cr 463	TSD
U7	01.9 +38	2	OC	C28 = NGC 752	TCO
U8	02.3 +60	2	OC	Stock 2	TSD
U9	02.3 +57	2	OC	NGC 869 = h Per Cluster	TCO/TMO
U10	02.4 +57	2	OC	NGC 884 = χ Per Cluster	TCO/TMO
U11	02.6 +56	2	OC	Tr 2	
U12	02.7 -00	4	Gal	M77 = NGC 1068	TMO
U13	03.2 +63	11/13	OC	Tr 3	
U14	03.3 +60	11/13	OC	Stock 23 = Pazmino's Cluster	
U15	03.4 +49	3	OC	Mel 20 = α Per Cluster	
U16	03.5 +37	13	OC	NGC 1342	TSD
U17	03.8 +24	15/A	OC	M45 = Mel 22 = Pleiades	TMO
U18	04.4 +16	15	OC	C41 = Mel 25 = Hyades	TCO
U19	04.8 +19	15	OC	NGC 1647	TCO/HT
U20	05.2 +17	14	OC	NGC 1807	TSD
U21	05.2 +17	14	OC	NGC 1817	TSD
U22	05.5 +36	12	OC	M38 = NGC 1912	TMO
U23	05.6 +34	12	OC	M36 = NGC 1960	TMO
U24	05.6 -05	16/B	BN	M42 = NGC 1976 = Orion Nebula	TMO
U25	05.6 -04	16/B	OC	NGC 1981	HT
U26	05.9 +33	12	OC	M37 = NGC 2099	TMO
U27	06.2 +24	14	OC	M35 = NGC 2168	TMO
U28	06.1 +14	14/25	OC	NGC 2169	HT
U29	06.5 -05	25/27	OC	NGC 2232	TSD
U30	06.5 +05	25/E	OC	C50 = NGC 2244	TCO
U31	06.7 +10	25/E	OC	NGC 2264	HT
U32	06.8 +41	23	OC	NGC 2281	TSD
U33	06.8 -21	27	OC	M41 = NGC 2287	TMO
U34	06.9 +00	25/27	OC	NGC 2301	HT
U35	07.0 -08	27	OC	M50 = NGC 2323	TMO
U36	07.5 +21	25	PN	C39 = NGC 2392 = Eskimo Nebula	TCO/TMO
U37	08.2 -13	26	OC	NGC 2539	HT
U38	08.2 -06	26	OC	M48 = NGC 2548	TMO
U39	08.7 +20	24	OC	M44 = NGC 2632 = Beehive Cluster	TMO
U40	08.9 +12	24	OC	M67 = NGC 2682	TMO
U41	09.9 +69	31	Gal	M81 = NGC 3031	TMO
U42	09.9 +70	31	Gal	M82 = NGC 3034	TMO
U43	10.4 -19	36/37	PN	C59 = NGC 3242 = Ghost of Jupiter	TCO/TMO
U44	12.4 +26	45	OC	Mel 111 = Coma Star Cluster	HT

Name	Position	PSA	Type	Other Names	DSC
U45	12.4 +13	45/C	Gal	M84 = NGC 4374	TMO
U46	12.4 +13	45/C	Gal	M86 = NGC 4406	TMO
U47	12.5 +12	45/C	Gal	M87 = NGC 4486	TMO
U48	12.7 -12	47	Gal	M104 = NGC 4594 = Sombrero Galaxy	TMO
U49	12.8 +41	43	Gal	M94 = NGC 4736	TMO
U50	12.9 +22	45	Gal	M64 = NGC 4826 = Black-Eye Galaxy	TMO
U51	13.7 +28	43/44	GC	M3 = NGC 5272	TMO
U52	15.3 +02	55/57	GC	M5 = NGC 5904	TMO
U53	16.4 -27	56/58	GC	M4 = NGC 6121	TMO
U54	16.7 +36	52	GC	M13 = NGC 6205 = Hercules Cluster	TMO
U55	16.7 +24	54	PN	NGC 6210	HT
U56	16.8 -02	54/56	GC	M12 = NGC 6218	TMO
U57	17.0 -04	54/56	GC	M10 = NGC 6254	TMO
U58	17.0 -30	58	GC	M62 = NGC 6266	TMO
U59	17.3 +43	52	GC	M92 = NGC 6341	TMO
U60	17.7 -32	58/69	OC	M6 = NGC 6405 = Butterfly Nebula	TMO
U61	17.8 +06	54	OC	IC 4665	HT
U62	17.9 -35	58/69	OC	M7 = NGC 6475	TMO
U63	18.1 -28	67	OC	NGC 6520	HT
U64	18.1 -24	67	BN	M8 = NGC 6523/6530 = Lagoon Nebula	TMO
U65	18.3 -16	67	BN	M17 = NGC 6618 = Omega Nebula	TMO
U66	18.5 +07	65	OC	NGC 6633	HT
U67	18.6 -24	67	GC	M22 = NGC 6656	TMO
U68	18.6 +05	65	OC	IC 4756	HT
U69	18.9 -06	65/67	OC	M11 = NGC 6705 = Wild Duck Cluster	TMO
U70	18.9 +10	65	OC	NGC 6709	HT
U71	18.9 +33	63	PN	M57 = NGC 6720 = Ring Nebula	TMO
U72	19.4 +20	64/65	OC	Cr 399 = Brocchi's Cluster	TMO/HT
U73	19.7 -14	66	PN	NGC 6818	HT
U74	19.7 +51	62	PN	NGC 6826	
U75	20.0 +23	64	PN	M27 = NGC 6853 = Dumbell Nebula	TMO
U76	20.4 +41	62	OC	NGC 6910	TSD
U77	20.6 +07	64	GC	NGC 6934	
U78	20.6 +28	62	OC	NGC 6940	HT
U79	21.1 -11	77	PN	NGC 7009	
U80	21.5 +12	75	GC	M15 = NGC 7078	TMO
U81	21.6 -01	75/77	GC	M2 = NGC 7089	TMO
U82	21.5 +48	73	OC	M39 = NGC 7092	TMO
U83	21.9 +63	71	OC	NGC 7160	TSD
U84	22.1 +47	73	OC	NGC 7209	TSD
U85	22.3 +50	73	OC	C16 = NGC 7243	TCO
U86	23.4 +43	72	PN	NGC 7662	
U87	23.9 +57	72	OC	NGC 7789	TCO/HT

U1

NGC 129

RA : 00.5^h PSA: 3

Dec: +60°

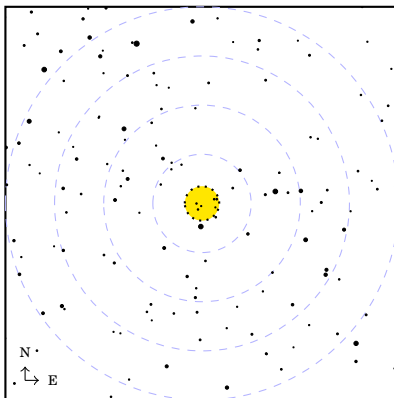
Con: And CSA: 2/7

Type: OC

Mag :

Size : 21'

See: DSC:TSD



U2

M32 = NGC 221

RA : 00.7^h PSA: 3

Dec: +41° OSA: N0

Con: And CSA: 2/7

Type: Gal

Mag : 8.2

Size : 11' × 7'

See: DSC:TMO OSA

Notes: M31/U3 is 0.5° N. M110 is 1° NW.

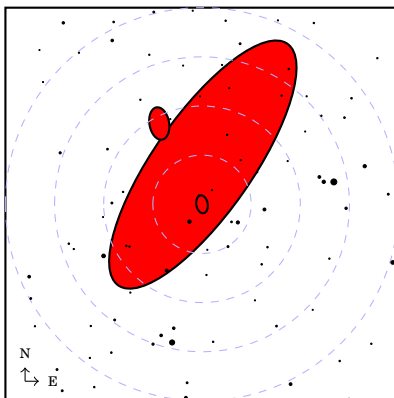
β And Mirach (mag 2.1)

→ 2.5° W

→ 3.0° N to μ And (mag 3.9)

→ 2.5° W

→ 2.5° N to M32/U2



U3

M31 = NGC 224 = Andromeda Galaxy

RA : 00.7^h PSA: 3

Dec: +41° OSA: N0

Con: And CSA: 2/7

Type: Gal

Mag : 3.4

Size : 180' × 60'

See: DSC:TMO OSA

Notes: M32/U2 is 0.5° S. M100 is 0.5° NW.

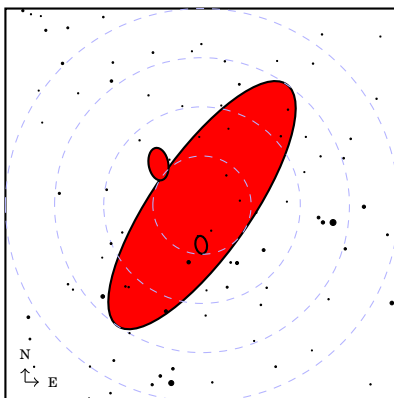
β And Mirach (mag 2.1)

→ 2.5° W

→ 3.0° N to μ And (mag 3.9)

→ 2.5° W

→ 3.0° N to M31/U3



U4

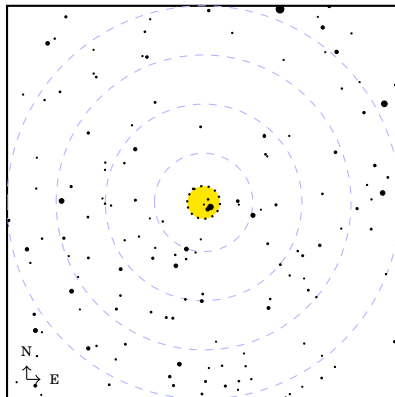
C13 = NGC 457 = E.T. Cluster = Owl Cluster

RA : 01.3^h PSA: 1/3 Type: OC

Dec: +58° OSA: N2 Mag : 6.4

Con: CSA: Size : 20'

See: DSC:TCO OSA



U5

C10 = NGC 663 = Horseshoe Cluster

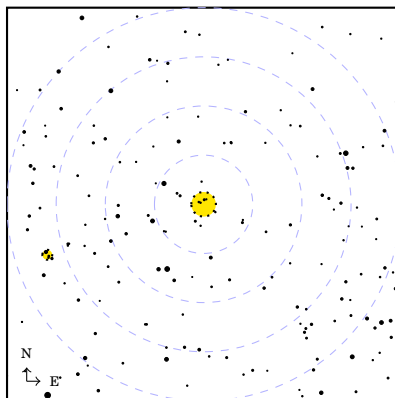
RA : 01.8^h PSA: 1/2 Type: OC

Dec: +61° OSA: N2 Mag : 6.7

Con: Cas CSA: 1/2 Size : 15'

See: DSC:TCO OSA

Notes: M103 is 1.7° WSW. C8 is 3° NW.



U6

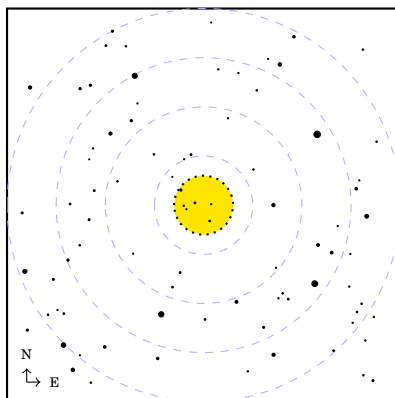
Cr 463

RA : 01.8^h PSA: 1 Type: OC

Dec: +72° OSA: Mag :

Con: CSA: Size : 36'

See: DSC:TSD



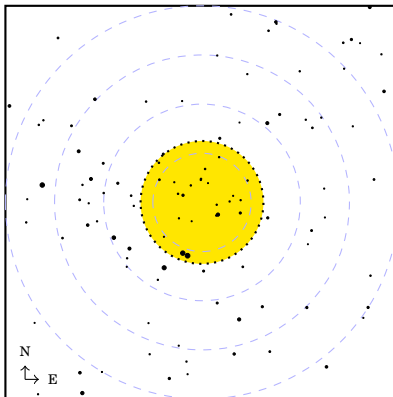
U7

C28 = NGC 752

RA : 01.9^h PSA: 2
Dec: +38° OSA: N0
Con: CSA:

Type: OC
Mag : 5.7
Size : 75'

See: DSC:TCO OSA



U8

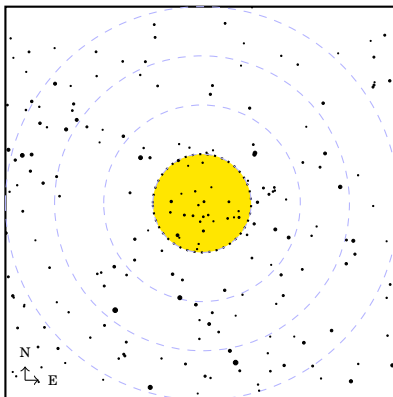
Stock 2

RA : 02.3^h PSA: 2
Dec: +60° OSA:
Con: CSA:

Type: OC
Mag :
Size : 60'

See: DSC:TSD

Notes: C14/U9/U10 is 2° S.



U9

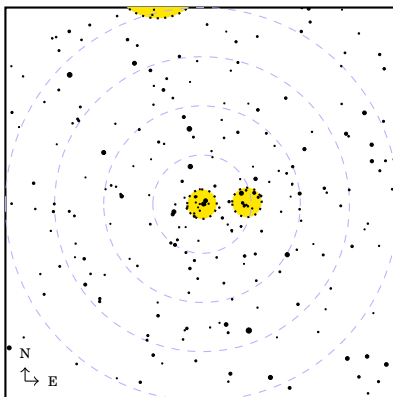
NGC 869 = h Per Cluster = half of C14

RA : 02.3^h PSA: 2
Dec: +57° OSA: N2
Con: CSA:

Type: OC
Mag :
Size : 18'

See: DSC:TCO DSC:TMO OSA

Notes: U9 and U10 form C14. U10 is 0.5° W. U8 is 2° N. U11 is 3.0° SE.



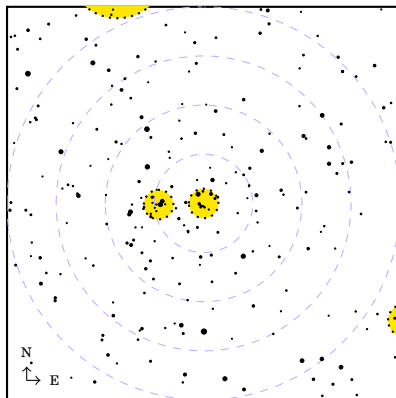
U10

NGC 884 = χ Per Cluster = half of C14

RA : 02.4 ^h	PSA: 2	Type: OC
Dec: +57°	OSA: N2	Mag :
Con: CSA:		Size : 18'

See: DSC:TCO DSC:TMO OSA

Notes: U9 and U10 form C14. U9 is 0.5° E. U8 is 2° N. U11 is 2.5° SE.



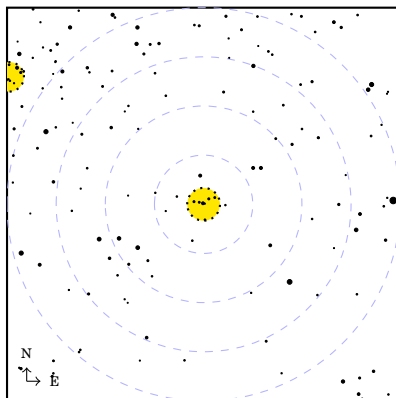
U11

Tr 2

RA : 02.6 ^h	PSA: 2	Type: OC
Dec: +56°	OSA:	Mag :
Con: CSA:		Size : 20'

See:

Notes: C14/U9/10 are 2.5° NW.

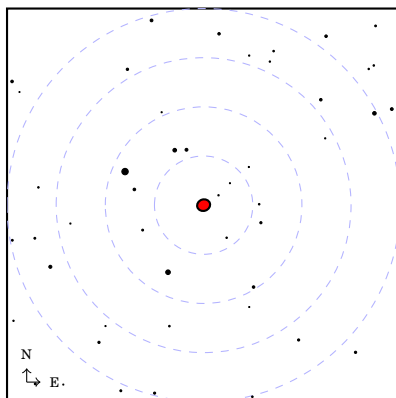


U12

M77 = NGC 1068

RA : 02.7 ^h	PSA: 4	Type: Gal
Dec: -00°	OSA: E0	Mag : 8.9
Con: Cet	CSA: 8	Size : 8' × 7'

See: DSC:TMO OSA



U13

Tr 3

RA : 03.2^h

PSA: 11/13

Type: OC

Dec: +63°

OSA:

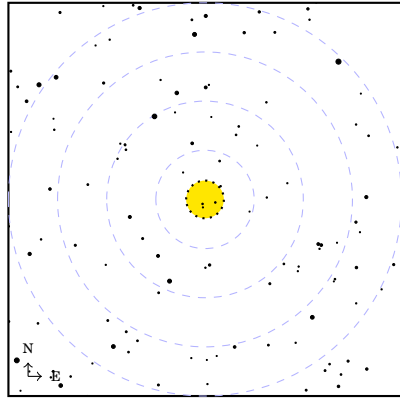
Mag :

Con:

CSA:

Size : 23'

See:



U14

Stock 23 = Pazmino's Cluster

RA : 03.3^h

PSA: 11/13

Type: OC

Dec: +60°

OSA:

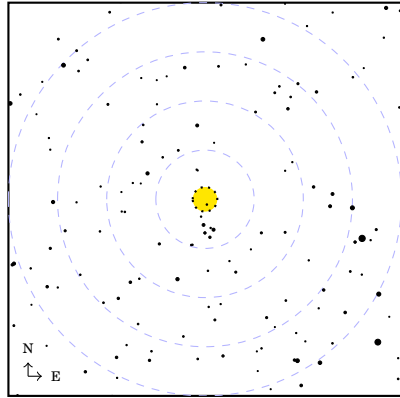
Mag :

Con:

CSA:

Size : 15'

See:



U15

Mel 20 = α Per Cluster

RA : 03.4^h

PSA: 3

Type: OC

Dec: +49°

OSA:

Mag :

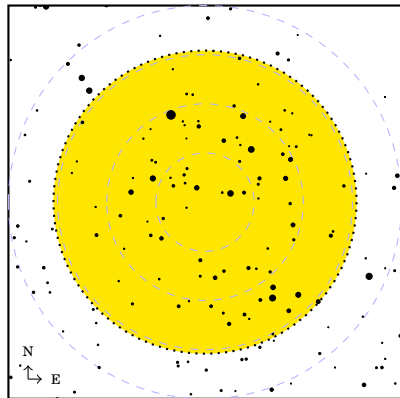
Con:

CSA:

Size : 185'

See:

Notes: Not shown in the PSA.



U16

NGC 1342

RA : 03.5^h PSA: 13

Dec: +37° OSA:

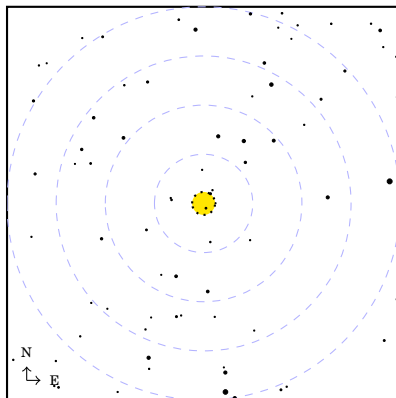
Con: CSA:

Type: OC

Mag :

Size : 14'

See: DSC:TSD



U17

M45 = Mel 22 = Pleiades

RA : 03.8^h PSA: 15/A

Dec: +24° OSA: E3

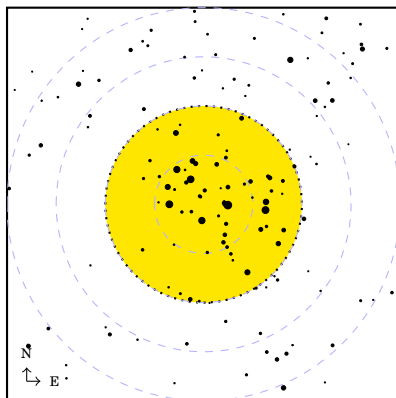
Con: Tau CSA: 2/3/8/9

Type: OC

Mag : 1.5

Size : 120'

See: DSC:TMO OSA



U18

C41 = Mel 25 = Hyades

RA : 04.4^h PSA: 15

Dec: +16° OSA: E3

Con: CSA:

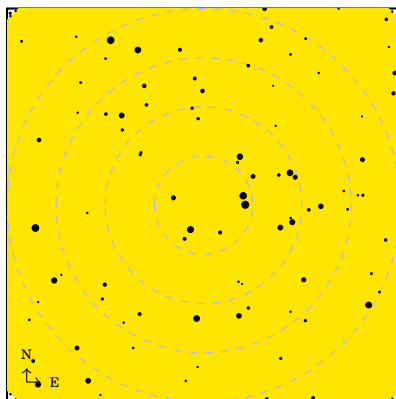
Type: OC

Mag : 0.5

Size : 330'

See: DSC:TCO OSA

Notes: Much bigger than the finder chart.



U19

NGC 1647

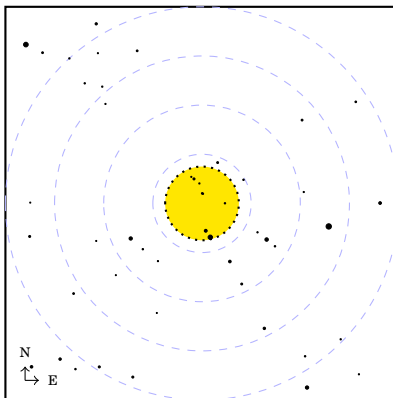
RA : 04.8 ^h	PSA: 15	Type: OC
Dec: +19°	OSA: E3	Mag :
Con:	CSA:	Size : 45'

See: DSC:TCO DSC:HT OSA

α Tau Betelgeuse (mag 0.3–0.9)

→ 2.5° N

→ 2.5° E to U19



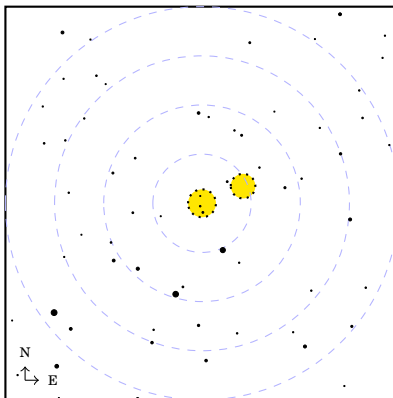
U20

NGC 1807

RA : 05.2 ^h	PSA: 14	Type: OC
Dec: +17°	OSA:	Mag :
Con:	CSA:	Size : 17'

See: DSC:TSD

Notes: U21 is 0.5° WSW.



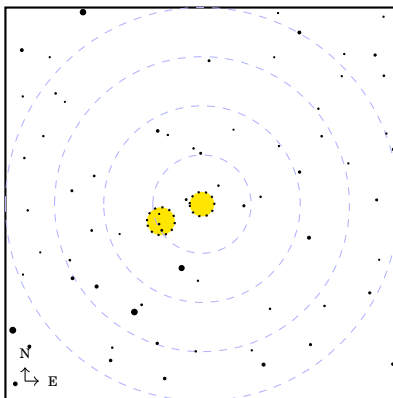
U21

NGC 1817

RA : 05.2 ^h	PSA: 14	Type: OC
Dec: +17°	OSA:	Mag :
Con:	CSA:	Size : 15'

See: DSC:TSD

Notes: U20 is 0.5° ENE.



U22

M38 = NGC 1912

RA : 05.5^h PSA: 12

Dec: +36° OSA: N6

Con: Aur CSA: 3

Type: OC

Mag : 6.4

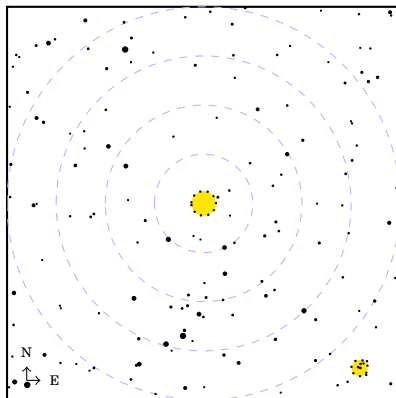
Size : 15'

See: DSC:TMO OSA

Notes: M36/U23 is 2.5° SE.

β Tau Elnath (mag 1.7)

→ 7° N to M38/U23



U23

M36 = NGC 1960

RA : 05.6^h PSA: 12

Dec: +34° OSA: N6

Con: Aur CSA: 3

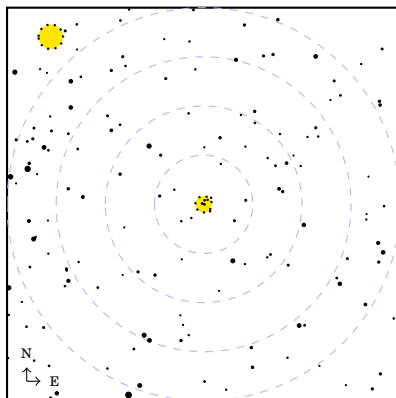
Type: OC

Mag : 6.0

Size : 10'

See: DSC:TMO OSA

Notes: M38/U22 is 2.5° NW.



U24

M42 = NGC 1976 = Orion Nebula

RA : 05.6^h PSA: 16/B

Dec: -05° OSA: E4

Con: Ori CSA: 9

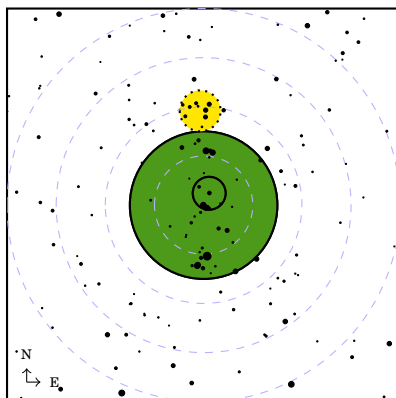
Type: BN

Mag : 3.7

Size : 90' × 60'

See: DSC:TMO OSA

Notes: M42/U24 surrounds M43. U25 is 1° N.



U25

NGC 1981

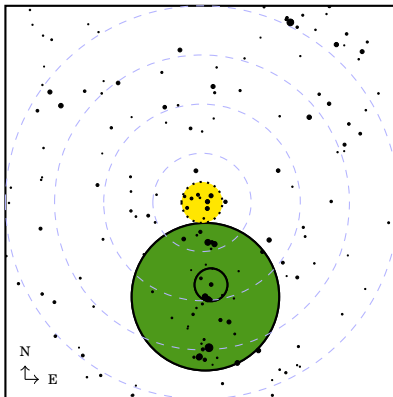
RA : 05.6^h PSA: 16/B Type: OC

Dec: -04° OSA: E4 Mag :

Con: CSA: Size : 25'

See: DSC:HT OSA

Notes: M42/U24 and M43 are 1° S.



U26

M37 = NGC 2099

RA : 05.9^h PSA: 12 Type: OC

Dec: +33° OSA: N6 Mag : 5.6

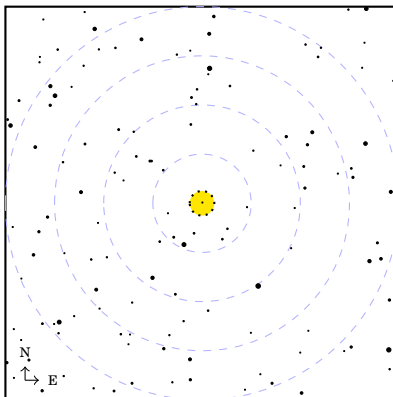
Con: Aur CSA: 3 Size : 15'

See: DSC:TMO OSA

β Tau Elnath (mag 1.7)

→ 6° W to point 1° N of 136 Tau

→ 4° N to M37/U26



U27

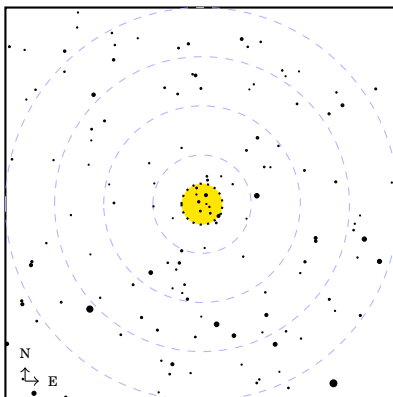
M35 = NGC 2168

RA : 06.2^h PSA: 14 Type: OC

Dec: +24° OSA: E7 Mag : 5.1

Con: Gem CSA: 3/9 Size : 25'

See: DSC:TMO OSA



U28

NGC 2169

RA : 06.1^h

PSA: 14/25

Type: OC

Dec: +14°

OSA:

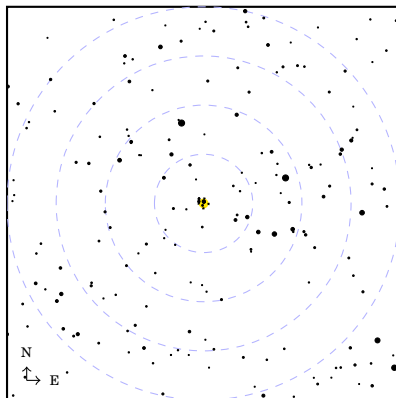
Mag :

Con:

CSA:

Size : 6'

See: DSC:HT



U29

NGC 2232

RA : 06.5^h

PSA: 25/27

Type: OC

Dec: -05°

OSA:

Mag :

Con:

CSA:

Size : 29'

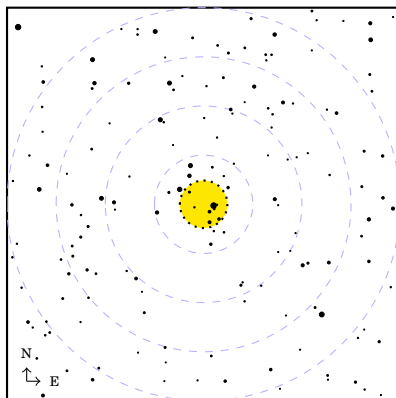
See: DSC:TSD

κ Ori Saiph (mag 2.1)

→ 3° N

→ 11° E passing γ Mon to β Mon

→ 2.5° N to U29



U30

C50 = NGC 2244

RA : 06.5^h

PSA: 25/E

Type: OC

Dec: +05°

OSA: E9

Mag : 4.4

Con:

CSA:

Size : 30'

See: DSC:TCO OSA

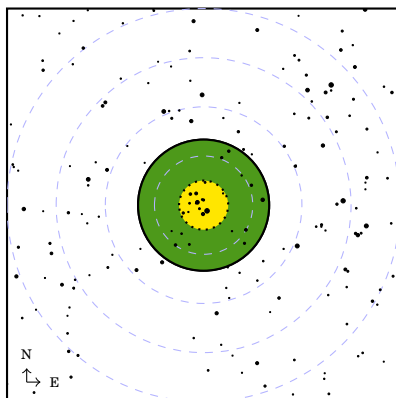
Notes: Surrounded by C49.

α Ori Betelgeuse (mag 0.3–0.9)

→ 3° S

→ 7° E to 8 Mon

→ 2° E to C50/U30



U31

NGC 2264

RA : 06.7^h PSA: 25/E Type: OC

Dec: +10° OSA: E7 Mag :

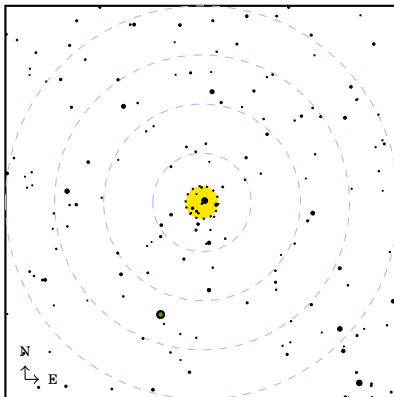
Con: CSA: Size : 20'

See: DSC:HT OSA

Notes: C46 is 1.25° SSW.

γ Gem Alhena (mag 1.9)

→ 7° S passing ξ Gem and 30 Gem to U31



U32

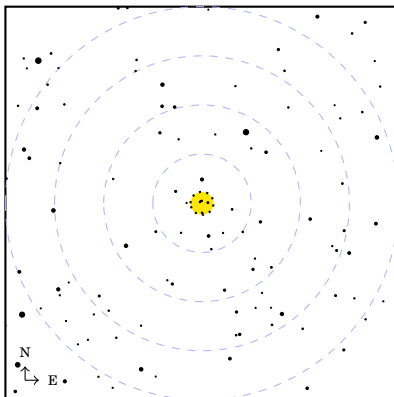
NGC 2281

RA : 06.8^h PSA: 23 Type: OC

Dec: +41° OSA: N6 Mag :

Con: CSA: Size : 14'

See: DSC:TSD OSA



U33

M41 = NGC 2287

RA : 06.8^h PSA: 27 Type: OC

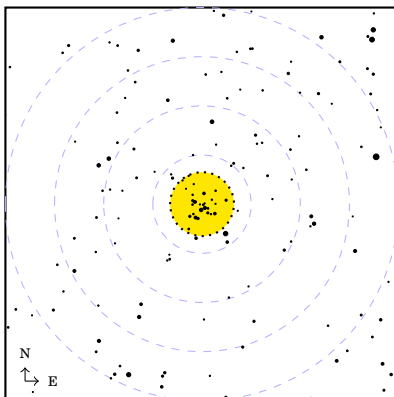
Dec: -21° OSA: E6 Mag : 4.5

Con: CMa CSA: 9/15 Size : 39'

See: DSC:TMO OSA

α CMa Sirius (mag -1.5)

→ 4° S to M41/U33



U34

NGC 2301

RA : 06.9^h PSA: 25/27

Dec: +00° OSA: E6

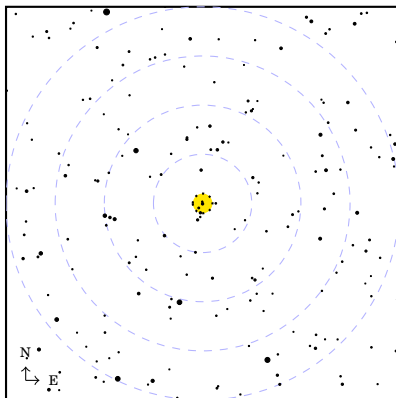
Con: CSA:

Type: OC

Mag :

Size : 12'

See: DSC:HT OSA



U35

M50 = NGC 2323

RA : 07.0^h PSA: 27

Dec: -08° OSA: E8

Con: Mon

CSA: 9

Type: OC

Mag : 5.9

Size : 15'

See: DSC:TMO OSA

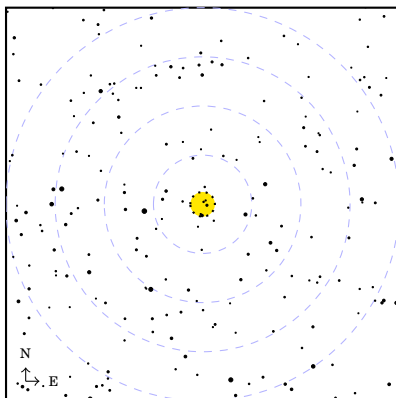
α CMa Sirius (mag -1.5)

→ 3° E to ι CMa

→ 5° N passing μ CMa to θ CMa

→ 4° N to asterism of three stars

→ 2° E to M50/U35



U36

C39 = NGC 2392 = Eskimo Nebula

RA : 07.5^h PSA: 25

Dec: +21° OSA: E7

Con:

CSA:

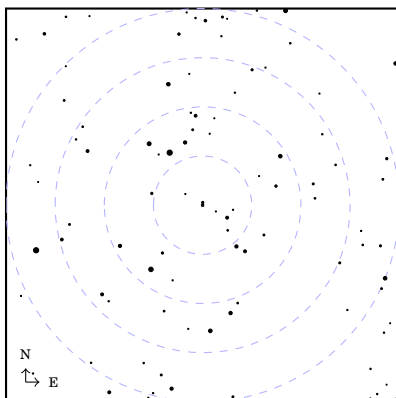
Type: PN

Mag : 9.2

Size : 0.8' × 0.7'

See: DSC:TCO DSC:TMO OSA

Notes: The open cluster NGC 2420 is 2° N (see DSC:TCO).



U37

NGC 2539

RA : 08.2^h PSA: 26

Dec: -13° OSA: E8

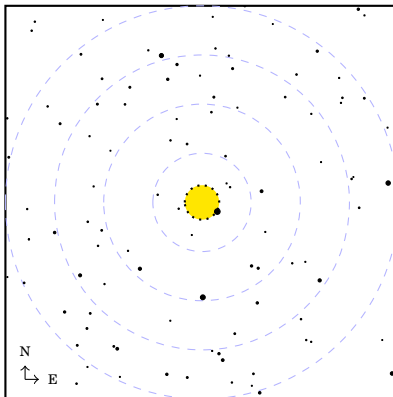
Con: CSA:

Type: OC

Mag :

Size : 21'

See: DSC:HT OSA



U38

M48 = NGC 2548

RA : 08.2^h PSA: 26

Dec: -06° OSA: E10

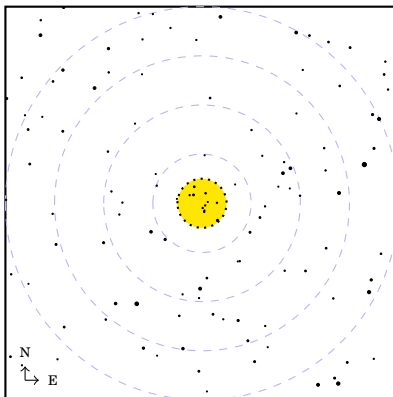
Con: Hya CSA: 9/10

Type: OC

Mag : 5.8

Size : 30'

See: DSC:TMO OSA



U39

M44 = NGC 2632 = Beehive Cluster

RA : 08.7^h PSA: 24

Dec: +20° OSA: E9

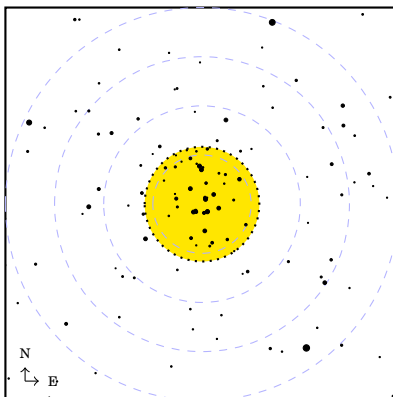
Con: Cnc CSA: 3/4/9/10

Type: OC

Mag : 3.1

Size : 70'

See: DSC:TMO OSA



U40

M67 = NGC 2682

RA : 08.9^h PSA: 24

Dec: +12° OSA: E9

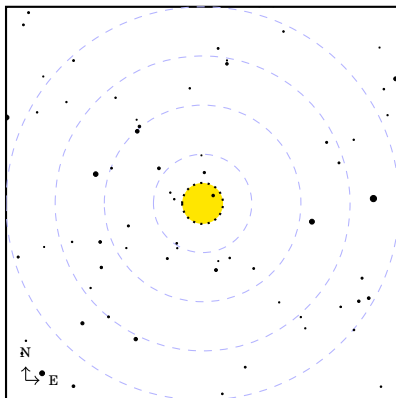
Con: Cnc CSA: 10

See: DSC:TMO OSA

Type: OC

Mag : 6.0

Size : 25'



U41

M81 = NGC 3031

RA : 09.9^h PSA: 31

Dec: +69° OSA: N8

Con: UMa CSA: 1/4

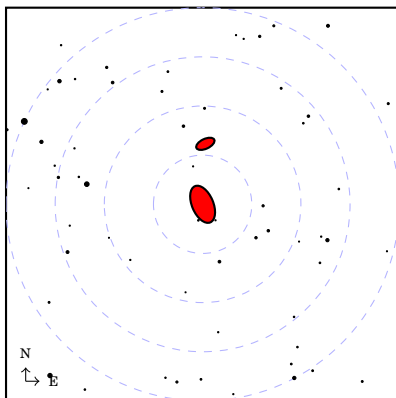
See: DSC:TMO OSA

Notes: M82/U42 is 0.6° N.

Type: Gal

Mag : 6.9

Size : 24' × 13'



U42

M82 = NGC 3034

RA : 09.9^h PSA: 31

Dec: +70° OSA: N8

Con: UMa CSA: 1/4

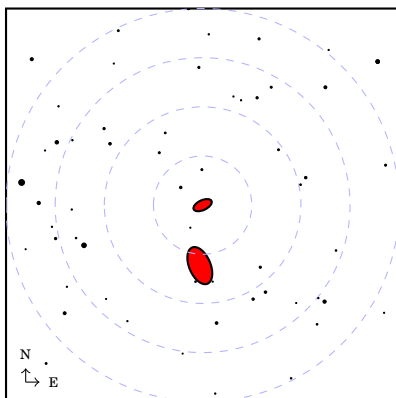
See: DSC:TMO OSA

Notes: M81/U41 is 0.6° S.

Type: Gal

Mag : 8.4

Size : 12' × 6'

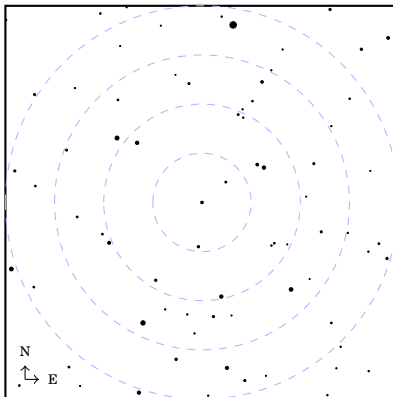


U43

C59 = NGC 3242 = Ghost of Jupiter

RA : 10.4 ^h	PSA: 36/37	Type: PN
Dec: -19°	OSA: E10	Mag : 7.3
Con:	CSA:	Size : 0.8' × 0.6'

See: DSC:TCO DSC:TMO OSA



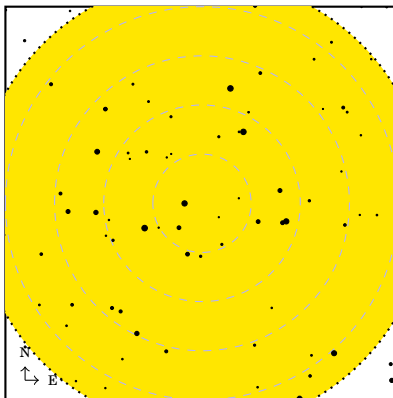
U44

Mel 111 = Coma Star Cluster

RA : 12.4 ^h	PSA: 45	Type: OC
Dec: +26°	OSA: E13	Mag :
Con:	CSA:	Size : 275'

See: DSC:HT OSA

Notes: Not shown in the PSA.



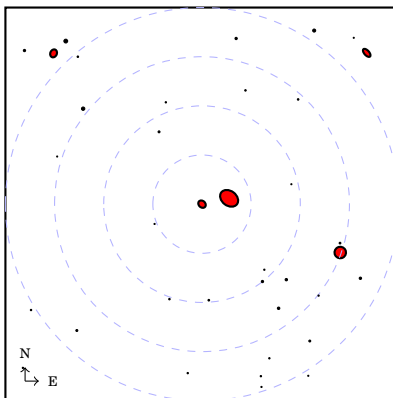
U45

M84 = NGC 4374

RA : 12.4 ^h	PSA: 45/C	Type: Gal
Dec: +13°	OSA: E14	Mag : 9.1
Con: Vir	CSA: 4/10/11	Size : 5' × 4'

See: DSC:TMO OSA

Notes: In the Virgo Cluster.



U46

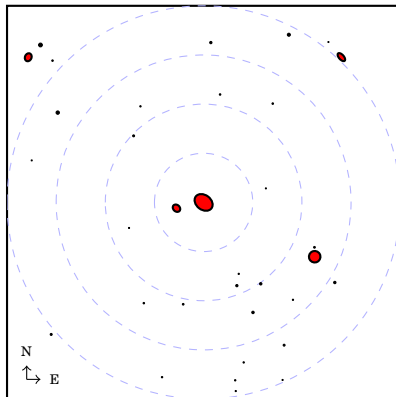
M86 = NGC 4406

RA : 12.4^h PSA: 45/C
Dec: +13° OSA: E14
Con: Vir CSA: 4/10/11

Type: Gal
Mag : 8.9
Size : 12' × 9'

See: DSC:TMO OSA

Notes: In the Virgo Cluster.



U47

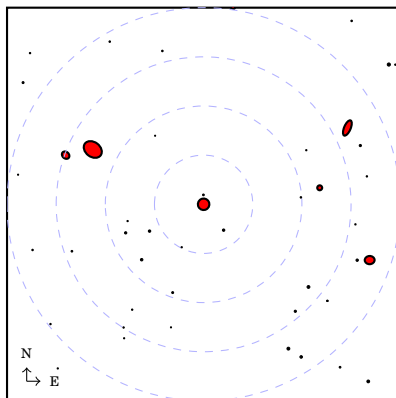
M87 = NGC 4486

RA : 12.5^h PSA: 45/C
Dec: +12° OSA: E14
Con: Vir CSA: 10/11

Type: Gal
Mag : 8.6
Size : 7'

See: DSC:TMO OSA

Notes: In the Virgo Cluster.



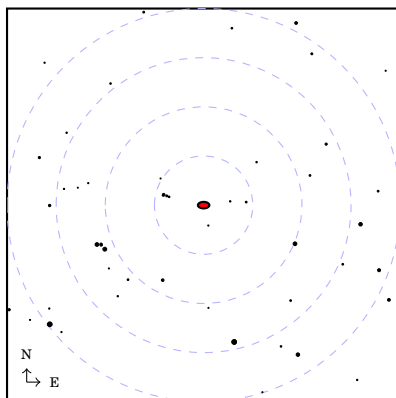
U48

M104 = NGC 4594 = Sombrero Galaxy

RA : 12.7^h PSA: 47
Dec: -12° OSA: E12
Con: Vir CSA: 10/11

Type: Gal
Mag : 8.0
Size : 7' × 4'

See: DSC:TMO OSA



U49

M94 = NGC 4736

RA : 12.8^h PSA: 43

Dec: +41° OSA: N12

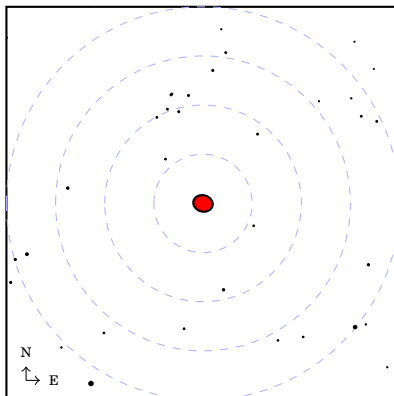
Con: CVn CSA: 4/5

Type: Gal

Mag : 8.2

Size : 12' × 10'

See: DSC:TMO OSA



U50

M64 = NGC 4826 = Black-Eye Galaxy

RA : 12.9^h PSA: 45

Dec: +22° OSA: E13

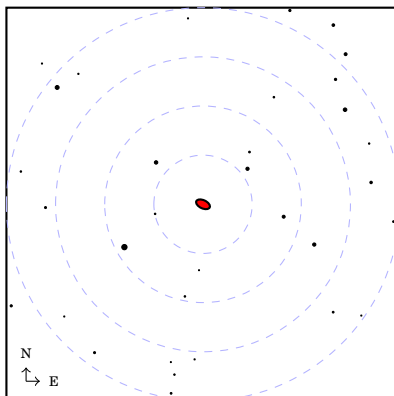
Con: Com CSA: 5/11

Type: Gal

Mag : 8.5

Size : 9' × 5'

See: DSC:TMO OSA



U51

M3 = NGC 5272

RA : 13.7^h PSA: 43/44

Dec: +28° OSA: E15

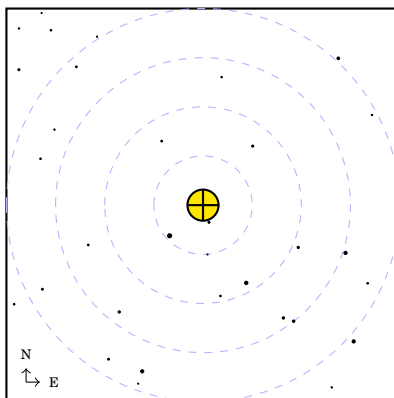
Con: CVn CSA: 5

Type: GC

Mag : 5.9

Size : 19'

See: DSC:TMO OSA



U52

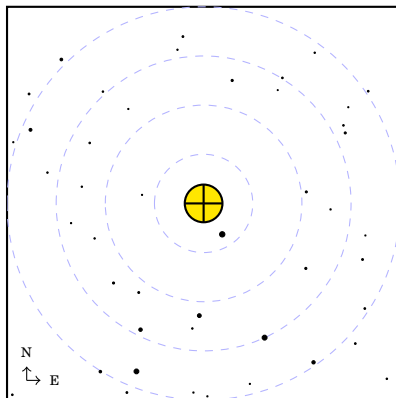
M5 = NGC 5904

RA : 15.3^h PSA: 55/57 Type: GC

Dec: +02° OSA: E15 Mag : 5.7

Con: Ser CSA: 11/12 Size : 23'

See: DSC:TMO OSA



U53

M4 = NGC 6121

RA : 16.4^h PSA: 56/58 Type: GC

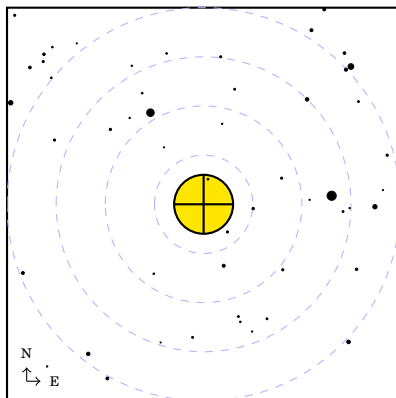
Dec: -27° OSA: E18 Mag : 5.4

Con: Sco CSA: 11/12/17/18 Size : 36'

See: DSC:TMO OSA

α Sco Antares (mag 0.9–1.1)

→ 1.2° W to M4



U54

M13 = NGC 6205 = Hercules Cluster

RA : 16.7^h PSA: 52 Type: GC

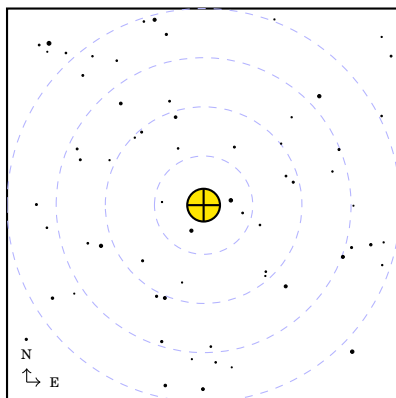
Dec: +36° OSA: N14 Mag : 5.3

Con: Her CSA: 5/6 Size : 20'

See: DSC:TMO OSA

ζ Her (mag 2.8)

→ 5° N (2/3 distance to η Her) to M13/U54



U55

NGC 6210

RA : 16.7^h

PSA: 54

Type: PN

Dec: +24°

OSA: E19

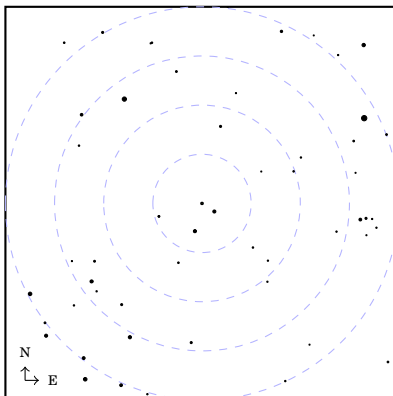
Mag :

Con:

CSA: 12

Size : 0.8'

See: DSC:HT OSA



U56

M12 = NGC 6218

RA : 16.8^h

PSA: 54/56

Type: GC

Dec: -02°

OSA: E17

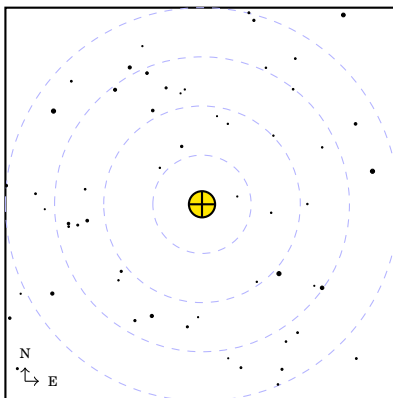
Mag : 6.1

Con: Oph

CSA: 12

Size : 16'

See: DSC:TMO OSA



U57

M10 = NGC 6254

RA : 17.0^h

PSA: 54/56

Type: GC

Dec: -04°

OSA: E17

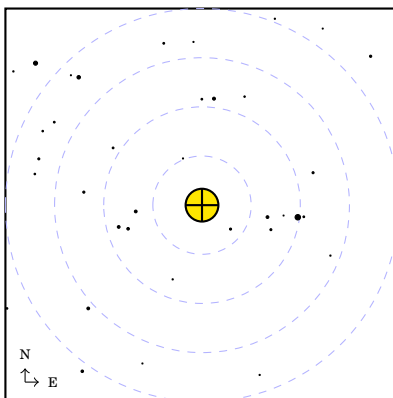
Mag : 6.6

Con: Oph

CSA: 12

Size : 20'

See: DSC:TMO OSA



U58

M62 = NGC 6266

RA : 17.0^h PSA: 58
Dec: -30° OSA: E18
Con: Oph CSA: 17/18

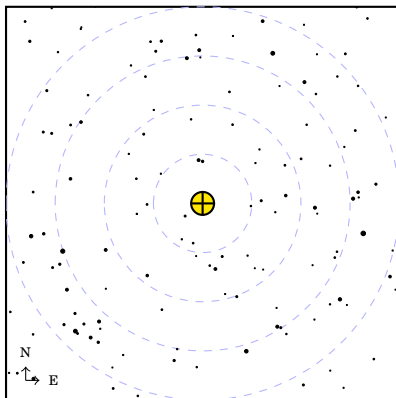
Type: GC
Mag : 6.7
Size : 14'

See: DSC:TMO OSA

ε Sco (mag 2.3)

→ 2° E

→ 4° N to M63



U59

M92 = NGC 6341

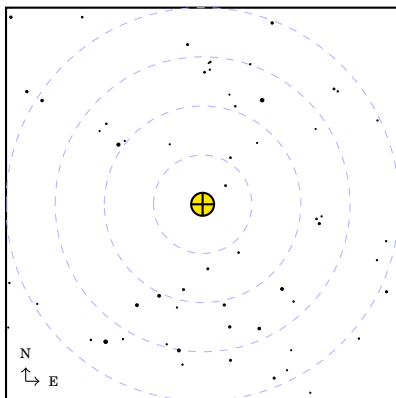
RA : 17.3^h PSA: 52
Dec: +43° OSA: N14
Con: Her CSA: 5/6

Type: GC
Mag : 6.5
Size : 14'

See: DSC:TMO OSA

π Her (mag 3.2)

→ 6.5° N to M92/U59



U60

M6 = NGC 6405 = Butterfly Nebula

RA : 17.7^h PSA: 58/69
Dec: -32° OSA: E18
Con: Sco CSA: 18

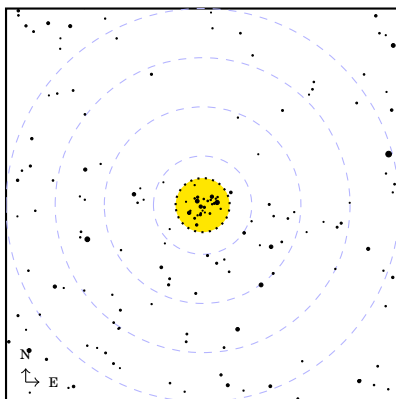
Type: OC
Mag : 4.2
Size : 33'

See: DSC:TMO OSA

λ Sco Shaula (mag 1.6)

→ 5° N

→ 1° E to M6/U60



U61

IC 4665

RA : 17.8^h

PSA: 54

Type: OC

Dec: +06°

OSA: E17

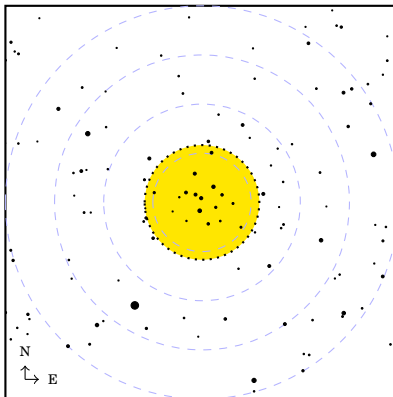
Mag :

Con:

CSA:

Size : 70'

See: DSC:HT OSA



U62

M7 = NGC 6475

RA : 17.9^h

PSA: 58/69

Type: OC

Dec: -35°

OSA: E18

Mag : 2.8

Con: Sco

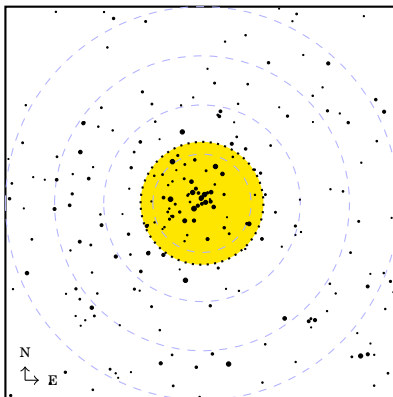
CSA: 18

Size : 75'

See: DSC:TMO OSA

ε Sgr Kaus Australis (mag 1.8)

→ 6° W to M7/U62



U63

NGC 6520

RA : 18.1^h

PSA: 67

Type: OC

Dec: -28°

OSA:

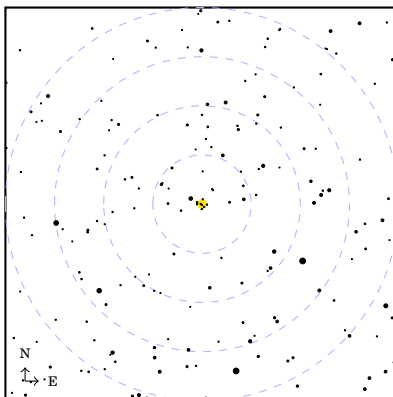
Mag :

Con:

CSA:

Size : 6'

See: DSC:HT



U64

M8 = NGC 6523/6530 = Lagoon Nebula

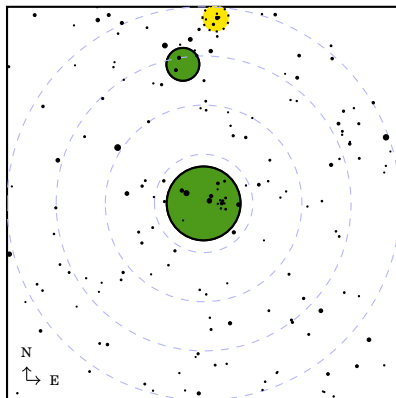
RA : 18.1 ^h	PSA: 67	Type: BN
Dec: -24°	OSA: E20	Mag : 3.0
Con: Sgr	CSA: 12/18	Size : 45' × 30'

See: DSC:TMO OSA

Notes: M20 is 1.5° N. M21 is 2.0° N.

γ Sgr Alnasl (mag 3.0)

→ 6° N to M8/U64



U65

M17 = NGC 6618 = Omega Nebula

RA : 18.3 ^h	PSA: 67	Type: BN
Dec: -16°	OSA: E20	Mag : 6.0
Con: Sgr	CSA: 12	Size : 20' × 15'

See: DSC:TMO OSA

Notes: M18 is 1° S. M24 is 3° SW.

ν Oph (mag 3.3)

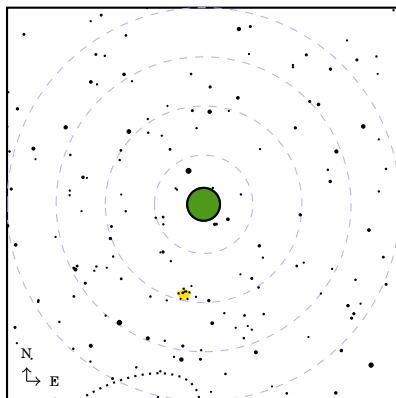
→ 5° E

→ 6° S to M17/U65

μ Sgr (mag 3.8)

→ 5° N passing M24 and M18

→ 2° E to M17

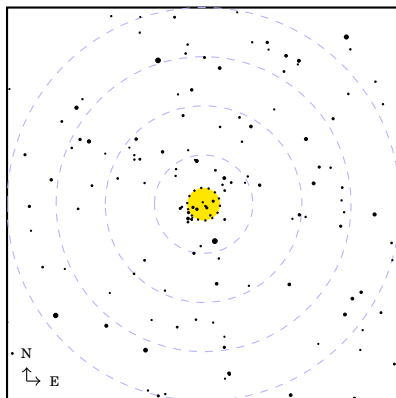


U66

NGC 6633

RA : 18.5 ^h	PSA: 65	Type: OC
Dec: +07°	OSA: E19	Mag :
Con:	CSA:	Size : 20'

See: DSC:HT OSA



U67

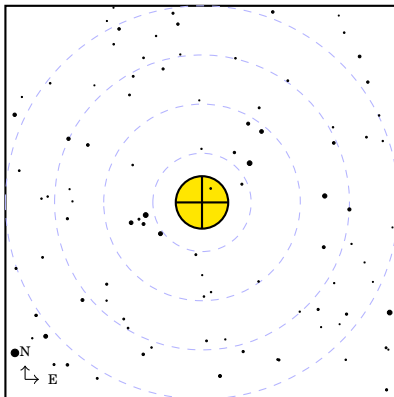
M22 = NGC 6656

RA : 18.6^h PSA: 67 Type: GC

Dec: -24° OSA: E20 Mag: 5.2

Con: Sgr CSA: 12/18 Size : 32'

See: DSC:TMO OSA



U68

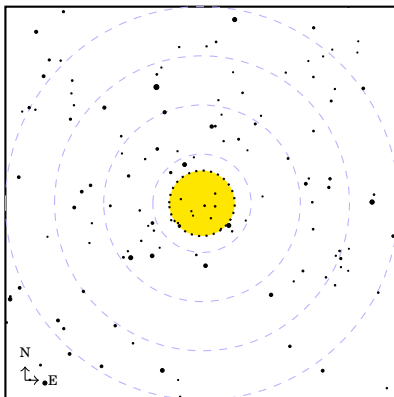
IC 4756

RA : 18.6^h PSA: 65 Type: OC

Dec: +05° OSA: E19 Mag :

Con: CSA: Size : 40'

See: DSC:HT OSA



U69

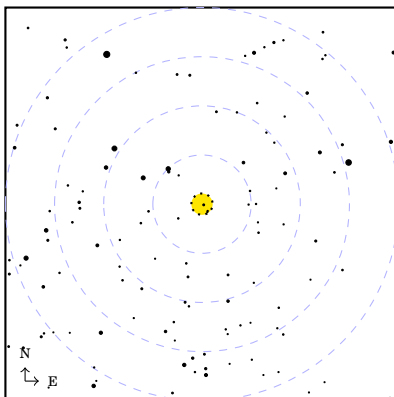
M11 = NGC 6705 = Wild Duck Cluster

RA : 18.9^h PSA: 65/67 Type: OC

Dec: -06° OSA: E19 Mag : 5.3

Con: Sct CSA: 12 Size : 13'

See: DSC:TMO OSA



U70

NGC 6709

RA : 18.9^h PSA: 65

Dec: +10° OSA:

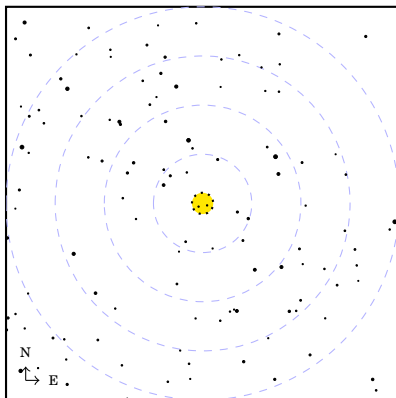
Con: CSA:

Type: OC

Mag :

Size : 13'

See: DSC:HT



U71

M57 = NGC 6720 = Ring Nebula

RA : 18.9^h PSA: 63

Dec: +33° OSA: N18

Con: Lyr CSA: 6/7

Type: PN

Mag : 8.8

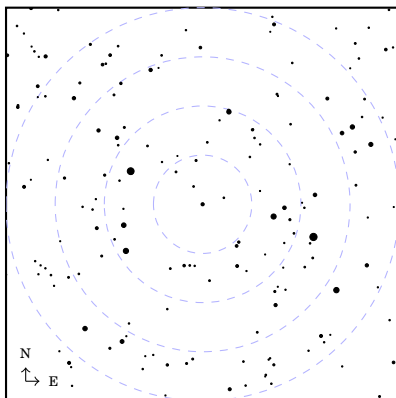
Size : 1.1'

See: DSC:TMO OSA

γ Lyr Sulaphat (mag 3.2)

→ 1.2° W

→ 0.4° N to M57/U71



U72

Cr 399 = Brocchi's Cluster

RA : 19.4^h PSA: 64/65

Dec: +20° OSA:

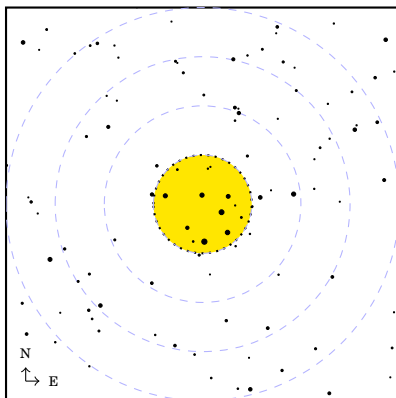
Con: CSA:

Type: OC

Mag :

Size : 60'

See: DSC:TMO DSC:HT



U73

NGC 6818

RA : 19.7^h PSA: 66

Dec: -14° OSA: E22

Con: CSA:

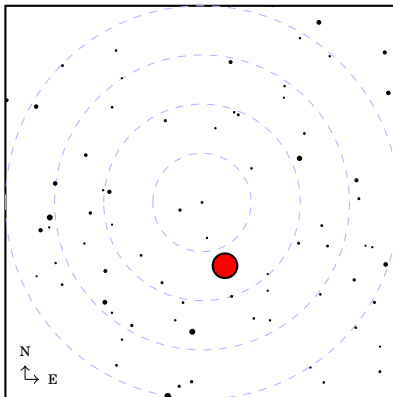
Type: PN

Mag :

Size : 0.4' × 0.3'

See: DSC:HT OSA

Notes: C57 is 0.7° SSE.



U74

NGC 6826

RA : 19.7^h PSA: 62

Dec: +51° OSA: N18

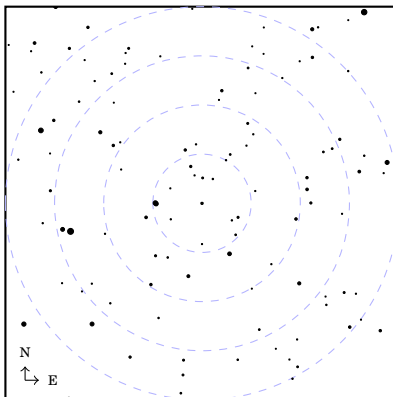
Con: CSA:

Type: PN

Mag :

Size : 0.5' × 0.4'

See: OSA



U75

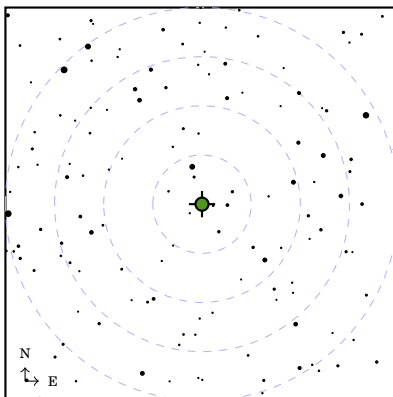
M27 = NGC 6853 = Dumbell Nebula

RA : 20.0^h PSA: 64 Type: PN

Dec: +23° OSA: E21 Mag : 7.3

Con: Vul CSA: 6/7/12/13 Size : 8' × 6'

See: DSC:TMO OSA



U76

NGC 6910

RA : 20.4^h PSA: 62

Dec: +41° OSA:

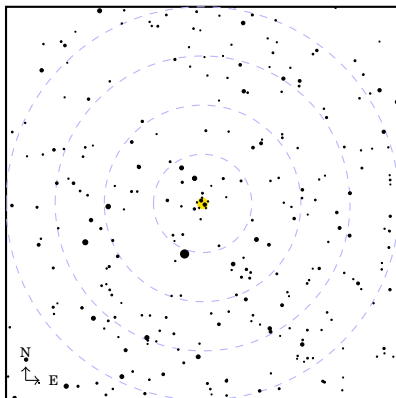
Con: CSA:

See: DSC:TSD

Type: OC

Mag :

Size : 7'



U77

NGC 6934

RA : 20.6^h PSA: 64

Dec: +07° OSA:

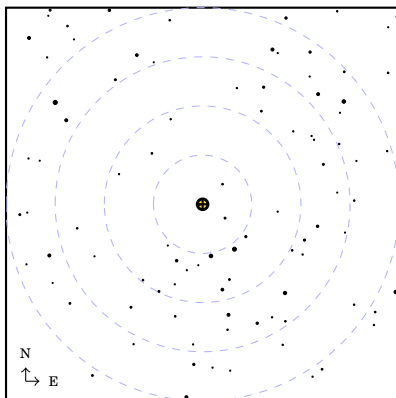
Con: CSA:

See:

Type: GC

Mag :

Size : 6'



U78

NGC 6940

RA : 20.6^h PSA: 62

Dec: +28° OSA: N20

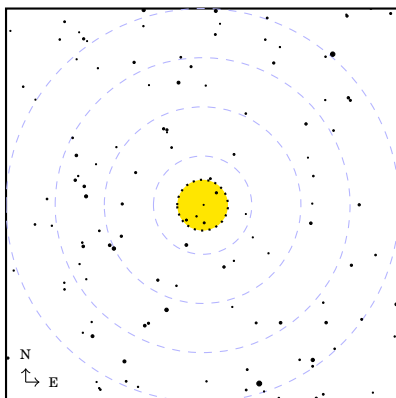
Con: CSA:

See: DSC:HT OSA

Type: OC

Mag :

Size : 31'



U79

NGC 7009

RA : 21.1^h

PSA: 77

Dec: -11°

OSA: E24

Con:

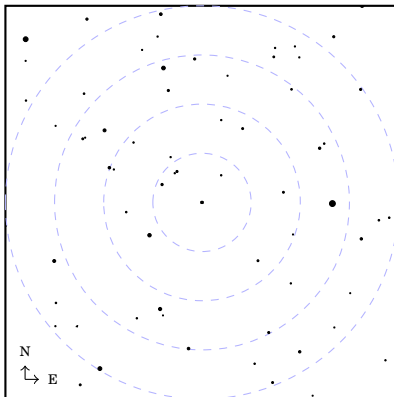
CSA:

Type: PN

Mag :

Size : 0.7' × 0.4'

See: OSA



U80

M15 = NGC 7078

RA : 21.5^h

PSA: 75

Dec: +12°

OSA: E23

Con: Peg

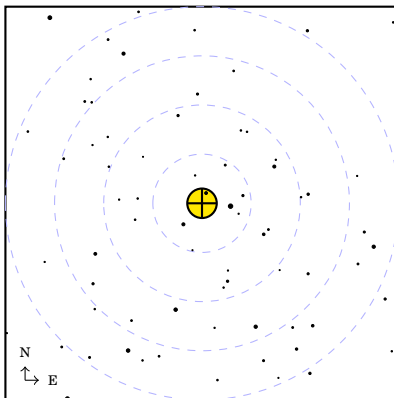
CSA: 13

Type: GC

Mag : 6.0

Size : 18'

See: DSC:TMO OSA



U81

M2 = NGC 7089

RA : 21.6^h

PSA: 75/77

Dec: -01°

OSA: E24

Con: Aqr

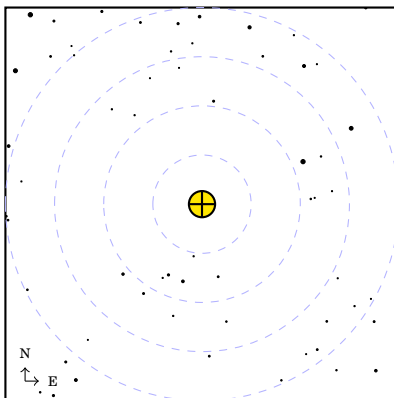
CSA: 13

Type: GC

Mag : 6.3

Size : 16'

See: DSC:TMO OSA



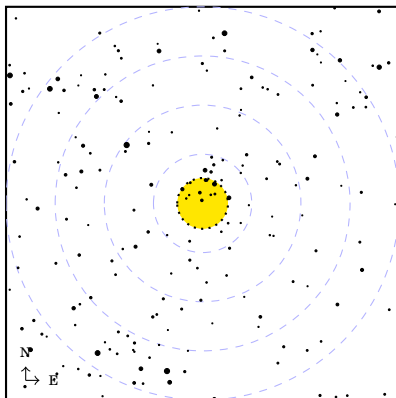
U82

M39 = NGC 7092

RA : 21.5^h PSA: 73
Dec: +48° OSA: N24
Con: Cyg CSA: 1/6/7

Type: OC
Mag : 4.6
Size : 31'

See: DSC:TMO OSA



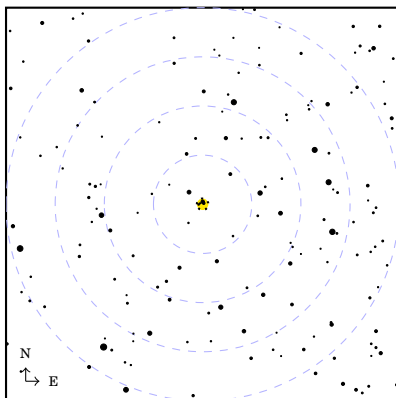
U83

NGC 7160

RA : 21.9^h PSA: 71
Dec: +63° OSA:
Con: CSA:

Type: OC
Mag :
Size : 7'

See: DSC:TSD



U84

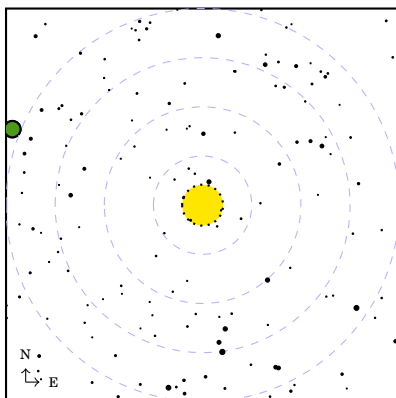
NGC 7209

RA : 22.1^h PSA: 73
Dec: +47° OSA: N24
Con: CSA:

Type: OC
Mag :
Size : 25'

See: DSC:TSD OSA

Notes: C19 is 2° WNW.



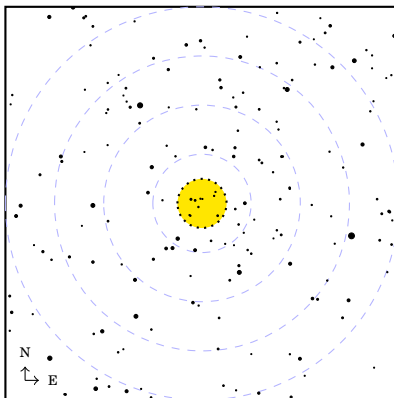
U85

C16 = NGC 7243

RA : 22.3^h PSA: 73
Dec: +50° OSA: N24
Con: CSA:

Type: OC
Mag: 6.4
Size : 30'

See: DSC:TCO OSA



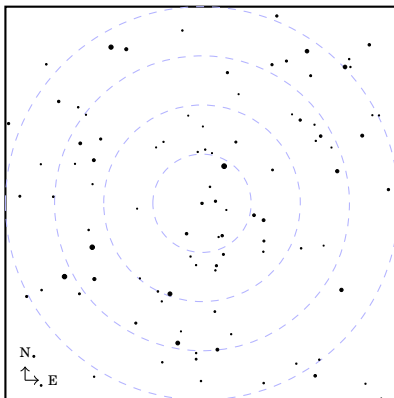
U86

NGC 7662

RA : 23.4^h PSA: 72
Dec: +43° OSA: N24
Con: CSA:

Type: PN
Mag :
Size : 0.5'

See: OSA



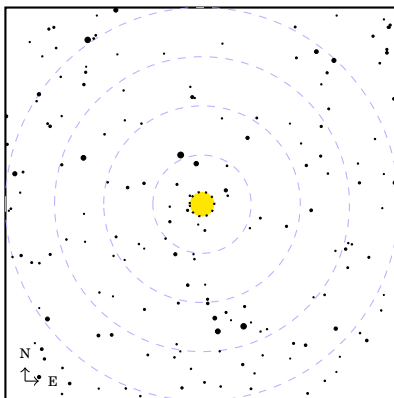
U87

NGC 7789

RA : 23.9^h PSA: 72
Dec: +57° OSA: N22
Con: CSA:

Type: OC
Mag :
Size : 15'

See: DSC:TCO DSC:HT OSA



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